

8. Electrical Sub-Block §

8.1 Electrical Specification Introduction §

Key attributes of the Electrical Specification Include:

- Support for NRZ signaling at 2.5, 5.0, 8.0, 16.0, 32.0 GT/s, and PAM4 signaling at 64.0 GT/s data rates
- Support for common and separate independent reference clock architectures
- Support for Spread Spectrum clocking
- Reduced swing mode for lower power Link operation
- In-band Receiver detection and electrical idle detection
- Channel compliance methodology
- Adaptive transmitter equalization and reference Receiver equalization allowing closed eye channel support at 8.0, 16.0, and 32.0 GT/s, and 64.0 GT/s
- Lane margining
- AC coupled channel

Please note that throughout this specification, the term GT/s is used to refer to the number of encoded bits transferred in a second on a direction of a lane. In PAM4 signaling, two bits are encoded in one UI with four voltage levels § Section 4.2.3.1.1. Consequently, the Nyquist frequency is 16 GHz for both 32.0 GT/s NRZ and 64.0 GT/s PAM4.

Because of four voltage levels and reduced amplitude for each voltage level, 64.0 GT/s PAM4 signaling is sensitive to noise and burst errors. The Bit Error Rate (BER), also referred as First Bit Error Rate (FBER) in § Chapter 4, for 64.0 GT/s is 10^{-6} . FBER refers to Bit Error Rate without accounting for any burst error. For 2.5, 5.0, 8.0, 16.0, and 32.0 GT/s data rates, BER of 10^{-12} implicitly assumes FBER of 10^{-12} and do not account for any types of burst error.

The 6.0 version of the PCI Express Electrical Specification is organized into separate sections for the Transmitter, Receiver, the Channel, and the Refclk. In this version most parameters have been regularized such that a common set of parameters is used to define compliance at all data rates.

8.2 Interoperability Criteria §

8.2.1 Data Rates §

A device must support 2.5 GT/s and is not permitted to skip support for any data rates between 2.5 GT/s and the highest supported rate.

8.2.2 Refclk Architectures §

PCIe supports two Refclk data architectures: Common Refclk, and Independent Refclk. These are described in detail in § Section 8.6. A PCIe device may support one or more of these architectures.

8.3 Transmitter Specification §

8.3.1 Measurement Setup for Characterizing Transmitters §

The PCI Express electrical specification references all measurements to the device's pin. However, the pin of a device under test (DUT) is not generally accessible, and the closest accessible point is usually a pair of microwave-type coaxial connectors separated from the DUT pins by several inches of PCB trace, called the breakout channel on a silicon test board. On a test board with many Lanes the minimum breakout channel length is constrained by the need to route to many coaxial connectors. Typically, this limitation holds true for both the Tx and the Rx pins. § Figure 8-1 illustrates a typical test connection to a DUT, showing a single Tx Lane breakout.

A low jitter Refclk source is used when the silicon supports using an external reference clock in order that the jitter measurements for the DUT include only contributions from the Transmitter.

When testing a Transmitter it is desirable to have as many other PCI Express Lanes sending or receiving the compliance pattern as is feasible. Similarly, if the device supports other I/O it should also be sending or receiving on these interfaces. The goal is to have the Tx test environment replicate that found in a real system as closely as possible.

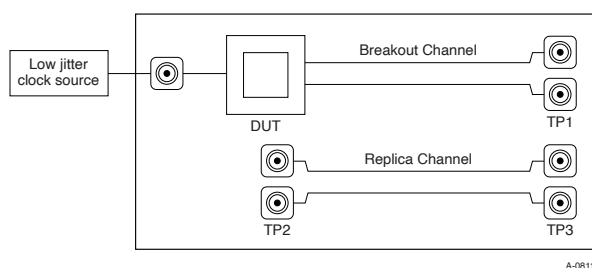


Figure 8-1 Tx Test Board for Non-Embedded Refclk §

The 6.0 version of the Tx specification also includes explicit support for Transmitters utilizing embedded Refclks. In this case the Tx under test is not driven with a low jitter clock source, and both the Tx data and Tx Refclk out must be sampled simultaneously by means of a 2-port measurement as shown in § Figure 8-2. For more details consult § Section 8.3.5.3. When an implementation is tested that is configured for the independent reference clock architecture only the data is sampled for both the Non-Embedded and Embedded reference clock cases.

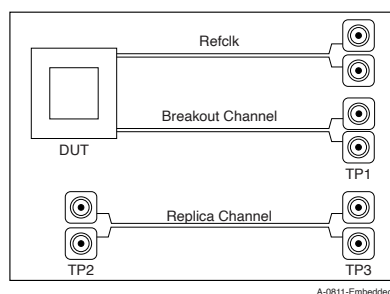


Figure 8-2 Tx Test board for Embedded Refclk §

8.3.1.1 Breakout and Replica Channels §

In order to specify a Transmitter with a uniform set of Tx parameters it is necessary to establish a one-to-one correspondence between what is measurable at TP1 and the corresponding Tx voltage or jitter at the pin. This may be achieved by means of a breakout channel and a replica channel. The replica channel reproduces the electrical characteristics of the breakout channel as closely as possible, matching its length, layer transitions, etc., making it possible to de-embed Tx measurements to the pin of the DUT. All voltage parameters are de-embedded to the pins unless otherwise specified. While the specification does not define precise electrical characteristics for the replica and breakout channels, it is advisable to adhere to the following guidelines:

- Breakout channels should be the same length for each Lane and routed on as few layers as possible, thereby reducing the number of replica channels that need to be built and measured.
- Each routing layer on a test board should have a separate breakout channel where the via and pad structures of the breakout and replica channels on respective layers match as closely as possible.
- Breakout and replica channels should be designed to have an insertion loss of less than 2 dB at the Nyquist frequency for the signaling rate (4 dB at Nyquist if the maximum signaling rate is 16.0 GT/s, 32.0 GT/s, and 64.0 GT/s) and a return loss of greater than 15 dB to Nyquist when measured from either TP2 or TP3, which may require use of low loss dielectric, wide signal traces and back-drilling of break-out vias or use of micro-via technology.
- The impedance targets for the breakout channel are 100 Ω differential and 50 Ω single-ended. For best accuracy the actual breakout channel impedance should be within $\pm 5\%$ of these values. For larger deviations a more complex de-embedding technique may be required.

8.3.2 Voltage Level Definitions §

A differential voltage is defined by taking the voltage difference between two conductors. In this specification, a differential signal or differential pair is comprised of a voltage on a positive conductor, V_{D+} , and a negative conductor, V_{D-} . The differential voltage (V_{DIFF}) is defined as the difference of the positive conductor voltage and the negative conductor voltage ($V_{DIFF} = V_{D+} - V_{D-}$). The Common Mode Voltage (V_{CM}) is defined as the average or mean voltage present on the same differential pair ($V_{CM} = (V_{D+} + V_{D-})/2$). This document's electrical specifications often refer to common mode peak-to-peak measurements or peak measurements, which are defined by the following equations.

$$V_{DIFFp-p} = (2 * \max | \underline{V_{D+}} - \underline{V_{D-}} |) \text{ (This applies to a symmetric differential swing)}$$

Equation 8-1 $V_{DIFFp-p}$ §

$$\underline{V_{TX-AC-CM-PP}} = \max(\underline{V_{D+}} + \underline{V_{D-}})/2 - \min(\underline{V_{D+}} + \underline{V_{D-}})/2$$

Equation 8-2 $V_{TX-AC-CM-PP}$ §

Note: The maximum value is calculated on a per unit interval evaluation with unit interval boundaries determined by the behavioral CDR. The maximum function as described is implicit for all peak-to-peak and peak common mode equations throughout the rest of this chapter.

In this section, DC is defined as all frequency components below $F_{DC} = 30$ kHz. AC is defined as all frequency components at or above $F_{DC} = 30$ kHz. These definitions pertain to all voltage and current specifications.

An example waveform is shown in § Figure 8-3. In this waveform the differential voltage (defined as $D+ - D-$) is approximately 800 mVPP, and the single-ended voltage for both $D+$ and $D-$ is approximately 400 mVPP for each. Note that while the center crossing point for both $D+$ and $D-$ is nominally at 200 mV, the corresponding crossover point for the differential voltage is at 0.0 V.

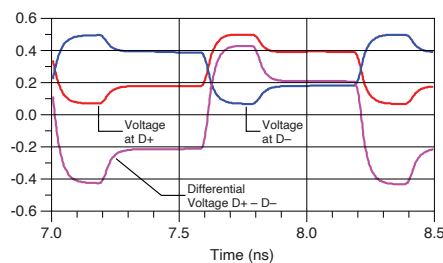


Figure 8-3 Single-ended and Differential Levels §

8.3.3 Tx Voltage Parameters §

Tx voltage parameters include equalization coefficients, equalization presets, and min/max voltage swings.

8.3.3.1 2.5 and 5.0 GT/s Transmitter Equalization §

Tx equalization at 2.5 and 5.0 GT/s is only de-emphasis. Tx equalization de-emphasis values at 2.5 and 5.0 GT/s are measured using the average ratio of transition to non-transition average eye amplitude at the 0.5 UI location using 500 repetitions of the compliance pattern.

8.3.3.2 8.0, 16.0, 32.0, and 64.0 GT/s Transmitter Equalization §

Tx voltage swing and equalization presets at 8.0, 16.0, 32.0, and 64.0 GT/s are measured by means of a low frequency pattern within the compliance pattern. The pattern consists of a sequence of 64 zeros followed by 64 ones, for 8.0, 16.0, and 32.0 GT/s and it consists of a sequence of 64 voltage level 0's followed by 64 voltage level 3's for 64.0 GT/s. The low frequency pattern permits an accurate measurement of voltage since ISI effects will have decayed and the signal will have approached a steady state. 8.0, 16.0, 32.0, and 64.0 GT/s transmitters must implement a coefficient-based equalization mode in order to support fine grained control over Tx equalization resolution. Additionally, a Transmitter must support a specified number of presets that give a coarser control over Tx equalization resolution. Both coefficient space and preset space are controllable via messaging from the Receiver via an equalization procedure. The equalization procedure operates on the same physical path as normal signaling and is implemented via extensions to the existing protocol Link layer.

All 8.0, 16.0, 32.0, and 64.0 GT/s Transmitters must implement support for the equalization procedure, whereas 8.0, 16.0, 32.0, and 64.0 GT/s Receivers may optionally make use of requests for the Transmitter on the Link partner to update Transmitter equalization. Details of the equalization procedure may be found in the Physical Layer Logical Block chapter.

Tx equalization coefficients for 8.0, 16.0, and 32.0 GT/s are based on the following FIR filter relationship as shown in § Figure 8-4. Equalization coefficients are subject to constraints limiting their max swing to $\pm$ unity with c_{-1} and c_{+1} being zero or negative. The inclusion of the unity condition means that only two of the three coefficients need to be specified to fully define $v_{out,n}$. In this specification for 8.0, 16.0, and 32.0 GT/s the two coefficients so specified are c_{-1} and c_{+1} , where c_0 is implied. Note that the coefficient magnitude is not the same as the Tx voltage swing magnitude.

Tx equalization coefficients for 64.0 GT/s are based on the following FIR filter relationship as shown in § Figure 8-5. Equalization coefficients are subject to constraints limiting their max swing to $\pm$ unity with c_{-2} being zero or positive, c_{-1} and c_{+1} being zero or negative.

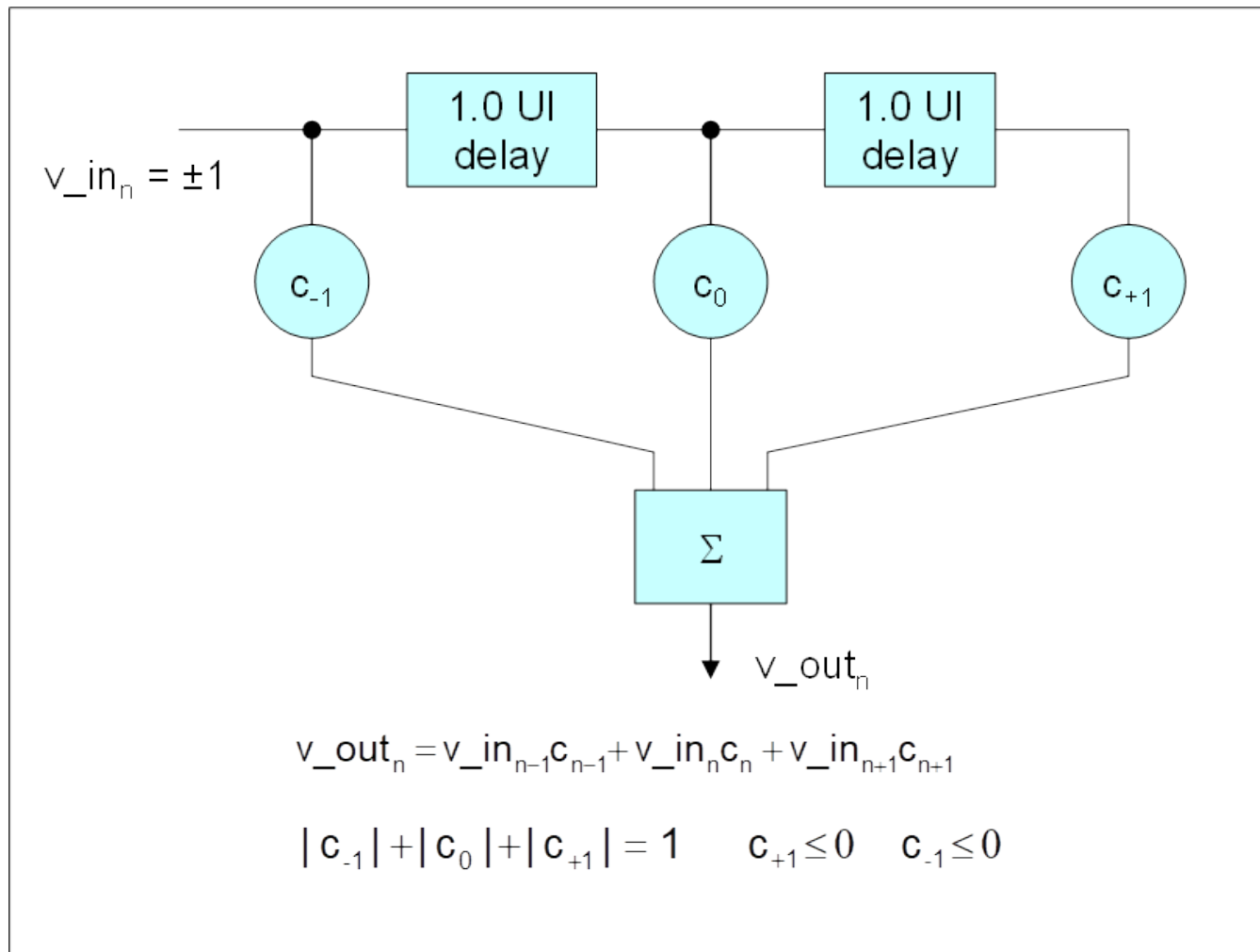


Figure 8-4 Tx Equalization FIR Representation for 8.0, 16.0, and 32.0 GT/s §

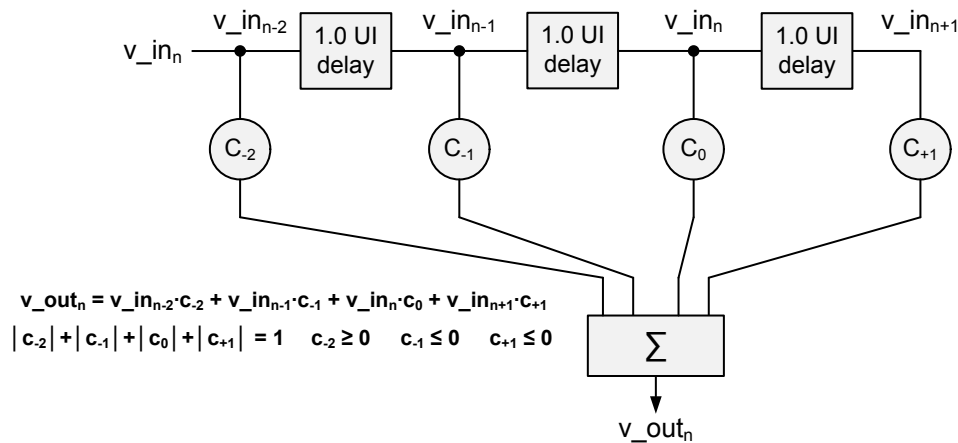


Figure 8-5 Tx Equalization FIR Representation for 64.0 GT/s §

8.3.3.3 Tx Equalization Presets for 8.0, 16.0, 32.0, and 64.0 GT/s §

When operating at 8.0 GT/s, 16.0 GT/s and 32.0 GT/s the Tx must support the full range of presets given in § Table 8-1. When operating at 64.0 GT/s, the Tx must support the full range of presets given in § Table 8-2. The data rate dependent encoding of presets has been defined in § Section 4.2.4.2. Presets are defined in terms of ratios, relating the pre-cursor and post-cursor equalization voltages. The pre-cursors (V_{c1}) and (V_{c2}) are referred to as pre-shoots, while the post-cursor (V_b) is referred to as de-emphasis. This convention permits the specification to retain the existing 2.5 GT/s and 5.0 GT/s definitions for Tx equalization, where only de-emphasis is defined, and it allows pre-shoots and de-emphasis to be defined such that each is independent of the other. The tolerances in § Table 8-1 also apply to 2.5 and 5.0 GT/s de-emphasis. The maximum swing, V_d , is also shown to illustrate that, when c_{+1} , c_{-2} , and c_{-1} are non-zero, the swing of V_a does not reach the maximum as defined by V_d . § Figure 8-6 is shown as an example of transmitter equalization, but it is not intended to represent the signal as it would appear for measurement purposes. The high frequency nature of PCIe signaling makes measurement of single UI pulse heights impractical.

The presets defined in § Table 8-1 and § Table 8-2 are numbered to match the designations in § Table 4-34.

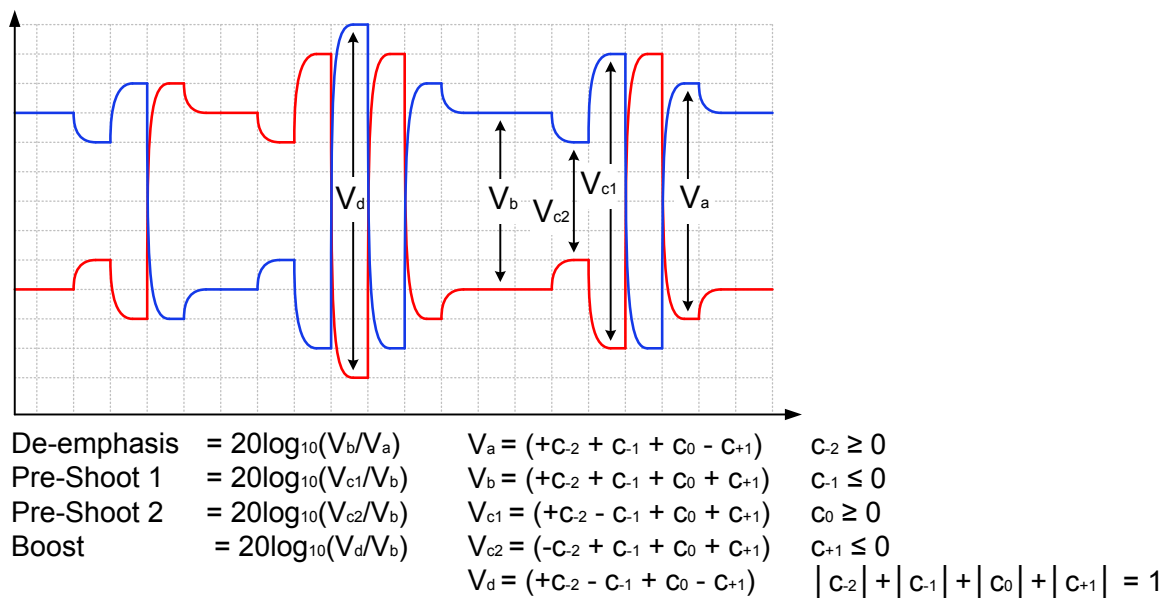


Figure 8-6 Definition of Tx Voltage Levels and Equalization Ratios §

\$ Table 8-1 lists the values for presets; at 8.0 GT/s, 16.0 GT/s and 32.0 GT/s. All preset values must be supported for full swing signaling.

Table 8-1 Tx Preset Ratios and Corresponding Coefficient Values for 8.0, 16.0, and 32.0 GT/s §

Preset #	Preshoot 2 (dB)	Preshoot 1 (dB)	De-emphasis (dB)	c_{-2}	c_{-1}	c_{+1}	Va/Vd	Vb/Vd	Vc1/Vd	Vc2/Vd
P4	0.0	0.0 ±1 dB	0.0 ±1 dB	0.000	0.000	0.000	1.000	1.000	1.000	1.000
P1	0.0	0.0 ±1 dB	-3.5 ±1 dB	0.000	0.000	-0.167	1.000	0.666	0.666	0.666
P0	0.0	0.0 ±1 dB	-6.0 ±1.5 dB	0.000	0.000	-0.250	1.000	0.500	0.500	0.500
P9	0.0	3.5 ±1 dB	0.0 ±1 dB	0.000	-0.167	0.000	0.666	0.666	1.000	0.666
P8	0.0	3.5 ±1 dB	-3.5 ±1 dB	0.000	-0.125	-0.125	0.750	0.500	0.750	0.500
P7	0.0	3.5 ±1 dB	-6.0 ±1.5 dB	0.000	-0.100	-0.200	0.800	0.400	0.600	0.400
P5	0.0	1.9 ±1 dB	0.0 ±1 dB	0.000	-0.100	0.000	0.800	0.800	1.000	0.800
P6	0.0	2.5 ±1 dB	0.0 ±1 dB	0.000	-0.125	0.000	0.750	0.750	1.000	0.750
P3	0.0	0.0 ±1 dB	-2.5 ±1 dB	0.000	0.000	-0.125	1.000	0.750	0.750	0.750
P2	0.0	0.0 ±1 dB	-4.4 ±1.5 dB	0.000	0.000	-0.200	1.000	0.600	0.600	0.600
P10	0.0	0.0 ±1 dB	Note 2	0.000	0.000	Note 2	1.000	Note 2	Note 2	Note 2

Notes:

Preset #	Preshoot 2 (dB)	Preshoot 1 (dB)	De-emphasis (dB)	c_{-2}	c_{-1}	c_{+1}	Va/Vd	Vb/Vd	Vc1/Vd	Vc2/Vd
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1. Reduced swing signaling must implement presets P4, P1, P9, P5, P6, and P3. Full swing signaling must implement all the above presets.
2. P10 boost limits are not fixed, since its de-emphasis level is a function of the LF level that the Tx advertises during training (see § Section 4.2.4.1). P10 is used for testing the boost limit of Transmitter at full swing. P1 is used for testing the boost limit of Transmitter at reduced swing.

§ Table 8-2 lists the values for presets at 64.0 GT/s. All preset values must be supported for full swing signaling.

Table 8-2 Tx Preset Ratios and Corresponding Coefficient Values for 64.0 GT/s §

Preset #	Preshoot 2 (dB)	Preshoot 1 (dB)	De-emphasis (dB)	c_{-2}	c_{-1}	c_{+1}	Va/Vd	Vb/Vd	Vc1/Vd	Vc2/Vd
Q0	0.0 ±0.5 dB	0.0 ±0.5 dB	0.0 ±0.5 dB	0.000	0.000	0.000	1.000	1.000	1.000	1.000
Q1	0.0 ±0.5 dB	1.6 ±0.5 dB	0.0 ±0.5 dB	0.000	-0.083	0.000	0.834	0.834	1.000	0.834
Q2	0.0 ±0.5 dB	3.5 ±0.5 dB	0.0 ±0.5 dB	0.000	-0.167	0.000	0.666	0.666	1.000	0.666
Q3	0.0 ±0.5 dB	0.0 ±0.5 dB	-1.6 ±0.5 dB	0.000	0.000	-0.083	1.000	0.834	0.834	0.834
Q4	0.0 ±0.5 dB	0.0 ±0.5 dB	-3.5 ±0.5 dB	0.000	0.000	-0.167	1.000	0.666	0.666	0.666
Q5	-1.3 ±0.5 dB	4.7 ±1.0 dB	0.0 ±0.5 dB	0.042	-0.208	0.000	0.584	0.584	1.000	0.500
Q6	-1.6 ±0.5 dB	3.5 ±0.5 dB	-3.5 ±0.5 dB	0.042	-0.125	-0.125	0.750	0.500	0.750	0.416
Q7	-2.9 ±0.5 dB	4.7 ±1.0 dB	0.0 ±0.5 dB	0.083	-0.208	0.000	0.584	0.584	1.000	0.418
Q8	-3.5 ±0.5 dB	6.0 ±1.0 dB	0.0 ±0.5 dB	0.083	-0.250	0.000	0.500	0.500	1.000	0.334
Q9	-4.4 ±1.0 dB	6.9 ±1.0 dB	-1.6 ±0.5 dB	0.083	-0.250	-0.042	0.500	0.416	0.916	0.250
Q10	0.0 ±0.5 dB	0.0 ±0.5 dB	Note 2	0.000	0.000	Note 2	1.000	Note 2	Note 2	Note 2

Notes:

1. Reduced swing signaling must implement presets Q0, Q1, Q2, Q3, and Q4. Full swing signaling must implement all the above presets.
2. Q10 boost limits are not fixed, since its de-emphasis level is a function of the LF level that the Tx advertises during training. Q10 is used for testing the boost limit of Transmitter at full swing. Q4 is used for testing the boost limit of Transmitter at reduced swing.

8.3.3.4 Measuring Tx Equalization for 2.5 GT/s and 5.0 GT/s §

Tx equalization de-emphasis values at 2.5 and 5.0 GT/s are measured using the average ratio of transition to non-transition eye heights at the 0.5 UI location using 500 repetitions of the compliance pattern.

8.3.3.5 Measuring Presets at 8.0, 16.0, 32.0, and 64.0 GT/s §

§ Figure 8-7 illustrate the methodology for measuring Tx equalization coefficients and presets. For a Tx preset to be measured, the DUT Tx transmits a Compliance Pattern with the corresponding Tx equalization coefficients. The

equalized Compliance Pattern is captured by a real-time oscilloscope and the post-processing software extracts an equalized step response waveform. The DUT Tx also transmits a Compliance Pattern with no Tx equalization. The unequalized Compliance Pattern is captured by the real-time oscilloscope and the post-processing software applies Tx equalization coefficients c_{-2} , c_{-1} , c_0 , and c_{+1} to construct an equalized step response waveform. The Tx preset coefficients are the best fit Tx equalization coefficients c_{-2} , c_{-1} , c_0 , and c_{+1} that minimize the Mean Square Error (MSE) between the measured equalized step response waveform and the reconstructed equalized step response waveform.

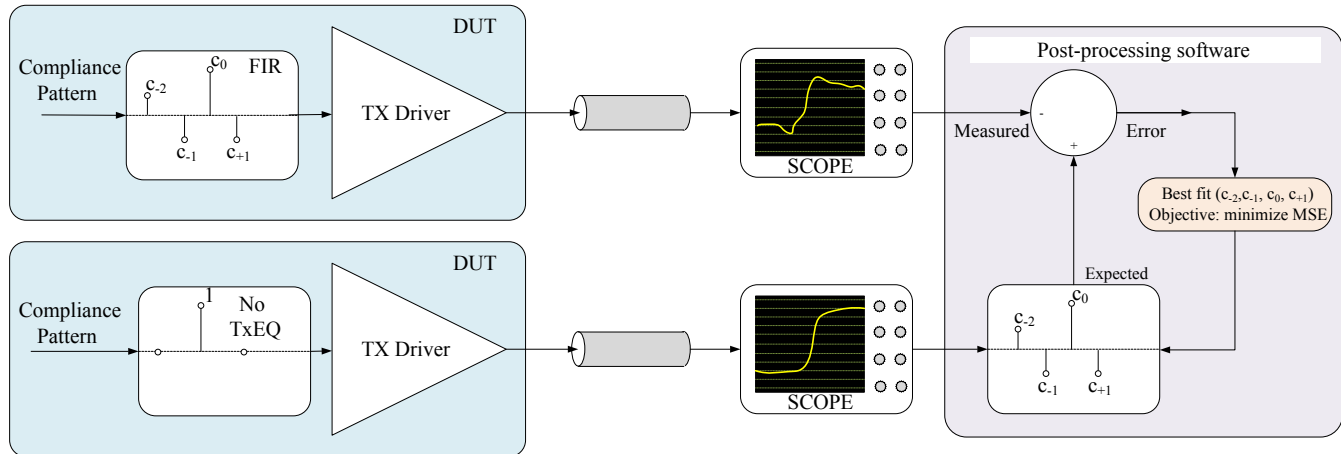


Figure 8-7 Methodology for measuring Tx equalization coefficients and presets §

8.3.3.6 Method for Measuring $V_{TX-DIFF-PP}$ at 2.5 GT/s and 5.0 GT/s §

$V_{TX-DIFF-PP}$ ($V_{TX-DIFF-PP-LOW}$ for reduced swing) at 2.5 GT/s and 5.0 GT/s are measured using the average transition eye amplitude at the 0.5 UI location using 500 repetitions of the compliance pattern.

8.3.3.7 Method for Measuring $V_{TX-DIFF-PP}$ at 8.0, 16.0, 32.0, and 64.0 GT/s §

The range for a Transmitter's output voltage swing, (specified by V_d) with no equalization is defined by $V_{TX-DIFF-PP}$ ($V_{TX-DIFF-PP-LOW}$ for reduced swing), and is obtained by setting c_{-2} , c_{-1} and c_{+1} to zero and measuring the peak-peak voltage on the 64-ones (64 PAM4 voltage level 3's at 64.0 GT/s)/64-zeros (64 PAM4 voltage level 0's at 64.0 GT/s) segment of the compliance pattern. The resulting signal effectively measures at the die pad, minus any low frequency package loss. ISI and switching effects are minimized by restricting the portion of the curve over which voltage is measured to the last few UI of each half cycle, as illustrated in § Figure 8-8. High frequency noise is mitigated by averaging over 500 repetitions of the compliance pattern.

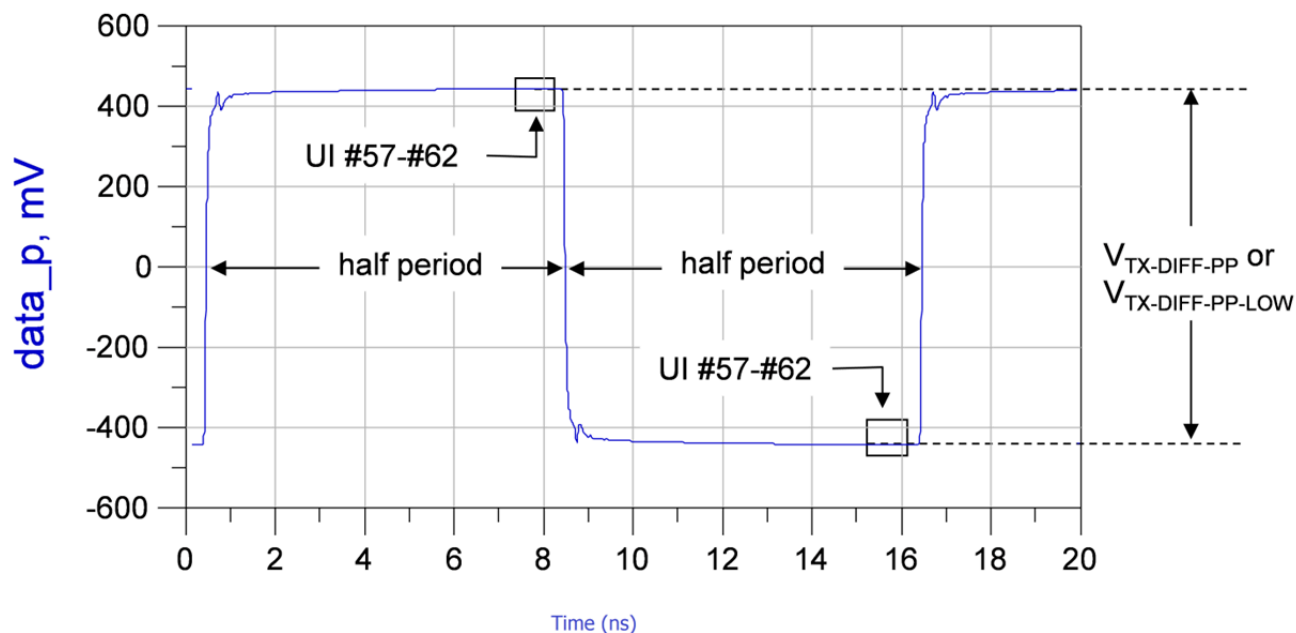


Figure 8-8 $V_{TX-DIFF-PP}$ and $V_{TX-DIFF-PP-LOW}$ Measurement §

8.3.3.8 Coefficient Range and Tolerance for 8.0, 16.0, 32.0, and 64.0 GT/s §

8.0, 16.0, 32.0, and 64.0 GT/s Transmitters are required to inform the Receiver of their coefficient range and tolerance. Coefficient range and tolerance are constrained by the following requirements.

- Coefficients must support all eleven presets and their respective tolerances as defined in § Table 8-1 and § Table 8-2
- All Transmitters must meet the full swing signaling $V_{TX-EIEOS-FS}$ limits.
- Transmitters may optionally support reduced swing, and if they do, they must meet the $V_{TX-EIEOS-RS}$ limits.
- The coefficients must meet the boost and resolution ($V_{TX-BOOST-FS}$, $V_{TX-BOOST-RS}$ and $EQ_{TX-COEFF-RES}$) limits defined in § Table 8-6.

When the above constraints are applied the resulting coefficient space for 8.0, 16.0, and 32.0 GT/s with pre-shoot2 coefficient $c_2 = 0$ may be mapped onto a triangular matrix, an example of which is shown in § Figure 8-9. The matrix may be interpreted as follows: pre-shoot1 and de-emphasis coefficients are mapped onto the Y-axis and X-axes, respectively. In both cases the maximum granularity of 1/24 is assumed. Each matrix cell, corresponding to a valid combination of pre-shoot1 and de-emphasis coefficients, has three entries corresponding to pre-shoot1 (PS1), de-emphasis (DE), and boost (as shown in the upper left-hand corner). Diagonal elements are defined by the maximum boost ratio. Those cells highlighted in blue are presets required for reduced swing, while cells in either blue or orange represent presets required for full swing signaling. Note that this figure is informative only and is not intended to imply any specific Tx implementation or to alter requirements for nominal preset equalization values and allowed ranges.

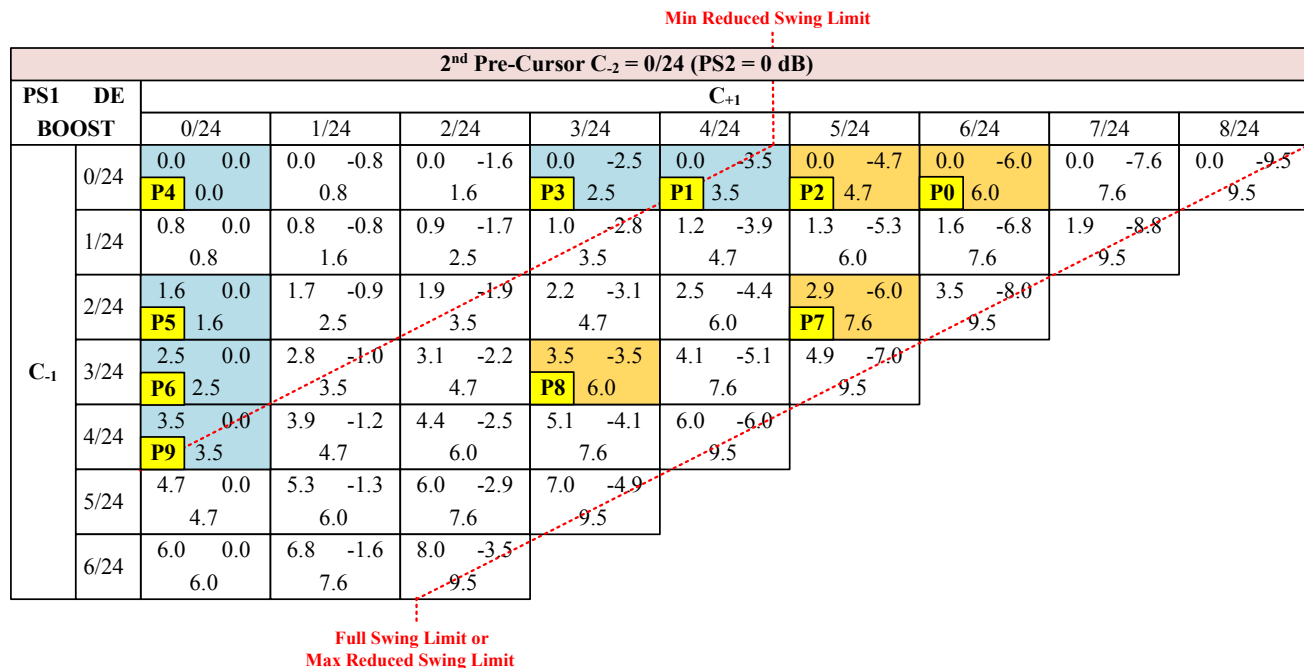


Figure 8-9 Transmit Equalization Coefficient Space Triangular Matrix Example for 8.0, 16.0, and 32.0 GT/s §

The coefficient space for 64.0 GT/s with each pre-shoot2 coefficient may be mapped onto a triangular matrix, an example of which is shown in § Figure 8-10. Maximum granularity of 1/24 is assumed for Tx equalization coefficient. Those cells highlighted in blue are presets required for reduced swing, while cells in either blue or orange represent presets required for full swing signaling. Note that this figure is informative only and is not intended to imply any specific Tx implementation or to alter requirements for nominal preset equalization values and allowed ranges.

2 nd Pre-Cursor C ₂ = 0/24												
PS2	PS1	DE	C ₊₁									
PRESET	BOOST		0/24	1/24	2/24	3/24	4/24	5/24	6/24	7/24	8/24	
C ₋₁	0/24	0.0 0.0 0.0	0.0 0.0 -0.8	0.0 0.0 -1.6	0.0 0.0 -2.5	0.0 0.0 -3.5	0.0 0.0 -4.7	0.0 0.0 -6.0	0.0 0.0 -7.6	0.0 0.0 -9.5		
		Q0 0.0	0.8	Q3 1.6	2.5	Q4 3.5	4.7	6.0	7.6	9.5		
	1/24	0.0 0.8 0.0	0.0 0.8 -0.8	0.0 0.9 -1.7	0.0 1.0 -2.8	0.0 1.2 -3.9	0.0 1.3 -5.3	0.0 1.6 -6.8	0.0 1.9 -8.8			
		0.8	1.6	2.5	3.5	4.7	6.0	7.6	9.5			
	2/24	0.0 1.6 0.0	0.0 1.7 -0.9	0.0 1.9 -1.9	0.0 2.2 -3.1	0.0 2.5 -4.4	0.0 2.9 -6.0	0.0 3.5 -8.0				
		Q1 1.6	2.5	3.5	4.7	6.0	7.6	9.5				
	3/24	0.0 2.5 0.0	0.0 2.8 -1.0	0.0 3.1 -2.2	0.0 3.5 -3.5	0.0 4.1 -5.1	0.0 4.9 -7.0					
		2.5	3.5	4.7	6.0	7.6	9.5					
	4/24	0.0 3.5 0.0	0.0 3.9 -1.2	0.0 4.4 -2.5	0.0 5.1 -4.1	0.0 6.0 -6.0						
		Q2 3.5	4.7	6.0	7.6	9.5						
	5/24	0.0 4.7 0.0	0.0 5.3 -1.3	0.0 6.0 -2.9	0.0 7.0 -4.9							
		4.7	6.0	7.6	9.5							
	6/24	0.0 6.0 0.0	0.0 6.9 -1.6	0.0 8.0 -3.5								
		6.0	7.6	9.5								

Full Swing Limit or
Max Reduced Swing Limit

Min Reduced Swing Limit

2 nd Pre-Cursor C ₂ = 1/24												
PS2	PS1	DE	C ₊₁									
PRESET	BOOST		0/24	1/24	2/24	3/24	4/24	5/24	6/24	7/24	8/24	
C ₋₁	0/24	-0.8 0.0 0.0	-0.8 0.0 -0.8	-0.9 0.0 -1.6	-1.0 0.0 -2.5	-1.2 0.0 -3.5	-1.3 0.0 -4.7	-1.6 0.0 -6.0	-1.9 0.0 -7.6	-2.5 0.0 -9.5		
		0.0	0.8	1.6	2.5	3.5	4.7	6.0	7.6	9.5		
	1/24	-0.8 0.8 0.0	-0.9 0.8 -0.8	-1.0 0.9 -1.7	-1.2 1.0 -2.8	-1.3 1.2 -3.9	-1.6 1.3 -5.3	-1.9 1.6 -6.8	-2.5 1.9 -8.8			
		0.8	1.6	2.5	3.5	4.7	6.0	7.6	9.5			
	2/24	-0.9 1.6 0.0	-1.0 1.7 -0.9	-1.2 1.9 -1.9	-1.3 2.2 -3.1	-1.6 2.5 -4.4	-1.9 2.9 -6.0	-2.5 3.5 -8.0				
		1.6	2.5	3.5	4.7	6.0	7.6	9.5				
	3/24	-1.0 2.5 0.0	-1.2 2.8 -1.0	-1.3 3.1 -2.2	-1.6 3.5 -3.5	-1.9 4.1 -5.1	-2.5 4.9 -7.0					
		2.5	3.5	4.7	Q6 6.0	7.6	9.5					
	4/24	-1.2 3.5 0.0	-1.3 3.9 -1.2	-1.6 4.4 -2.5	-1.9 5.1 -4.1	-2.5 6.0 -6.0						
		3.5	4.7	6.0	7.6	9.5						
	5/24	-1.3 4.7 0.0	-1.6 5.3 -1.3	-1.9 6.0 -2.9	-2.5 7.0 -4.9							
		Q5 4.7	6.0	7.6	9.5							
	6/24	-1.6 6.0 0.0	-1.9 6.9 -1.6	-2.5 8.0 -3.5								
		6.0	7.6	9.5								

Full Swing Limit or
Max Reduced Swing Limit

Min Reduced Swing Limit

2 nd Pre-Cursor C ₂ = 2/24												
PS2	PS1	DE	C ₊₁									
PRESET	BOOST		0/24	1/24	2/24	3/24	4/24	5/24	6/24	7/24	8/24	
C ₋₁	0/24	-1.6 0.0 0.0	-1.7 0.0 -0.8	-1.9 0.0 -1.6	-2.2 0.0 -2.5	-2.5 0.0 -3.5	-2.9 0.0 -4.7	-3.5 0.0 -6.0	-4.4 0.0 -7.6	-6.0 0.0 -9.5		
		0.0	0.8	1.6	2.5	3.5	4.7	6.0	7.6	9.5		
	1/24	-1.7 0.8 0.0	-1.9 0.8 -0.8	-2.2 0.9 -1.7	-2.5 1.0 -2.8	-2.9 1.2 -3.9	-3.5 1.3 -5.3	-4.4 1.6 -6.8	-6.0 1.9 -8.8			
		0.8	1.6	2.5	3.5	4.7	6.0	7.6	9.5			
	2/24	-1.9 1.6 0.0	-2.2 1.7 -0.9	-2.5 1.9 -1.9	-2.9 2.2 -3.1	-3.5 2.5 -4.4	-4.4 2.9 -6.0	-6.0 3.5 -8.0				
		1.6	2.5	3.5	4.7	6.0	7.6	9.5				
	3/24	-2.2 2.5 0.0	-2.5 2.8 -1.0	-2.9 3.1 -2.2	-3.5 3.5 -3.5	-4.4 4.1 -5.1	-6.0 4.9 -7.0					
		2.5	3.5	4.7	6.0	7.6	9.5					
	4/24	-2.5 3.5 0.0	-2.9 3.9 -1.2	-3.5 4.4 -2.5	-4.4 5.1 -4.1	-6.0 6.0 -6.0						
		3.5	4.7	6.0	7.6	9.5						
	5/24	-2.9 4.7 0.0	-3.5 5.3 -1.3	-4.4 6.0 -2.9	-6.0 7.0 -4.9							
		Q7 4.7	6.0	7.6	9.5							
	6/24	-3.5 6.0 0.0	-4.4 6.9 -1.6	-6.0 8.0 -3.5								
		Q8 6.0	Q9 7.6	9.5								

Full Swing Limit or
Max Reduced Swing Limit

Figure 8-10 Transmit Equalization Coefficient Space Triangular Matrix Example for 64.0 GT/s §

8.3.3.9 EIEOS and $V_{TX-EIEOS-FS}$ and $V_{TX-EIEOS-RS}$ Limits §

EIEOS signaling is defined for 5.0, 8.0, 16.0, 32.0, and 64.0 GT/s only.

- At 5.0 GT/s the $K_{28.7}$ Symbol is used to exit Electrical Idle.
- At 8.0 GT/s the EIEOS pattern consists of 8 consecutive zeros followed by the same number of consecutive ones, where the pattern is repeated for a total of 128 UI.
- At 16.0 GT/s the EIEOS pattern consists of 16 consecutive zeros followed by the same number of consecutive ones, where the pattern is repeated for a total of 128 UI.
- At 32.0 GT/s the EIEOS pattern consists of 32 consecutive zeros followed by the same number of consecutive ones, where the pattern is repeated for a total of 128 UI.
- At 64.0 GT/s the EIEOS pattern consists of 32 UI consecutive voltage level 0's followed by 32 UI consecutive voltage level 3's (see § Section 4.2.5.3).

A transmitter sends an EIEOS to cause an exit of Electrical Idle at the Receiver. This pattern guarantees the Receiver will properly detect the EI Exit condition with its squelch exit detect circuit, something not otherwise guaranteed by scrambled data. The Tx EIEOS launch voltage is defined by $V_{TX-EIEOS-FS}$ for full swing signaling and by $V_{TX-EIEOS-RS}$ for reduced swing signaling. $V_{TX-EIEOS-RS}$ is smaller than $V_{TX-EIEOS-FS}$ to reflect the fact that reduced swing is typically supported only for lower loss channels where there is less attenuation at the EIEOS signaling rate.

$V_{TX-EIEOS-FS}$ and $V_{TX-EIEOS-RS}$ are measured using the EIEOS sequence contained within the compliance pattern for 8.0, 16.0, 32.0, and 64.0 GT/s.

For full swing signaling $V_{TX-EIEOS-FS}$ is measured with a preset number P10 for 8.0, 16.0, and 32.0 GT/s, and with a preset number Q10 for 64.0 GT/s. This is equivalent to a maximum nominal boost of 9.5 dB and represents the maximum boost attainable in coefficient space. When a tolerance of ± 1.5 dB is factored in this yields the minimum boost limit of 8.0 dB appearing in § Table 8-6. For reduced swing signaling $V_{TX-EIEOS-RS}$ is measured with preset P1 for 8.0, 16.0, and 32.0 GT/s, and with a preset number Q4 for 64.0 GT/s.

A Transmitter is not always permitted to generate the maximum boost level noted above. A Transmitter that cannot drive significantly more than 800 mVPP is limited by the need to meet $V_{TX-EIEOS-FS}$. The Tx must reject any adjustments to its presets or coefficients that would violate the $V_{TX-EIEOS-FS}$ or $V_{TX-EIEOS-RS}$ limits. The EIEOS voltage limits are imposed to guarantee the EIEOS threshold of 175 mVPP at the Rx pin.

§ Figure 8-11 illustrates the de-emphasis peak as observed at the pin of a Tx for $V_{TX-EIEOS-FS}$. At the far end of a lossy channel the de-emphasis peak will be attenuated; this is why the measurement interval includes only the middle five UI at 8.0 GT/s, UI number 5-14 at 16.0 GT/s, and UI number 9-28 at 32.0 and 64.0 GT/s. The voltage is averaged over this interval for both the negative and positive halves of the waveform over 500 repetitions of the compliance pattern. $V_{TX-EIEOS-FS}$ and $V_{TX-EIEOS-RS}$ are defined as the difference between the negative and positive waveform segment averages. UI boundaries are defined with respect to the edge of the recovered data clock.

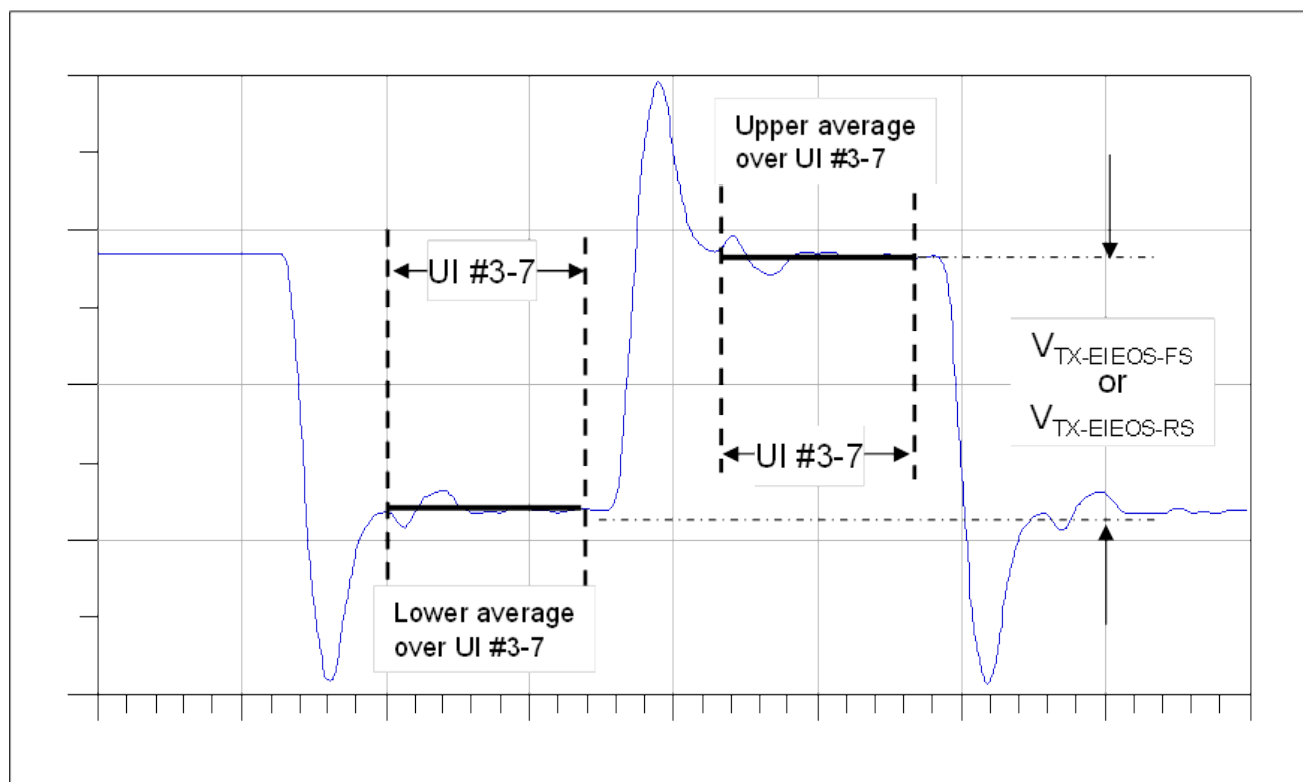


Figure 8-11 Measuring $V_{TX-EIEOS-FS}$ and $V_{TX-EIEOS-RS}$ at 8.0 GT/s §

8.3.3.10 Reduced Swing Signaling §

PCI Express Transmitters may optionally support a reduced swing signaling. It is left as an implementation option to define the maximum reduced swing voltage value for $V_{TX-DIFF-PP-LOW}$ anywhere up to the maximum full swing voltage. The minimum for $V_{TX-DIFF-PP-LOW}$ is captured indirectly by the constraint imposed by $V_{TX-EIEOS-RS}$, so there is no need to define a separate minimum limit for $V_{TX-DIFF-PP-LOW}$. Reduced swing limits the range of presets and the maximum boost. The boost for reduced swing must be in the region shown in § Figure 8-9 and § Figure 8-10 between the Max reduced swing limit and minimum reduced swing boost limit.

Form factors are permitted to disallow, optionally allow, or require Reduced Swing Signaling, depending on the channel requirements for the form factor. When Reduced Swing Signaling is allowed or required it is required that form factor specifications provide any additional details necessary to support interoperability.

8.3.3.11 Effective Tx Package Loss at 8.0, 16.0, 32.0, and 64.0 GT/s §

Package loss (including silicon driver bandwidth) is represented by the $ps21_{TX}$ parameter. Since both package IL and driver bandwidth affect the signal as observed at the Tx pin, the $ps21_{TX}$ parameter has the advantage of representing both of these effects, while permitting the measurement to be made at a point (TP1) that can easily be probed. It is necessary to include a package loss parameter in the Tx specification, since the voltage swing parameters ($V_{TX-DIFF-PP}$ and $V_{TX-DIFF-PP-LOW}$) are defined at an equivalent pulse frequency of 1/128 UI and purposely do not capture high frequency driver or package loss effects.

At 16.0 GT/s and 32.0 GT/s, separate $ps21_{TX}$ parameters are defined for packages containing Root Ports (Root Package) and for all other packages (Non-Root Package), based on the assumption that the former tend to be large and require socketing, while the latter are smaller and usually not socketed.

The $ps21_{TX}$ parameter is informative for 16.0 GT/s Root Package devices and for Non-Root Package devices that only support PCI Express standard form factors (i.e., CEM, M.2, etc.). The $ps21_{TX}$ parameter is normative for all 8.0, 32.0, and 64.0 GT/s devices and for 16.0 GT/s Non-Root Package devices that support captive channels. (See § Table 8-3 below).

Table 8-3 Cases that the Reference Packages and $ps21_{TX}$ Parameter are Normative §

	8.0, 32.0, and 64.0 GT/s		16.0 GT/s	
	Root Package Device	Non-Root Package Device	Root Package Device	Non-Root Package Device
Device supports captive channels	Normative	Normative	Informative	Normative
Device does not support captive channels	Normative	Normative	Informative	Informative

All implementations of PCI Express standard form factors must still meet form factor requirements. Devices for which the $ps21_{TX}$ parameter is informative, as defined above must provide a package model for use in channel compliance if they do not meet the informative $ps21_{TX}$ parameter.

Package loss is measured by comparing the 64-zeros/64-ones voltage swing (V_{111}) against a 1010 pattern (V_{101}). Tx package loss measurement is made with c_{-2} , c_{-1} , and c_{+1} set to zero. Measurements shall be made averaging over 500 repetitions of the compliance pattern.

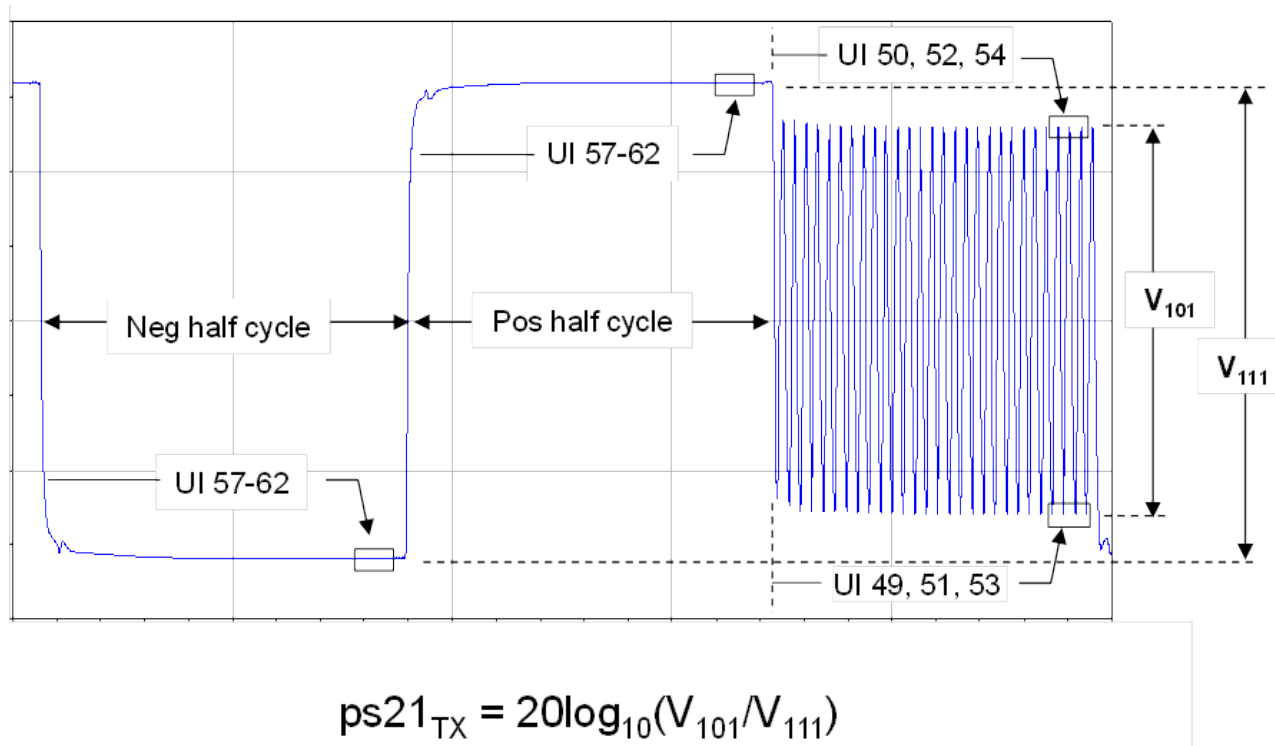


Figure 8-12 Compliance Pattern and Resulting Package Loss Test Waveform §

Measurement of V_{101} and V_{111} is made towards the end of each interval to minimize ISI and low frequency effects. V_{101} is defined as the peak-peak voltage between minima and maxima of the clock pattern. V_{111} is defined as the average voltage difference between the positive and negative levels of the two half cycles. The measurement should be averaged over 500 repetitions of the compliance pattern.

At 32.0 GT/s only the $ps21_{TX}$ parameter is calculated by filtering the captured voltage waveforms normally used for $ps21_{TX}$ measurements as follows:

- V_{111} is measured from a filtered voltage waveform with a first order (20 dB/decade) low pass filter with a -3 dB corner frequency at 1 GHz applied.
- V_{101} is measured from a filtered voltage waveform with a first order (20 dB/decade) high pass filter with a -3 dB corner frequency at 7 GHz and a low-pass filter with zero phase, no more than 0.1 dB attenuation at 16 GHz, no less than 40 dB of attenuation at frequencies above 48 GHz and cutoff frequency between 32 GHz to 40 GHz applied. If the low-pass filter causes the $ps21_{TX}$ parameter to fail the specification, then V_{101} can be remeasured without the low-pass filter.

Since the Nyquist frequency for 64.0 GT/s PAM4 signaling is 16.0 GHz, no additional measurement is necessary for 64.0 GT/s $ps21_{TX}$ spec compliance. For a PCIe 6.0 device that supports maximum data rate of 64.0 GT/s, the $ps21_{TX}$ spec parameter measured for 32.0 GT/s must be lower than the $ps21_{TX}$ spec values provided under the 64.0 GT/s column in § Table 8-6.

8.3.3.12 Linear Fit Pulse Response §

Several PCIe measurements are based on a Linear Fit Pulse Response (LFPR) model. This section provides a description of the general method. The basic LFPR is a linear model $y(k) = x(n) * p$, where “*” denotes convolution, $x(n)$ is an input stream of test pattern symbols where symbols {0, 1, 2, 3} are represented using {-1, -1/3, 1/3, 1} respectively, $y(k)$ is an output stream of waveform samples, and p is the linear fit pulse response which maps ideal symbols to measured samples.

A waveform is captured from a DUT (device under test) with an effective sample rate that is M times the signaling rate of the DUT, where M is an integer, and resampling may be used to meet this requirement. The waveform is then averaged using multiple pattern repetitions. Thus, if the pattern length is N and the number of pattern repetitions is R_{PAT} , then $R_{PAT} \cdot N \cdot M$ samples are averaged into a single pattern repetition of length $N \cdot M$ samples. The samples of this waveform are denoted as $y(k)$. The symbols $x(n)$ are aligned with $y(k)$ such that the first M samples of $y(k)$ correspond to the first symbol of the test pattern $x(n)$, the second M samples of $y(k)$ to the second symbol, and so on. A linear system of equations is solved for vector p . An efficient way to solve the system is to organize it as the matrix equation $PX = Y$. The M -by- N waveform matrix Y is defined as:

$$Y = \begin{bmatrix} y(1) & y(M+1) & \vdots & y(M(N-1)+1) \\ y(2) & y(M+2) & \vdots & y(M(N-1)+2) \\ \vdots & \vdots & \vdots & \vdots \\ y(M) & y(2M) & \vdots & y(MN) \end{bmatrix}$$

Equation 8-3 Y §

Rotate the symbols vector x by a specified pulse delay D_p to yield x_r as shown in [§ Equation 8-4](#).

$$x_r = \begin{bmatrix} x(D_p + 1) & x(D_p + 2) & \vdots & x(N) & x(1) & \vdots & x(D_p) \end{bmatrix}$$

[Equation 8-4 \$x_r\$](#) §

Define the matrix X to be an N -by- N matrix derived from x_r as shown in [§ Equation 8-5](#).

$$X = \begin{bmatrix} x_r(1) & x_r(2) & \vdots & x_r(N) \\ x_r(N) & x_r(1) & \vdots & x_r(N-1) \\ \vdots & \vdots & \vdots & \vdots \\ x_r(2) & x_r(3) & \vdots & x_r(1) \end{bmatrix}$$

[Equation 8-5 \$X\$](#) §

Define the matrix X_1 to be the first N_p rows of X concatenated with a row vector of ones of length N , where N_p is a specified pulse length. The M -by- (N_p+1) coefficient matrix, P , corresponding to the linear fit pulse response is then defined by [§ Equation 8-6](#). The superscript “ T ” denotes the matrix transpose operator.

$$P = YX_1^T(X_1X_1^T)^{-1}$$

[Equation 8-6 \$P\$](#) §

The pulse fit vector $p(k)$ is read column-wise from the first N_p columns of P . The error waveform, $e(k)$, is read column-wise from the elements of E as shown in [§ Equation 8-7](#)

$$E = PX_1 - Y = \begin{bmatrix} e(1) & e(M+1) & \vdots & e(M(N-1)+1) \\ e(2) & e(M+2) & \vdots & e(M(N-1)+2) \\ \vdots & \vdots & \vdots & \vdots \\ e(M) & e(2M) & \vdots & e(MN) \end{bmatrix}$$

[Equation 8-7 \$E\$](#) §

The distortion term, also called fitting error, σ_e , is defined as the root mean squared (RMS) value of the error waveform $e(k)$.

8.3.3.13 Transmitter Signal-to Noise and Distortion Ratio (SNDR_{TX}) for 64.0 GT/s §

Signal-to-noise and distortion ratio (SNDR) is measured at the transmitter output using the Compliance Pattern (see § Section 4.2.14) with preset Q_0 (no Tx equalization), and the lanes not under test also transmitting the Compliance Pattern with preset Q_0 . The recorded waveform must have a minimum of 250 repetitions (R_{PAT}) of the compliance pattern. Measurements should be made with a 4th order Bessel-Thomson filter with a roll-off from DC value by 3 dB at 33 GHz to minimize the impact of scope high-frequency noise. The minimum scope bandwidth is 50 GHz.

§ Section 8.3.3.12 describes the Linear Fit Pulse Response method where a single pulse response is created from an averaged transmitter waveform. The Multi Pulse Response Fit (MPRF) method is introduced as the terminology for creating more than one linear fit pulse responses. For the SNDR_{TX} measurement, two pulse responses (even and odd) are used to mitigate the impact of duty cycle errors. Another variant of the MPRF will be used in § Section 8.3.3.14 with four pulse responses, one for each PAM4 signal level. Let x be the input stream of transmitted symbols and y be the output stream of captured waveform. In this method, two pulse responses are defined, denoted as p_E and p_O , to provide the best possible approximation to § Equation 8-8.

$$y = p_E * (x \cdot \mathbf{1}_E) + p_O * (x \cdot \mathbf{1}_O)$$

Equation 8-8 y §

where '*' represents convolution. Here, $\mathbf{1}_E$ is an indicator function, or mask, with a value of 1 for even UIs and 0 for odd UIs, while $\mathbf{1}_O$ has a value of 1 for odd UIs and 0 for even UIs. The notations $(x \cdot \mathbf{1}_E)$ and $(x \cdot \mathbf{1}_O)$ define pointwise multiplication, where each element in x is multiplied by its corresponding element in either $\mathbf{1}_E$ or $\mathbf{1}_O$.

To solve the system, define the matrix X_{NP} to be the first N_p rows of X from § Equation 8-5 and construct the $(2N_p+1)$ -by- N block linear system

$$X_{EO} = \begin{bmatrix} X_{NP} \cdot \mathbf{1}_E \\ X_{NP} \cdot \mathbf{1}_O \\ 1 \ 1 \ \dots \ 1 \end{bmatrix}$$

Equation 8-9 X_{NP} §

where masks $\mathbf{1}_E$ and $\mathbf{1}_O$ are organized here as a 2D array (but play the same role), and $1 \ 1 \ \dots \ 1$ is a row of ones. Next, solve the linear system by computing,

$$P_{EO} = Y X_{EO}^T (X_{EO} X_{EO}^T)^{-1}$$

Equation 8-10 P_{EO} §

with Y as defined in § Equation 8-3. The pulse fit vector p_E is read column-wise from the first N_p columns of P_{EO} , while p_O is read column-wise from the next N_p columns.

The pulse responses p_E and p_O each have a length of N_p UI, and when oversampling by a factor of M (the number of samples per PAM4 symbol), the array sizes for p_E , p_O , and e become $N_p M$. For PCIe specific case, N_p is set to 600, the pulse delay D_p is 4, and M should be a minimum of 32.

The error vector, e , is defined by [§ Equation 8-11](#)

$$e = p_E * (x \cdot \mathbf{1}_E) + p_O * (x \cdot \mathbf{1}_O) - y$$

[Equation 8-11 \$e\$](#) §

and can be computed using P_{EO} and X_{EO} in [§ Equation 8-10](#). The standard deviation of the error vector, denoted as σ_e , is obtained from the measured PRBS portion of the compliance pattern.

The parameter σ_n measures the uncorrelated RMS amplitude noise of each symbol level (including random noise and uncorrelated bounded noise effects), while not including ISI and jitter effects. Noise for each of the four PAM4 voltage levels, σ_L , is measured by using the PAM4 symbol 61 of the 64-UI long slow pattern for the corresponding voltage level that appears once in every repeat of the Compliance Pattern. When measuring σ_L , an adjustment for uncorrelated random noise contributed by the instrumentation such as uncorrelated random scope noise shall be applied. Equivalent oscilloscope settings used for noise characterization shall be consistent with the oscilloscope settings used for waveform capture when calculating SNDR_{TX} . For each voltage level L (where $L = 0, 1, 2, 3$), the σ_L measurement is the result of eight independent measurements on eight evenly spaced sample points within the Unit Interval of symbol 61 in the run of 64 identical symbols. Each of the eight measurements denoted as $\sigma_{L,i}$ (where $i=1..8$) is calculated by using the following equations.

$$\sigma_{L,i}^2 = \frac{1}{N_k} \sum_{k=1}^{N_k} (V_{L,i,k} - \mu_{L,i})^2$$

[Equation 8-12 \$\sigma_{L,i}\$](#) §

$$\mu_{L,i} = \frac{1}{N_k} \sum_{k=1}^{N_k} V_{L,i,k}$$

[Equation 8-13 \$\mu_{L,i}\$](#) §

In the above equations, N_k is the number of repetitions of the compliance pattern in the recorded waveform, $V_{L,i,k}$ is the waveform voltage at the i_{th} data sample location within the 61st symbol UI in the k_{th} repetition of the compliance pattern for each voltage level, and $\mu_{L,i}$ is the mean of N_k waveform voltage samples for the i_{th} data sample location for the corresponding voltage level.

σ_L is obtained via the following equation:

$$\sigma_L = \sqrt{\frac{\sum_{i=1}^8 \sigma_{L,i}^2}{8}}$$

Equation 8-14 σ_L §

σ_n is the average of the four σ_L measurements, one for each PAM4 voltage level, denoted as

$$\sigma_n = \frac{1}{4}(\sigma_0 + \sigma_1 + \sigma_2 + \sigma_3)$$

Equation 8-15 σ_n §

The Tx SNDR_{TX} is defined by § Equation 8-16

$$SNDR = 10 \times \log_{10} \left(\frac{\sum_{k \in S_E} P_E(k)^2 + \sum_{k \in S_O} P_O(k)^2}{2(\sigma_e^2 + \sigma_n^2)} \right)$$

Equation 8-16 SNDR §

Sets S_E and S_O are associated with pulse response p_E and p_O respectively. To obtain set S from a pulse response p , follow these steps:

1. Find the index of pulse response peak, j : $P(j) = p_{max}$
2. Pick all indexes of p that have the same within-UI offset as the peak index j .
3. Mathematically, set S is defined as:

$$S = \{k \mid k = j + nM, n \in \mathbb{Z} \text{ and } 0 \leq k < N_p M\}$$

Note that j may not be the same for p_E and p_O .

8.3.3.14 Transmitter Ratio of Level Mismatch (R_{LM-TX}) for 64.0 GT/s §

Transmitter linearity is defined as a function of the PAM4 signal levels (V_0 , V_1 , V_2 , and V_3) transmitted for PAM4 2-bit symbols (see § Section 4.2.3.1.1). The ratio of level mismatch, R_{LM} , is defined as shown below:

$$V_{mid} = (V_0 + V_3) / 2$$

$$ES_1 = (V_1 - V_{mid}) / (V_0 - V_{mid})$$

$$ES_2 = (V_2 - V_{mid}) / (V_3 - V_{mid})$$

$$R_{LM} = \min\left(3 \times ES_1, 3 \times ES_2, (2 - 3 \times ES_1), (2 - 3 \times ES_2)\right)$$

Equation 8-17 R_{LM} §

The PAM4 signal levels (V_L where $L = 0, 1, 2$, and 3) described above are measured by following a variant of the Multi Pulse Response Fit method. The method of obtaining the V_L values described in this section only applies to calculation of R_{LM-TX} .

8.3.3.14.1 Multi Pulse Response Fit (MPRF) – PAM4 Voltage Variant §

To quantify the linearity, the levels of symbols 0 and 3 are used as reference levels. The deviation of levels of symbol 1 and 2 from even spacing between the reference levels is then measured. For the Linear Fit Pulse Response (multiple pulses) method used for PAM4 signaling, three pulses are extracted: one normalized pulse representing both symbol 0 and symbol 3, one normalized pulse for symbol 1 and one normalized pulse for symbol 2.

The steps in § Section 8.3.3.12 describing the Linear Fit Pulse Response (single pulse) method will be followed through § Equation 8-5 using M greater than or equal to 32, at least 250 pattern repeats (R_{PAT}), the pulse delay D_p of 4, and a pulse length (N_p) of 600.

Define the matrix X_{N_p} to be the first N_p rows of X .

Define the following three N_p -by- N switch matrices for the three normalized pulses as

$$W_{sym0_3} = \left(X_{N_p} == -1 \right) \text{ or } \left(X_{N_p} == 1 \right)$$

$$W_{sym1} = \left(X_{N_p} == -1 \mid 3 \right)$$

$$W_{sym2} = \left(X_{N_p} == 1 \mid 3 \right)$$

Equation 8-18 W_{sym0_3} §

Where the “==” operation returns Boolean value. If the Boolean value in an element in the matrix is true where the “==” condition is satisfied, then the corresponding element in W_{sym} is set to 1. If the Boolean value is false where the “==” condition is not satisfied, the element in W_{sym} is set to 0. These matrices serve the same purpose as the indicator function, or mask, functions in § Section 8.3.3.13 .

Denote

$$X_{sym} = \begin{bmatrix} X_{N_p} \cdot W_{sym0_3} \\ X_{N_p} \cdot W_{sym1} \\ X_{N_p} \cdot W_{sym2} \\ 1 \ 1 \ \dots \ 1 \end{bmatrix}$$

Equation 8-19 X_{sym} §

The notation $(X_{N_p} \cdot W_{sym*})$ defines pointwise multiplication, where each element in X_{N_p} is multiplied by its corresponding element in W_{sym*} . The 1 1 1 ... 1 notation denotes a row vector of ones.

Calculate the following equation to get:

$$P_{sym} = YX_{sym}^T \left(X_{sym} X_{sym}^T \right)^{-1}$$

Equation 8-20 P_{sym} §

Let P_{sym0_3} be the first N_p columns of P_{sym} , then P_{sym1} be the next N_p columns of P_{sym} , and P_{sym2} be the next N_p columns. The normalized linear fit pulse response for the symbol 0 and symbol 3, $p_{sym0_3}(k)$, is then read column-wise from the elements of P_{sym0_3} . The normalized linear fit pulse response for symbol 1, $p_{sym1}(k)$, is read column-wise from the elements of P_{sym1} . The normalized linear fit pulse response for symbol 2, p_{sym2} , is read column-wise from the elements of P_{sym2} .

For an example waveform, its symbol pulses are obtained as shown in § Figure 8-13:

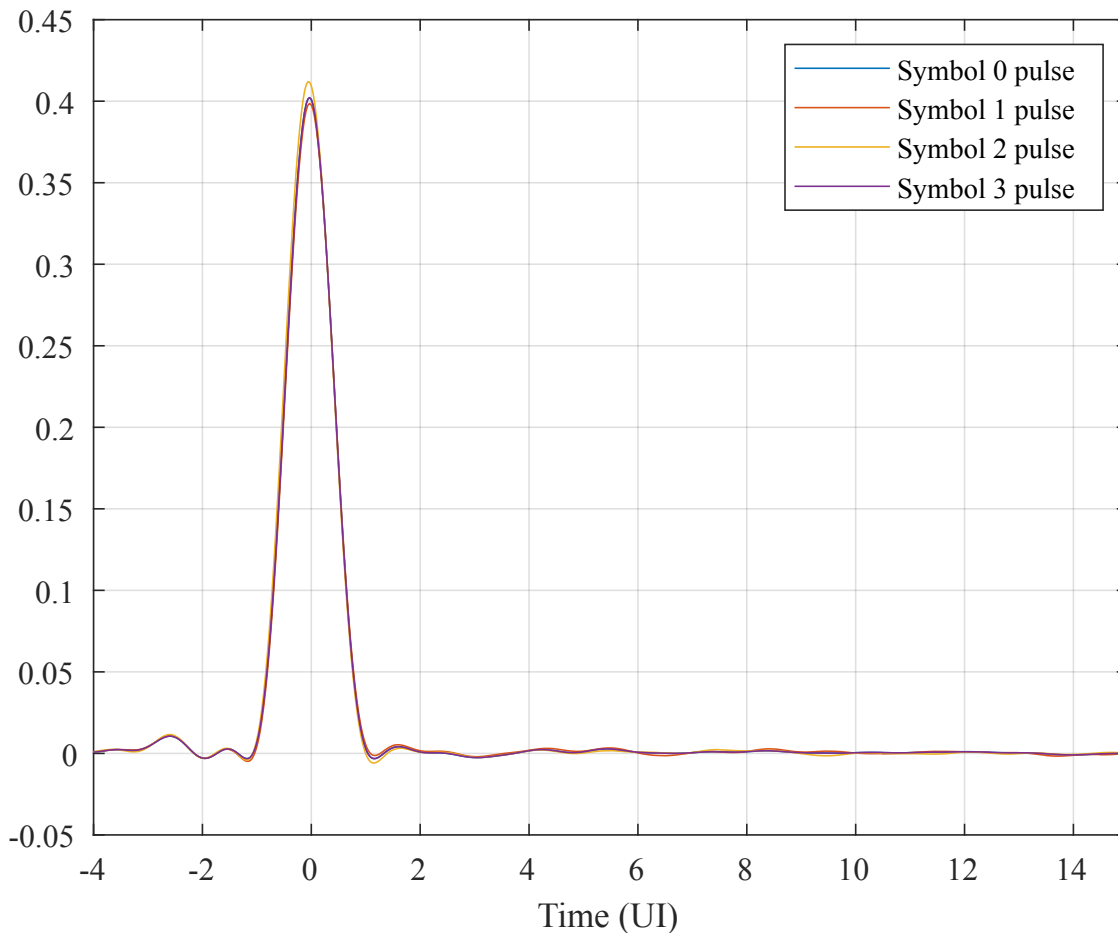


Figure 8-13 Example of Normalized Four Symbol Linear Pulse Responses §

The normalized pulses for all symbols are like each other as shown in § Figure 8-13, make it easier to compare the pulses in the same scale. The normalized pulses can be un-normalized:

$$p_{\text{sym0_un_normalized}} = -p_{\text{sym0_3}}$$

$$p_{\text{sym1_un_normalized}} = -\frac{1}{3}p_{\text{sym1}}$$

$$p_{\text{sym2_un_normalized}} = \frac{1}{3}p_{\text{sym2}}$$

$$p_{\text{sym3_un_normalized}} = p_{\text{sym0_3}}$$

Equation 8-21 $p_{\text{sym_un_normalized}}$ §

The un-normalized pulses are shown in § Figure 8-14.

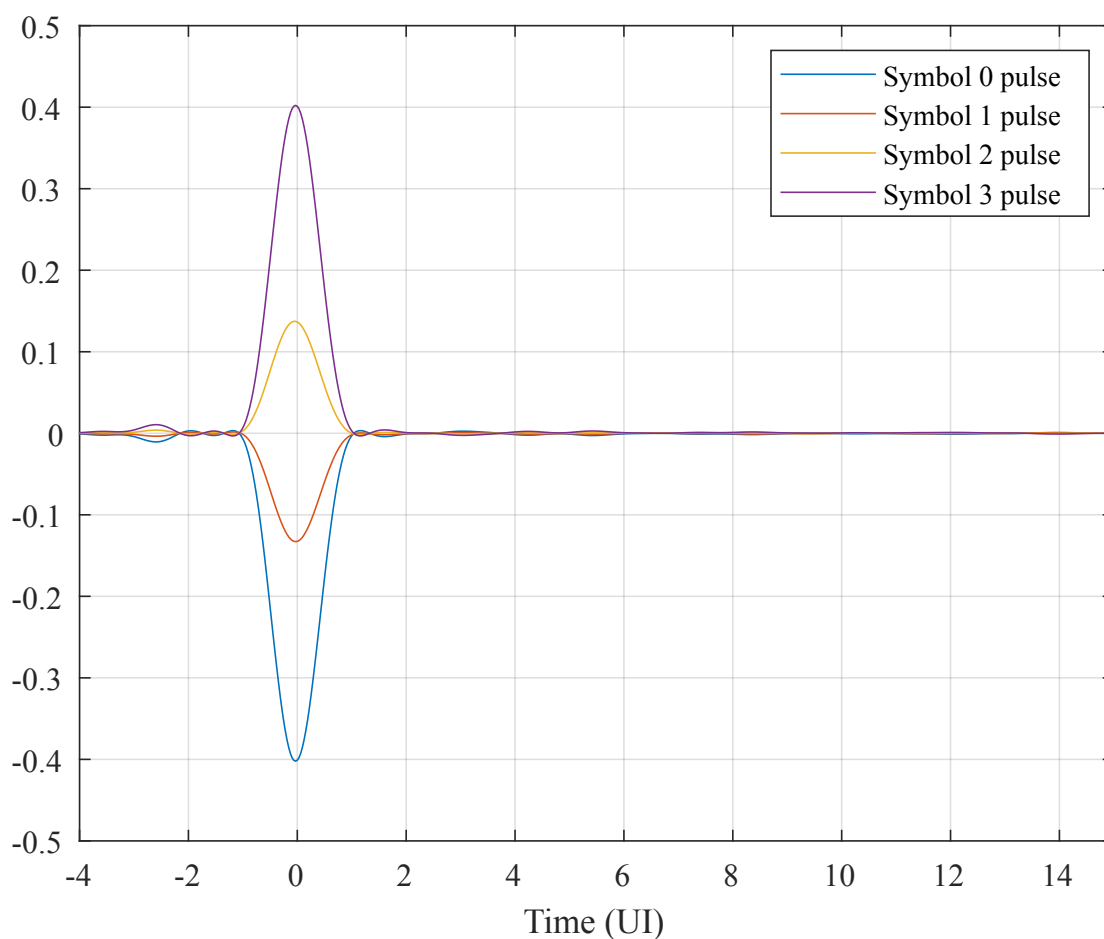


Figure 8-14 Example of Un-normalized Four Symbol Linear Pulse Responses §

The 4 pulses can be used to perform the $R_{\text{LM-TX}}$ measurement § Equation 8-17, by defining values V_L as the extrema of the un-normalized pulse fit vectors.

The multiple pulse linear fitting process is summarized as following:

1. Acquire the waveform from the transmitter.
2. Perform software clock recovery if needed, or hardware-based clock recovery.
3. Calculate the averaged data pattern waveform.
4. Get the data pattern sequence through pattern detection or obtained the known pattern sequence. Align the pattern sequence with the pattern waveform.
5. Construct the matrix X_{N_p} and Y based on the data pattern sequencer and the pattern waveform.
6. Construct the switch matrices for multiple pulse fitting.
7. Construct the matrices for the fitting solution based on the X_{N_p} and switch matrices.
8. Calculate the matrix P that contains pulses information.
9. Read the pulse samples $P_{pulse\ i}(k)$ from the matrix P .
10. Compute the un-normalized pulses using § Equation 8-21.
11. Find the peaks of the un-normalized pulses which are used for V_0 , V_1 , V_2 , and V_3 .
12. Compute R_{LM-TX} using § Equation 8-17.

8.3.4 Transmitter Margining §

Transmitters shall implement a margining procedure that allows the Tx launch voltage to be adjusted. Margining is enabled by programming a register set. Due to the larger range of Transmitter equalization, 8.0, 16.0, 32.0, and 64.0 GT/s Tx margining is subject to additional constraints: Tx margining at these speeds shall not require any coefficient or preset resolution finer than can be generated with 1/24 coefficient resolution defined for normal operation, and shall not require more Tx accuracy or capability than is required to support normal operation. It is acceptable that V_b fall below the limit set by $V_{TX-EIEOS-FS}$ or $V_{TX-EIEOS-RS}$, although proper end to-end operation is no longer guaranteed. Transmitter equalization accuracy requirements do not need to be met during margining. A Transmitter is not required to change the FS/LF values it sends in TS1 Ordered Sets during margining from the values used in normal operation.

There are 8 encoded values for transmit margin from 000b to 111b. Encoding 000b represents the normal operating range. For all supported data rates and Tx signaling mode (full swing or reduced swing), encoding 001b must produce a $V_{TX-DIFF-PP}$ compliant with the specification limits. At least three additional encodings with monotonically decreasing values for $V_{TX-DIFF-PP}$ must be supported for each data rate and Tx swing mode. For full swing signaling there must be at least one encoding with index 100b or higher that produces a $V_{TX-DIFF-PP}$ between 200 and 400 mV. For reduced swing signaling there must be at least one encoding with value 100b or higher that produces a $V_{TX-DIFF-PP}$ between 100 and 200 mV.

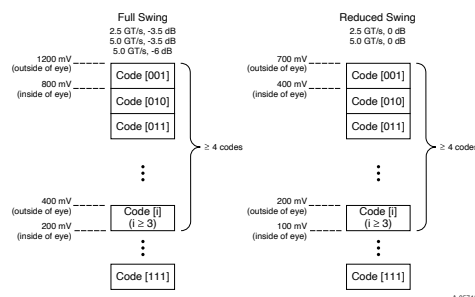


Figure 8-15 2.5 and 5.0 GT/s Transmitter Margining Voltage Levels and Codes §

8.3.5 Tx Jitter Parameters §

Jitter limits are defined identically for all data rates, although their respective values will vary with data rate. Jitter is measured at the zero-crossing point at full speed using the Compliance Pattern for 2.5, 5.0, 8.0, and 16.0 GT/s. When measuring jitter, the preset yielding the lowest jitter value should be selected. At 32.0 GT/s, the Tx under test must transmit Jitter Measurement Pattern (see § Section 4.2.13) with no Tx equalization. At 64.0 GT/s, the Tx under test must transmit Jitter Measurement Pattern (see § Section 4.2.16) with no Tx equalization for measuring the uncorrelated total jitter and deterministic jitter for all twelve transitions between the four PAM4 voltage levels. At 64.0 GT/s, the Tx under test must transmit High Swing Toggle Pattern (see § Section 4.2.17) with no Tx equalization for measuring the uncorrelated total pulse width jitter and deterministic pulse width jitter for the transitions between voltage level 0 and voltage level 3. When measuring a particular Tx Lane, it is necessary to ensure that all other PCI Express Lanes are transmitting Compliance Pattern in order to capture Tx die and package crosstalk effects. When measuring Tx jitter, it is required for the DUT to drive as many of its outputs as would occur during normal operation in a system environment. The minimum oscilloscope bandwidth for Tx jitter measurements at 32.0 and 64.0 GT/s is 50 GHz. The jitter measurements at 64.0 GT/s should be made with a 4th order Bessel-Thomson filter with a roll-off from DC value by 3 dB at 33 GHz to minimize the impact of scope high-frequency noise.

8.3.5.1 Post Processing Steps to Extract Jitter §

Measured Tx jitter is referenced to the Tx pin, and depending on what type of jitter is being measured and what reference clock architecture is being tested, is subsequently referenced to a recovered data clock, an embedded reference clock captured simultaneously with the data, or to a data edge. Data captured at TP1 requires post processing in order to remove the effects of the breakout channel and to regenerate a data clock (when an embedded reference clock is not captured simultaneously with the data).

8.3.5.2 Applying CTLE or De-embedding §

Direct probing at a Transmitter's pins is not generally feasible, so data is instead measured at TP1 of the breakout channel. By means of the replica channel it is possible to determine the loss vs. frequency characteristics of the breakout channel and de-embed this channel, resulting in measurements that are effectively referenced to the DUT's pins. Note that since de-embedding amplifies HF noise there is a practical frequency cutoff limit to de-embedding. As de-embedding amplifies HF channel and measurement noise, an HF cutoff limit must be applied to de-embedding, depending on data rate as shown in § Table 8-4.

*Table 8-4 Recommended De-embedding
Cutoff Frequency §*

Data Rate	HF Cutoff limit for de-embedding
8.0 GT/s	8 GHz - 12 GHz
16.0 GT/s	20 GHz

Jitter is decomposed into data dependent and uncorrelated terms. This separation process effectively separates the jitter caused by package effects from that caused by silicon effects such as PLL jitter and power supply noise that cannot be mitigated by equalization. As a result, the uncorrelated jitter terms define jitter as it would appear at the die pad.

As an alternative to de-embedding at 16.0 GT/s the -12 dB CTLE in the reference equalizer can be applied to the data measured at TP1 for measuring all uncorrelated jitter parameters (not DDJ).

It is recommended that s-parameters for de-embedding are measured to at least 3 times the Nyquist frequency with a frequency step size of 10 MHz.

If both de-embedding and CTLE approaches are used and given different answers only the lower values for the uncorrelated jitter parameters are used.

For 32.0 GT/s, Jitter Measurement Pattern (see § Section 4.2.13) with no Tx equalization is used to minimize the breakout channel ISI impact and avoid pessimism in jitter measurement from de-embedding and associated high frequency scope noise amplification. For further reduction of channel loss impact on jitter at 32.0 GT/s, any CTLE curve in the reference equalizer or no CTLE curve can be applied to the data measured at TP1 for measuring all uncorrelated jitter parameters (not DDJ). The CTLE or no CTLE curve that gives the lowest result for $T_{TX-UPW-TJ}$ is used.

For 64.0 GT/s, the Jitter Measurement Pattern (see § Section 4.2.16) with no Tx equalization is used for measuring the uncorrelated total jitter and the uncorrelated deterministic jitter for all twelve transitions between the four PAM4 voltage levels. The 64.0 GT/s Jitter Measurement Pattern is 52-UI long and all 12 PAM4 transitions repeat four times within the pattern resulting in 48 edge transitions. Jitter must be measured on each of the 48 edges individually and then averaged. The Q-scale associated with the 64.0 GT/s Jitter Measurement Pattern for BER of 10^{-6} is 4.8759. For mitigating channel ISI impact on jitter at 64.0 GT/s, any CTLE curve in the reference equalizer or no CTLE curve can be applied to the data measured at TP1 for measuring all uncorrelated jitter parameters (not DDJ). The CTLE or no CTLE curve that gives the lowest result for the average T_{TX-RJ} of the 48 edges is used. The uncorrelated jitter parameters for all 48 transitions must be measured and corrected for the scope noise impact. The average uncorrelated total jitter and the average uncorrelated deterministic jitter are obtained by averaging the jitter parameters of all 48 edge transitions.

For 64.0 GT/s, the High Swing Toggle Pattern (see § Section 4.2.17) with no Tx equalization is used for measuring the uncorrelated total pulse width jitter and deterministic pulse width jitter for transitions between voltage level 0 and voltage level 3. The Q-scale associated with the 64.0 GT/s High Swing Toggle Pattern for BER of 10^{-6} is 4.8916. The High Swing Toggle Pattern minimizes the breakout channel ISI impact and avoids pessimism in jitter measurement from de-embedding and associated high frequency scope noise amplification. For further reduction of channel loss impact on pulse width jitter at 64.0 GT/s, any CTLE curve in the reference equalizer or no CTLE curve can be applied to the data measured at TP1 for measuring all uncorrelated jitter parameters (not DDJ). The CTLE or no CTLE curve that gives the lowest result for $T_{TX-UPW-TJ}$ is used. The uncorrelated pulse width jitter parameters must be corrected for the scope noise impact.

8.3.5.3 Independent Refclk Measurement and Post Processing §

A Transmitter may operate in the Independent Refclk (IR) mode, in which case the Transmitter may not provide a Refclk output. In this case a single-port jitter measurement is required. The post processing algorithm must employ the appropriate model CDR for the reference clock architecture being tested.

8.3.5.4 Embedded and Non-Embedded Refclk Measurement and Post Processing §

When the transmitting PCIe device is driven from an external source to its Refclk pin it permits the Tx under test to be driven with a clean Refclk as shown in § Figure 8-1.

The specification now explicitly supports the complete matrix of Refclk options, including where the Refclk is embedded, where the reference clock is external, where the reference clock is available at the DUT's pins, and where the reference clock is not available at the DUT's pins. § Table 8-5 lists the post processing requirements for each of the four possible combinations. Embedded Refclk with Refclk available at the DUT's pin represents a special case where any jitter common to both the Refclk and the data must be removed via a two-port measurement.

If the DUT supports multiple Refclk modes, as described in § Table 8-5 then the Tx needs to be tested in each of the Refclk modes it supports.

Table 8-5 Tx Measurement and Post Processing For Different Refclks §

	Embedded Refclk	Non-Embedded Refclk
Refclk available at DUT pin and not testing SRIS mode	2-port measurement, CC CDR, PLL ¹ and 10 ns transport delay ²	1-port measurement, CC CDR, clean external Refclk
Refclk not available at DUT pin or testing SRIS mode	1-port measurement, SRIS CDR	1-port measurement, CC CDR, clean external Refclk

Notes:

1. PLL characteristics are defined in Refclk section for each data rate.
2. Refer to § Section 8.6.6 for a discussion of the transport delay

8.3.5.5 Behavioral CDR Characteristics §

A behavioral CDR filter is applied to reject low frequency jitter that would normally be tracked by the CDR in a Receiver. As such, the behavioral CDR represents a bounding function for actual CDR implementations. Roll-off characteristics of the behavioral CDR are dependent on whether the corresponding DUT supports an embedded vs. non-embedded Refclk, is operating in CC or IR mode, and whether the Refclk pin is available for probing (see § Table 8-5). In all cases the behavioral CDR represents a high pass filter function where the corner frequency depends on the Tx data rate. § Figure 8-16 shows the CC first-order CDR transfer functions for an f_{3dB} of 1.5 MHz, 5.0 MHz, and 10 MHz that corresponds to 2.5 GT/s, 5.0 GT/s, and 8.0 GT/s, respectively. The 10 MHz behavioral CDR is also used for CC Transmitter and CC Reference Clock testing for 16.0 GT/s and, optionally, for Receiver Stressed Eye calibration when 32.0 GT/s is not supported.

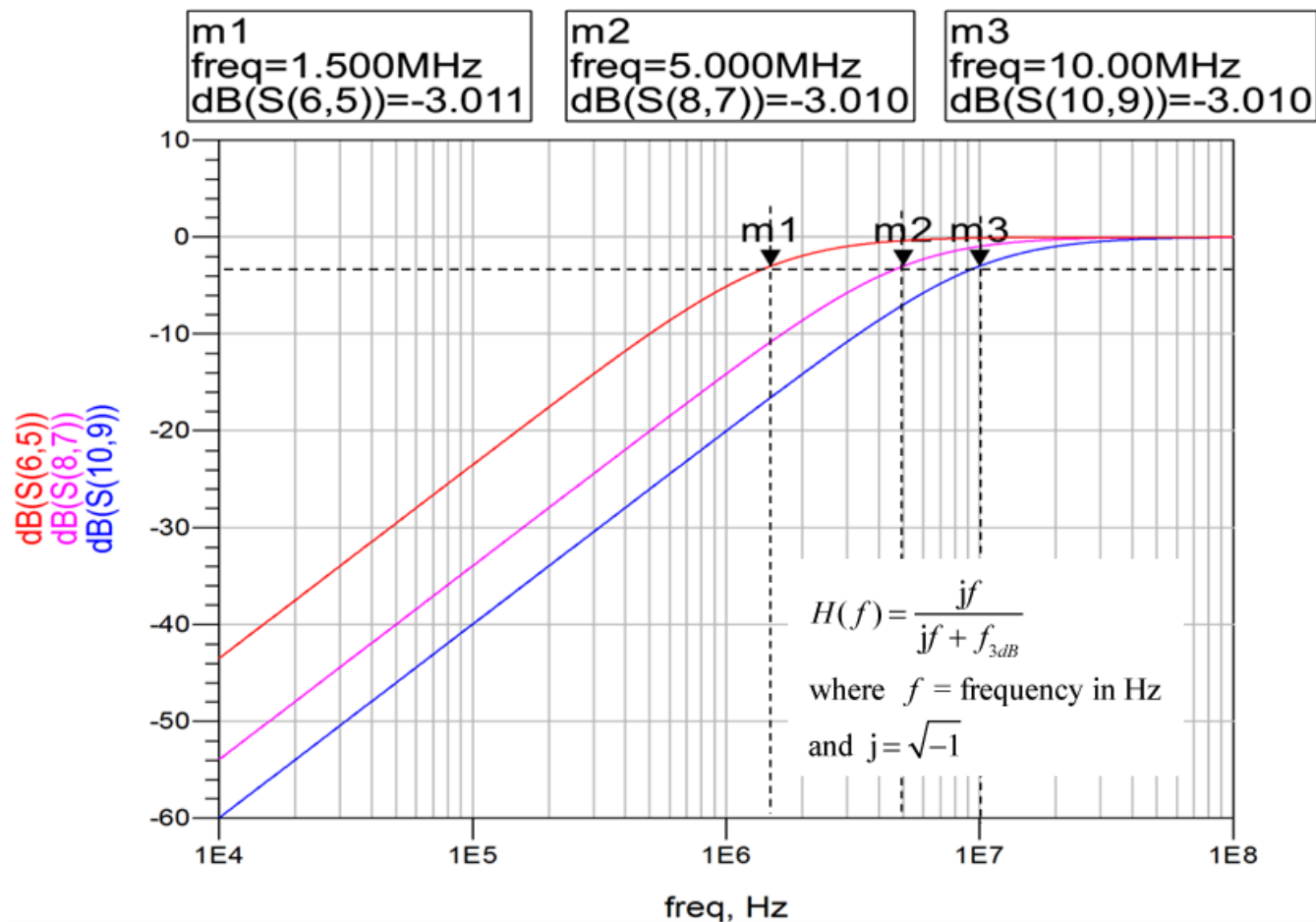


Figure 8-16 First Order CC Behavioral CDR Transfer Functions §

§ Figure 8-17 illustrates second order CDR transfer functions corresponding to 2.5 GT/s and 5.0 GT/s. These functions are defined by a ζ of 0.707 and an f_{3dB} of 1.5 MHz and 5.0 MHz, respectively. Behavioral transfer functions for 8.0 GT/s, 16.0 GT/s, 32.0 GT/s, and 64.0 GT/s approximate the piecewise linear sinusoidal jitter (Sj) masks shown in § Section 8.4.2.2.1. SRIS capable Transmitters must be evaluated using these behavioral transfer functions.

Note: The common clock (CC) and independent reference clock (IR) architectures are not interoperable - although it is possible to design a single Receiver that meets both sets of electrical requirements.

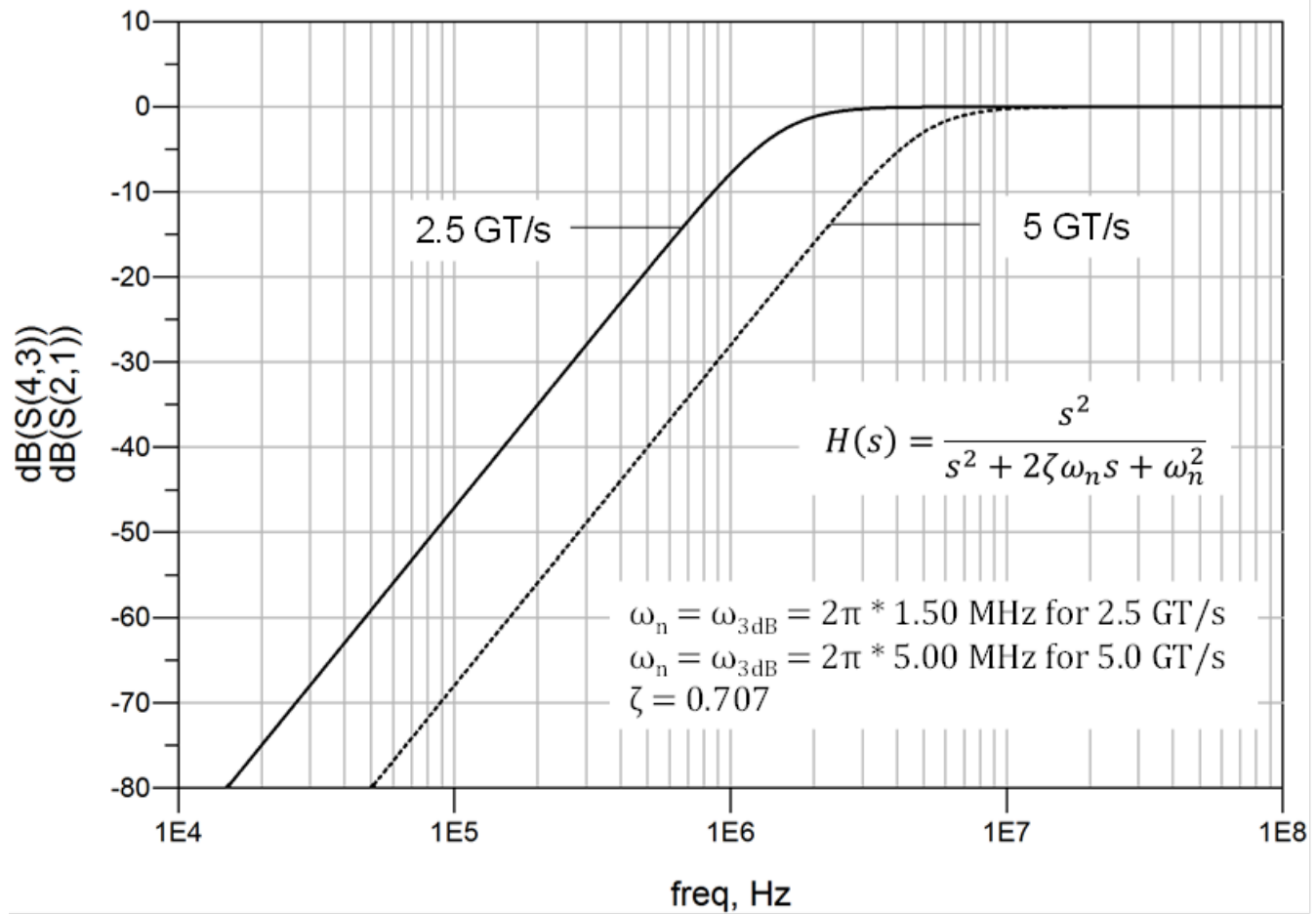


Figure 8-17 2nd Order Behavioral SRIS CDR Transfer Functions for 2.5 GT/s and 5.0 GT/s §

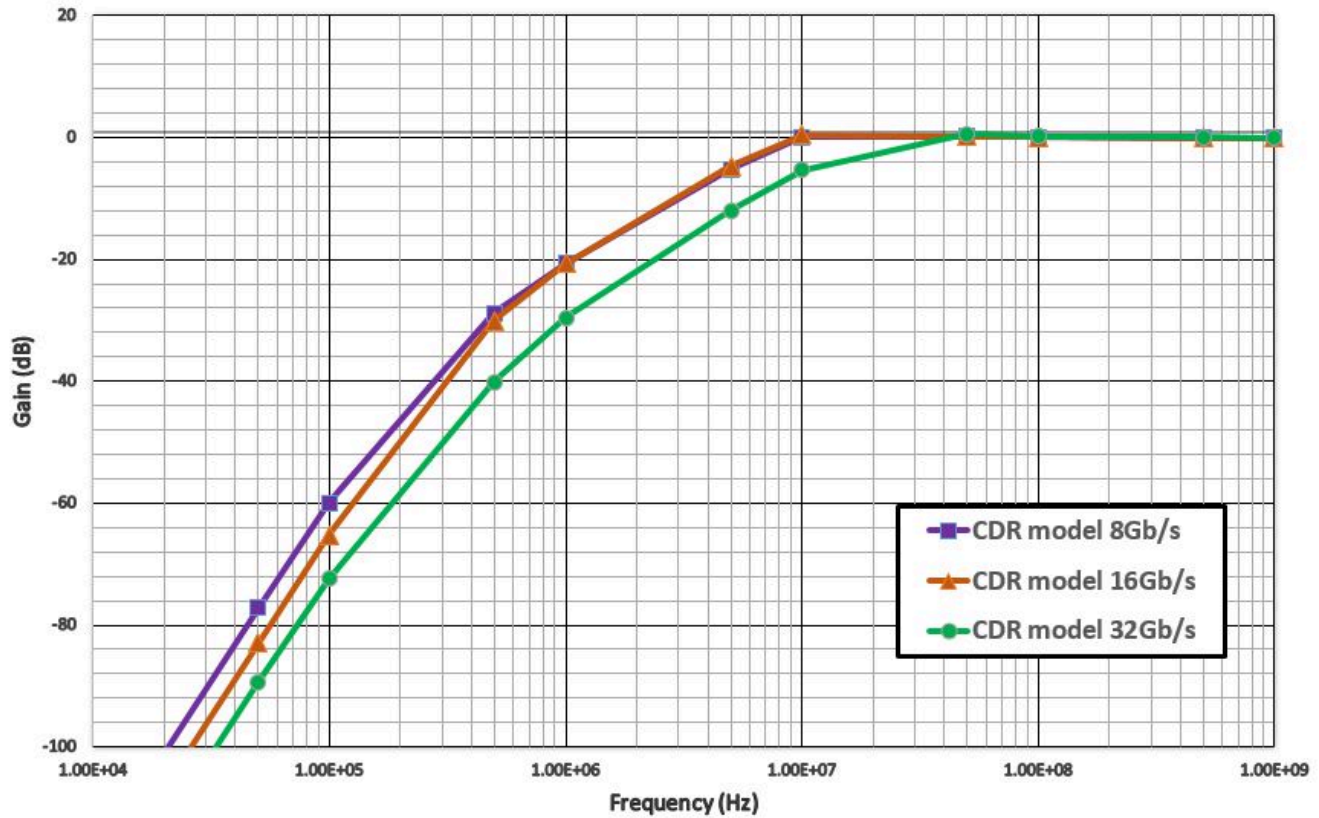


Figure 8-18 Behavioral SRIS CDR Function for 8.0 GT/s, and SRIS and CC CDR for 16.0 and 32.0 GT/s §

$$H(s) = \frac{s^2}{s^2 + sA + B} \times \frac{s^2 + 2\zeta_2\omega_0s + \omega_0^2}{s^2 + 2\zeta_1\omega_0s + \omega_0^2} \times \frac{s}{s + \omega_1}$$

$$\zeta_1 = \frac{1}{\sqrt{2}}$$

$$\zeta_2 = 1$$

$$\omega_0 = 10^7 \times 2\pi$$

$$\omega_1 = 4 \times 10^5 \times 2\pi$$

Equation 8-22 Behavioral SRIS CDR at 8.0 GT/s and SRIS and CC Behavioral CDR at 16.0 GT/s §

$$A = 10^7 \times 2\pi$$

$$B = 2.2 \times 10^{12} \times (2\pi)^2$$

Equation 8-23 SRIS Behavioral CDR Parameters at 8.0 GT/s §

$$A = 9.5 \times 10^6 \times 2\pi$$

$$B = 4.36 \times 10^{12} \times (2\pi)^2$$

Equation 8-24 SRIS and CC Behavioral CDR Parameters at 16.0 GT/s §

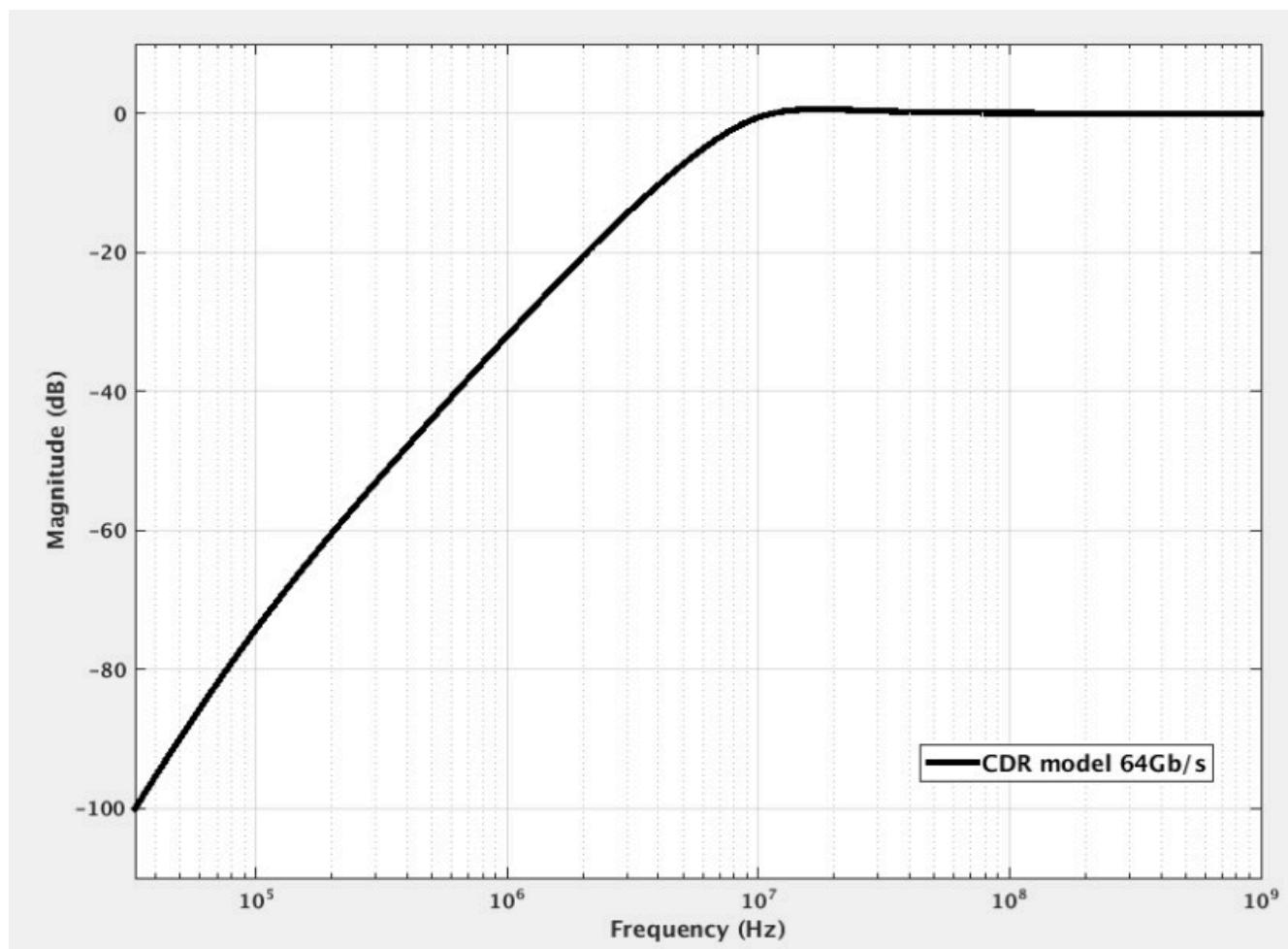


Figure 8-19 Behavioral SRIS and CC CDR for 64.0 GT/s §

$$H(s) = \frac{s^2}{(s + \omega_0)(s + \omega_1)} \times \frac{s^2 + 2\zeta_2\omega_0s + \omega_0^2}{s^2 + 2\zeta_1\omega_0s + \omega_0^2} \times \frac{s}{s + \omega_{LF}}$$

$$\zeta_1 = \frac{1}{\sqrt{2}}$$

$$\zeta_2 = 1$$

32.0 GT/s

64.0 GT/s

$$\omega_0 = 20 \times 10^6 \times 2\pi$$

$$\omega_0 = 10 \times 10^6 \times 2\pi$$

$$\omega_1 = 1.1 \times 10^6 \times 2\pi$$

$$\omega_1 = 3.88 \times 10^6 \times 2\pi$$

$$\omega_{LF} = 160 \times 10^3 \times 2\pi$$

$$\omega_{LF} = 87 \times 10^3 \times 2\pi$$

Equation 8-25 SRIS and CC Behavioral CDR Parameters at 32.0 and 64.0 GT/s §

8.3.5.6 Data Dependent and Uncorrelated Jitter §

Measured at TP1 and de-embedded back to the pin, a Transmitter's jitter contains both data dependent and uncorrelated components. The data dependent components occur principally due to package loss and reflection. Uncorrelated jitter sources include PLL jitter, power supply noise, and crosstalk. The specification separates jitter into uncorrelated and data dependent bins, because such a separation matches well with the Tx and Rx equalization capabilities. Uncorrelated jitter is not mitigated by Tx or Rx equalization and represents timing margin that cannot be recovered via equalization. It is important that margin recoverable by means of equalization (data dependent) is not budgeted as non-recoverable jitter.

Once data dependent jitter has been removed from the Tx measurement it becomes possible to resolve the remaining jitter into T_j and deterministic jitter (Dual Dirac Model) (DJDD) components. High frequency jitter (which is subject to jitter amplification in the channel) is accounted for by separate T_{TX-UPW-DJDD} and T_{TX-UPW-TJ} parameters.

8.3.5.7 Data Dependent Jitter §

While DDJ is not explicitly defined as a parameter in the specification, it is necessary to separate DDJ in order to eliminate package loss effects and reference the jitter parameters of interest to the Tx die pad. Separation of jitter into data dependent and uncorrelated components may be achieved by averaging techniques; for example, by having the Tx repeatedly drive the compliance test pattern which is a repeating pattern.

§ Figure 8-20 illustrates the relation between Tx data, recovered clock, and the data's PDF. Data dependent jitter is defined as the time delta between the PDF's mean for each zero-crossing point and the corresponding recovered clock edge. A sufficient number of repeated patterns must be accumulated to yield stable mean values and PDF profiles for each transition. These PDFs are then utilized to extract uncorrelated jitter parameters.

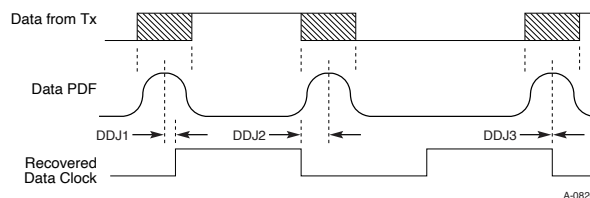


Figure 8-20 Relation Between Data Edge PDFs and Recovered Data Clock §

8.3.5.8 Uncorrelated Total Jitter and Deterministic Jitter (Dual Dirac Model) (T_{TX-UTJ} and $T_{TX-UDJDD}$) §

Uncorrelated Total Jitter (UTJ) and uncorrelated deterministic jitter (Dual Dirac model) (UDJDD) are referenced to a recovered data clock generated by means of a CDR tracking function. Uncorrelated jitter may be derived after removing the DDJ component from each PDF and combining the PDFs for all edges in the pattern. By appropriately converting the PDF to a Q-scale it is possible to obtain the graphical relation shown in § Figure 8-21, from which T_{TX-UTJ} and $T_{TX-UDJDD}$ may be derived. In § Figure 8-21 note that the two PDF curves are identical but that the fitted slopes, defined by $1/RJ_{LH}$ and $1/RJ_{RH}$, may differ.

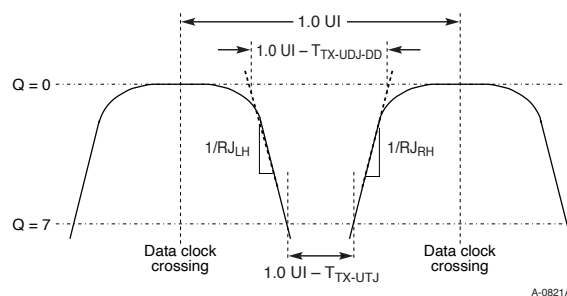


Figure 8-21 Derivation of T_{TX-UTJ} and $T_{TX-UDJDD}$ §

8.3.5.9 Random Jitter (T_{TX-RJ}) (informative) §

Random jitter is uncorrelated with respect to data dependent jitter. T_{TX-RJ} may be obtained by subtracting $T_{TX-UDJDD}$ from T_{TX-UTJ} and is included in the specification as an informative parameter only. It is typically used as a benchmark to characterize PLL performance.

8.3.5.10 Uncorrelated Total and Deterministic PWJ ($T_{TX-UPW-TJ}$ and $T_{TX-UPW-DJDD}$) §

Pulse width jitter is defined as an edge to edge phenomenon on consecutive edges nominally 1.0 UI apart. § Figure 8-22 illustrates how PWJ is defined, showing that it is typically present on both data edges of consecutive UI. To accurately quantify PWJ it is first necessary to remove the ISI contributions to PWJ. The shaded areas on either side of the unjittered edges represent the maximum amount of jitter about that edge. Note the jitter for one edge is assumed to be independent from the other.

An equivalent description of PWJ may be obtained by referencing to a fixed leading edge and having jitter contributions from both edges appear at the trailing edge. This approach yields a single PDF as shown below. Each 1 UI wide pulse in the pattern will have a different median for this PDF which is caused by ISI and F/2 jitter. The average of the medians for 1 UI wide pulses at odd and even UI numbers within the pattern are calculated, and the odd and even PDF's are normalized to the appropriate average of medians and summed to form an odd UI PDF and an even UI PDF. The final PDF is calculated from the sum of the summed odd and even UI PDFs. The key idea here is that the final PDF for uncorrelated PWJ should include F/2 or odd/even UI jitter

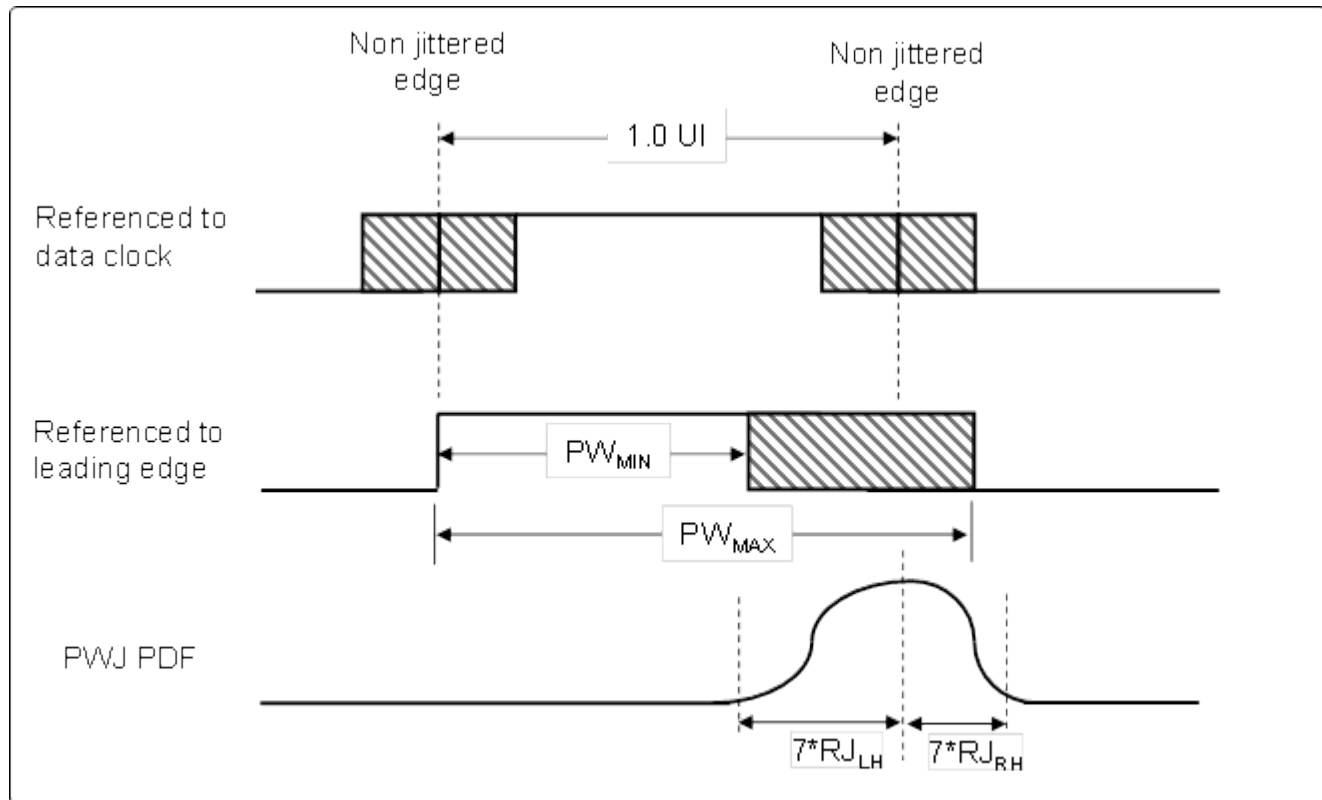


Figure 8-22 PWJ Relative to Consecutive Edges 1 UI Apart §

The PDF of jitter around each non-jittered edge may be converted into the Q-scale (see § Figure 8-23) from which $T_{TX-UPW-TJ}$ and $T_{TX-UPW-DJDD}$ may be derived in a manner analogous to T_{TX-UTJ} and $T_{TX-UDJDD}$. Note that the PDF may not be symmetric, and the tail of interest is RJ_{LH} , since it represents pulse compression.

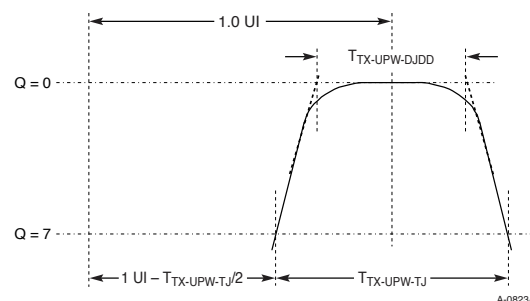


Figure 8-23 Definition of $T_{TX-UPW-DJDD}$ and $T_{TX-UPW-TJ}$ Data Rate Dependent Transmitter Parameters §

8.3.6 Data Rate Dependent Parameters §

Note: the jitter margins for 2.5 GT/s and 5.0 GT/s were previously defined at the device's pins. For 2.5 GT/s the jitter was defined via a single parameter that lumped DDJ, UDJDD, UTJ, PWJ-DDJ and PWJ-TJ into a single quantity. Consequently, it is necessary to first remove the DDJ jitter component. Since there was no previous UDJ-UTJ separation T_{TX-UTJ} and $T_{TX-UDJDD}$ are set equal to each other. Similarly, there was no UTJ-PWJ separation, so it is necessary to assume that the entirety of the uncorrelated jitter is PWJ that occurs oppositely on consecutive edges of a 1 UI wide pulse.

For 5.0 GT/s a similar removal of DDJ must be performed to obtain UTJ. However, [PCIe-3.0] for 5.0 GT/s did specify Rj, so a distinct value for $T_{TX-UDJDD}$ can be obtained. Similarly, [PCIe-3.0] for 5.0 GT/s defined a minimum pulse width, assumed to be 100% Dj, from which $T_{TX-UPW-TJ}$ and $T_{TX-UPW-DJDD}$ may be derived. For 64.0 GT/s PAM4 signaling, the voltage parameters such as differential peak-to-peak Tx voltage swing correspond to the swing between PAM4 voltage level 0 and voltage level 3.

Table 8-6 Data Rate Dependent Transmitter Parameters §

Symbol	Parameter description	2.5 GT/s	5.0 GT/s	8.0 GT/s	16.0 GT/s	32.0 GT/s	64.0 GT/s	Units	Notes
UI (Tx)	Unit Interval	(min) 399.88 (max) 400.12 (300 PPM)	(min) 199.94 (max) 200.06 (300 PPM)	(min) 124.9625 (max) 125.0375 (300 PPM)	(min) 62.48125 (max) 62.51875 (300 PPM)	(min) 31.246875 (max) 31.253125 (100 PPM)	(min) 31.246875 (max) 31.253125 (100 PPM)	ps	Does not include SSC variations
BW_{TX-PKG-PLL1}	Tx PLL bandwidth corresponding to <u>PKG_{TX-PLL1}</u>	(min) 1.5 (max) 22.0	(min) 8.0 (max) 16.0	(min) 0.5 (max) 4.0	(min) 0.5 (max) 4.0	(min) 0.5 (max) 1.8	(min) 0.5 (max) 1.0	MHz	Second order PLL jitter transfer bounding function. Notes 1, 2, 9.
BW_{TX-PKG-PLL2}	Tx PLL bandwidth corresponding to <u>PKG_{TX-PLL2}</u>	N/A	(min) 5.0 (max) 16.0	(min) 0.5 (max) 5.0	(min) 0.5 (max) 5.0	N/A	N/A	MHz	2.5 and 32.0 GT/s specify only one combination of PLL BW and jitter. Notes 1, 2, 9.
PKG_{TX-PLL1}	Tx PLL peaking corresponding to <u>BW_{TX-PKG-PLL1}</u>	(max) 3.0	(max) 3.0	(max) 2.0	(max) 2.0	(max) 2.0	(max) 2.0	dB	Second order PLL jitter transfer bounding function. Notes 1, 2.
PKG_{TX-PLL2}	Tx PLL peaking corresponding to <u>BW_{TX-PKG-PLL2}</u>	N/A	(max) 1.0	(max) 1.0	(max) 1.0	N/A	N/A	dB	2.5 and 32.0 GT/s specify only one combination of PLL BW and jitter. Notes 1, 2.
V_{TX-DIFF-PP}	Differential peak-peak Tx voltage swing for full swing operation	(min) 800 (max) 1000	(min) 800 (max) 1000	(min) 800 (max) 1000	(min) 800 (max) 1000	(min) 800 (max) 1000	(min) 800 (max) 1000	mVPP	As measured with compliance test load. Defined as $2 \times V_{TXD+} - V_{TXD-} $ Note 3.
V_{TX-DIFF-PP-LOW}	Differential peak-peak Tx	(min) 400	(min) 400	(min) 400	(min) 400	(min) 400	(min) 400	mVPP	As measured with compliance test

Symbol	Parameter description	2.5 GT/s	5.0 GT/s	8.0 GT/s	16.0 GT/s	32.0 GT/s	64.0 GT/s	Units	Notes
	voltage swing for low swing operation	(max) 1000	(max) 1000	(max) 1000	(max) 1000	(max) 1000	(max) 1000		load. Defined as $2 \times V_{TXD+} - V_{TXD-} $ Note 3.
$V_{TX-EIEOS-FS}$	Minimum voltage swing during EIEOS for full swing signaling	N/A	N/A	250 (min)	250 (min)	250 (min)	250 (min)	mVPP	Note 4
$V_{TX-EIEOS-RS}$	Minimum voltage swing during EIEOS for reduced swing signaling	N/A	N/A	232 (min)	232 (min)	232 (min)	232 (min)	mVPP	Note 4
$ps21_{TX-ROOT-DEVICE}$	Pseudo package loss of a device containing root port	N/A	N/A	(max) 3.0	(max) 5.0	(max) 8.5	(max) 7.5	dB	Note 5.
$ps21_{TX-NON-ROOT-DEVICE}$	Pseudo package loss for all devices not containing root ports	N/A	N/A	(max) 3.0	(max) 3.0	(max) 3.7	(max) 3.7	dB	Note 5.
$IL_{fit_{TX-ROOT-DEVICE}}$	Fitted insertion loss at Nyquist	N/A	N/A	N/A	N/A	(max) 9.0	(max) 8.0	dB	Note 8
$IL_{fit_{TX-NON-ROOT-DEVICE}}$	Fitted insertion loss at Nyquist	N/A	N/A	N/A	N/A	(max) 4.0	(max) 4.0	dB	Note 8
$V_{TX-BOOST-FS}$	Maximum nominal Tx boost ratio for full swing	N/A	N/A	8.0	8.0 (min)	8.0 (min)	8.0 (min)	dB	Nominal boost beyond 8.0 dB is limited to guarantee that $ps21_{TX}$ limits are satisfied.
$V_{TX-BOOST-RS}$	Maximum nominal Tx boost ratio for reduced swing	N/A	N/A	2.5	~2.5 (min)	~2.5 (min)	~2.5 (min)	dB	Assumes ± 1.0 dB tolerance from diagonal elements in § Figure 8-9.
$EQ_{TX-COEFF-RES}$	Tx coefficient resolution	N/A	N/A	1/(min) 63 1/(max) 24	1/(min) 63 1/(max) 24	1/(min) 63 1/(max) 24	1/(min) 63 1/(max) 24	N/A	

Symbol	Parameter description	2.5 GT/s	5.0 GT/s	8.0 GT/s	16.0 GT/s	32.0 GT/s	64.0 GT/s	Units	Notes
$V_{TX-DE-RA-TIO-3.5dB}$	Tx de-emphasis ratio for 2.5 and 5.0 GT/s	(min) 2.5 (max) 4.5	(min) 2.5 (max) 4.5	N/A	N/A	N/A	N/A	dB	
$V_{TX-DE-RA-TIO-6dB}$	Tx de-emphasis ratio for 5.0 GT/s	N/A	(min) 4.5 (max) 7.5	N/A	N/A	N/A	N/A	dB	
T_{TX-UTJ}	Tx uncorrelated total jitter	(max) 100	(max) 50	(max) 27.55	(max) 11.8	(max) 6.25	(max) 4.00 at 10^{-6}	ps PP at 10^{-12}	See § Section 8.3.5.1 and § Section 8.3.5.8 for details.
$T_{TX-UTJ-SRIS}$	Tx uncorrelated total jitter when testing for the IR clock mode with SSC	(max) 100	(max) 66.51	(max) 33.83	(max) 15.85	(max) 7.15	(max) 4.389 at 10^{-6}	ps PP at 10^{-12}	See § Section 8.3.5.1 and § Section 8.3.5.8 for details.
$T_{TX-UDJDD}$	Tx uncorrelated Dj for non-embedded Refclk	(max) 100	(max) 30	(max) 12	(max) 6.25	(max) 3.125	(max) 1.563	ps PP	See § Section 8.3.5.1 and § Section 8.3.5.8 for details.
$T_{TX-UPW-TJ}$	Total uncorrelated pulse width jitter	N/A	(max) 40	(max) 24	(max) 12.5	(max) 6.25	(max) 4.00 at 10^{-6}	ps PP at 10^{-12}	See § Section 8.3.5.1 and § Section 8.3.5.8 for details.
T_{TX-RJ}	Tx Random jitter	N/A	1.4 - 3.6	1.17 - 1.97	0.40 - 0.84	0.23 - 0.45	0.26 - 0.42	ps RMS	Informative parameter only. Range of Rj possible with zero to maximum allowed $T_{TX-UDJDD}$.
$T_{TX-UPW-DJDD}$	Deterministic DjDD uncorrelated pulse width jitter	N/A	(max) 40	(max) 10	(max) 5	(max) 2.5	(max) 1.25	ps PP	See § Section 8.3.5.1 and § Section 8.3.5.8 for details.
R_{LM-TX}	Level Separation Mismatch Ratio	N/A	N/A	N/A	N/A	N/A	(min) 0.95		See § Section 8.3.3.14 for details
$SNDR_{TX}$	Signal-to-Noise-Distortion Ratio	NA	NA	NA	NA	NA	(min) 34	dB	See § Section 8.3.3.13 for details
$V_{TX-AC-CM-PP}$	Tx AC peak-peak common mode voltage over 0.03-500 MHz range	(max) 150	(max) 100	(max) 50	(max) 50	(max) 50	(max) 25	mVPP	Tx ACCM noise measurement analysis is done without any de-embedding.

Symbol	Parameter description	2.5 GT/s	5.0 GT/s	8.0 GT/s	16.0 GT/s	32.0 GT/s	64.0 GT/s	Units	Notes
$V_{TX-AC-CM-PP-filtered}$	Tx AC peak-peak common mode voltage filtered with a simple low-pass filter (-3 dB roll-off at Nyquist frequency)	(max) 150	(max) 150	(max) 150	(max) 150	(max) 150	(max) 75	mVPP	Tx ACCM noise measurement analysis is done without any de-embedding.
$L_{TX-SKEW}$	Lane-to-Lane Output Skew	(max) 2.5	(max) 2.0	(max) 1.5	(max) 1.25	(max) 1.25	(max) 1.25	ns	Between any two Lanes within a single Transmitter.
$RL_{TX-DIFF}$	Tx package plus die differential return loss	See § Figure 8-26					See § Figure 8-26	dB	Note 6
RL_{TX-CM}	Tx package plus die common mode return loss	See § Figure 8-25					See § Figure 8-27	dB	Note 6

Notes:

1. A single combination of PLL BW and peaking is specified for 2.5, 32.0, and 64.0 GT/s implementations. For other data rates, two combinations of PLL BW and peaking are specified to permit designers to make a tradeoff between the two parameters.
2. The Tx PLL Bandwidth must lie between the min and max ranges given in the above table. PLL peaking must lie below the value listed above. Note: the PLL B/W extends from zero up to the value(s) specified in the above table. The PLL BW is defined at the point where its transfer function crosses the -3dB point.
3. See § Section 8.3.3.6 and § Section 8.3.3.7 for measurement details.
4. $V_{TX-EIEOS-FS}$ and $V_{TX-EIEOS-RS}$ are measured at the device pin and include package loss. Voltage limits comprehend both full swing and reduced swing modes. A Transmitter must advertise a value for LF during TS1 at 8.0, 16.0, 32.0, and 64.0 GT/s that ensures that these parameters are met.
5. The numbers above consider measurement error. For some Tx package/driver combinations ps_{21TX} may be greater than 0 dB. The channel compliance methodology at 2.5 and 5.0 GT/s assumes the 8.0 GT/s package model.
6. The DUT must be powered up and DC isolated, and its data+/data- outputs must be in the low-Z state at a static value.
7. The reference plane for all parameters at 2.5 and 5.0 GT/s is the package pins.
8. These are design parameter requirements - a specific test methodology for them is not defined.
9. For PCIe 6.0 devices that do not support 32.0 and 64.0 GT/s have the option to use 2 MHz as min of $BW_{TX-PKG-PLL1}$ and $BW_{TX-PKG-PLL2}$ for both 8.0 and 16.0 GT/s. The corresponding T_{TX-UTJ} max value is 31.25 ps at 8.0 GT/s and 12.5 ps at 16.0 GT/s. The range of T_{TX-RJ} is 1.4-2.2 ps at 8.0 GT/s and 0.45-0.89 ps at 16.0 GT/s. Such devices also have the option to use 1st-order, 10 MHz CDR filter for testing Tx, Reference clock, and CC Rx.

8.3.7 Tx and Rx Return Loss for 2.5, 5.0, 8.0, 16.0, and 32.0 GT/s §

Return loss measurements for the Tx and Rx are essentially identical, so both are included in the Transmitter section. Return loss measurements are made at the end of the respective breakout channels and require that the breakout channel's contribution to RL be de-embedded, thereby associating the return loss with the Tx or Rx pin. Return loss

measurements are made with a reference impedance of 50 ohms. § Figure 8-24 defines the pass/fail mask for differential return loss for 2.5, 5.0, 8.0, 16.0, and 32.0 GT/s. Both differential and common mode are defined over a frequency range of 50 MHz to 16.0 GHz.

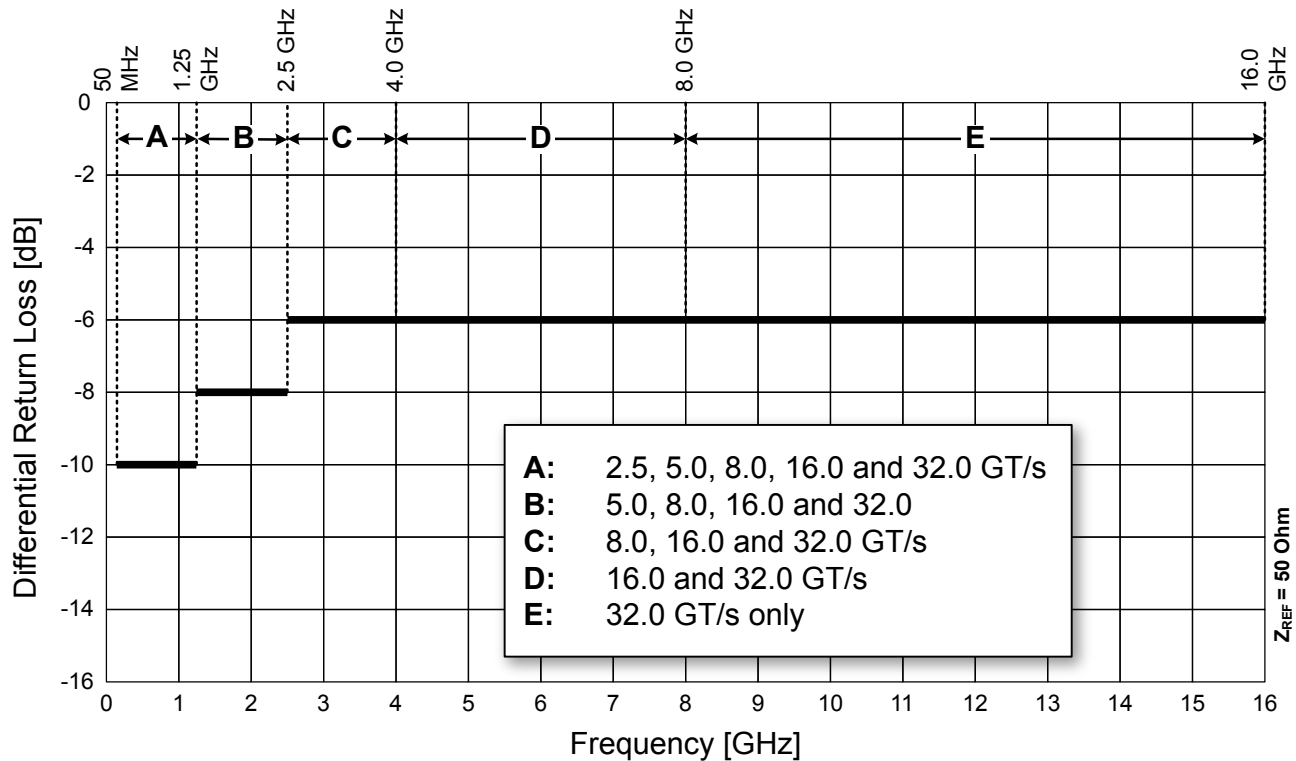


Figure 8-24 Tx, Rx Differential Return Loss Mask with 50 Ohm Reference §

The pass/fail mask for common mode return loss is shown in § Figure 8-25 for 2.5, 5.0, 8.0, 16.0, and 32.0 GT/s. Return loss measurements require that both the Tx and Rx are powered up and that their respective termination circuits are enabled.

Microprobing the package may be required to measure RL accurately.

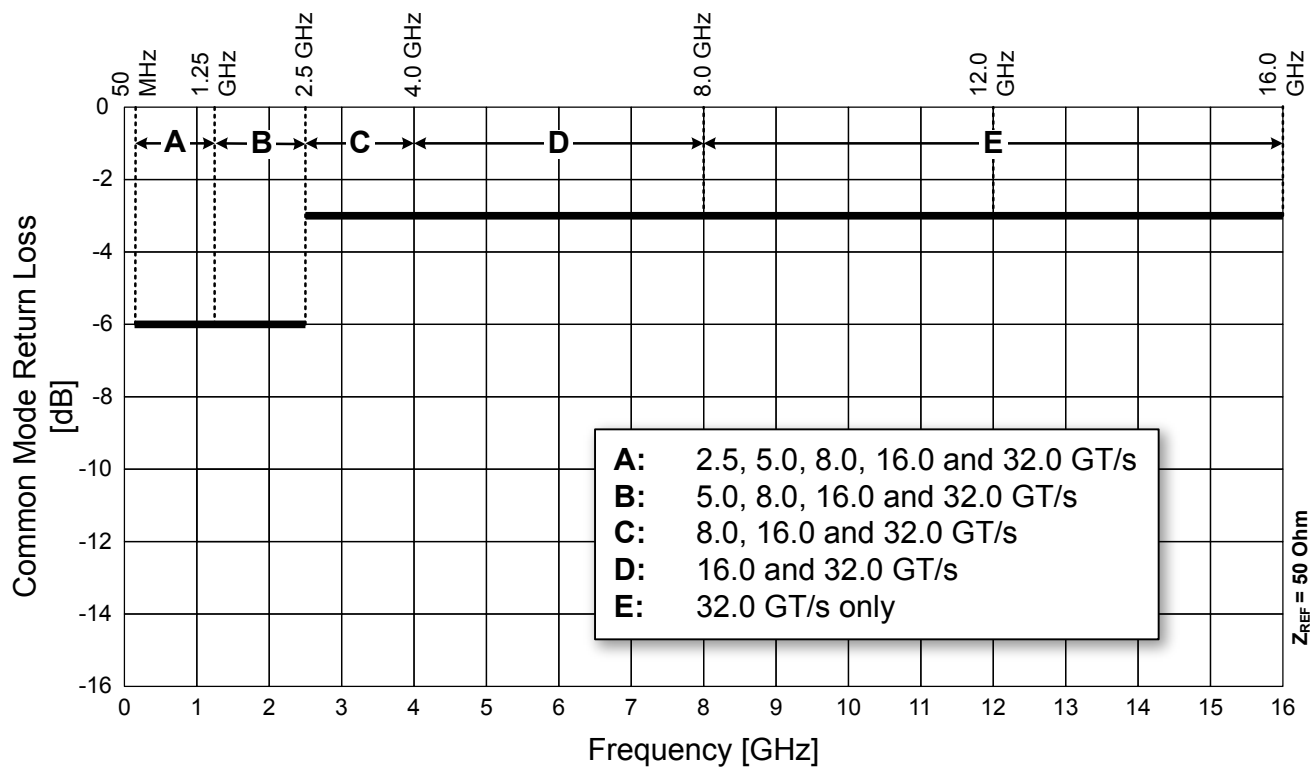


Figure 8-25 Tx, Rx Common Mode Return Loss Mask with 50 Ohm Reference §

8.3.8 Tx and Rx Return Loss for 64.0 GT/s §

§ Figure 8-26 defines the pass/fail mask for differential return loss for 64.0 GT/s with a single-ended reference impedance of 50 ohms.

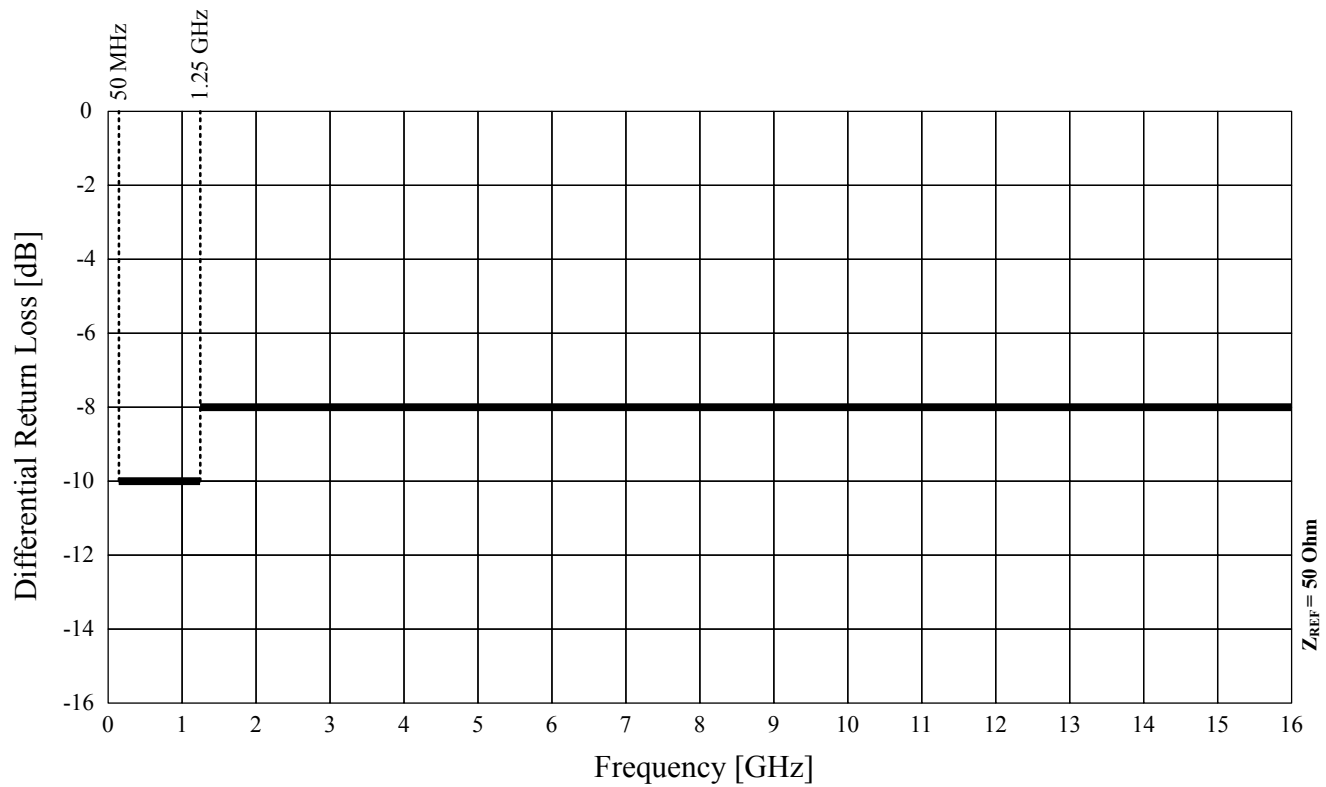


Figure 8-26 64.0 GT/s Tx, Rx Differential Return Loss Mask with 50 Ohm Reference §

§ Figure 8-27 defines the pass/fail mask for 64.0 GT/s common mode return loss with a single-ended reference impedance of 50 ohms. Return loss measurements require that both the Tx and Rx are powered up and that their respective termination circuits are enabled. Microprobing the package may be required to measure RL accurately.

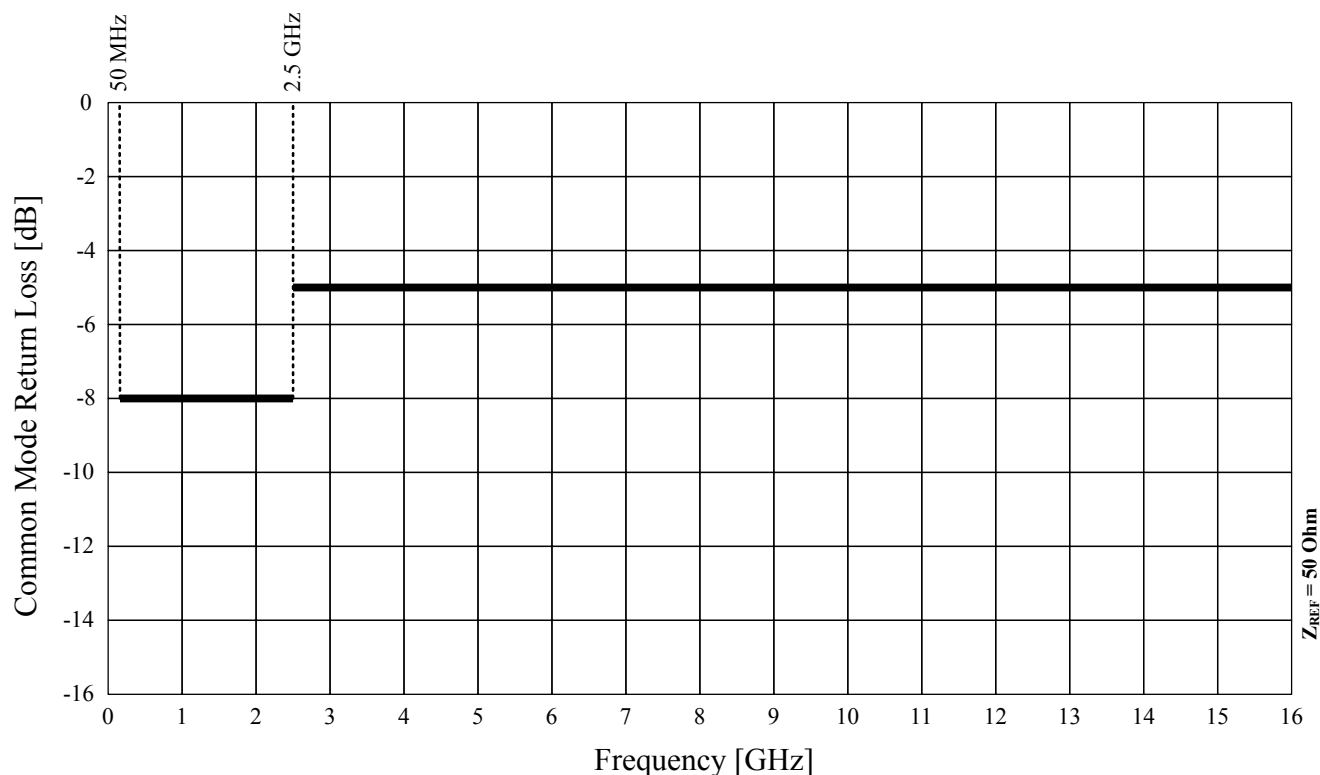


Figure 8-27 64.0 GT/s Tx, Rx Common Mode Return Loss Mask with 50 Ohm Reference §

8.3.9 Transmitter PLL Bandwidth and Peaking §

8.3.9.1 2.5 GT/s and 5.0 GT/s Tx PLL Bandwidth and Peaking §

PLL bandwidth and peaking are defined for both the Transmitter and Receiver in order to place an upper limit on the amount of Refclk jitter that is propagated to the transmitted data and to the CDR. Defining PLL BW and peaking limits also guarantees a minimum degree of Tx/Rx jitter tracking in those systems utilizing a Common Refclk Rx architecture.

The 2.5 GT/s PLL characteristics have been moved from the 3.0 CEM spec to the Electrical Base Spec. A single PLL bandwidth range from 1.5 to 22 MHz is given, which is identical to that defined in the CEM spec. No range of peaking was given in the CEM spec for the 2.5 GT/s PLL. However, for the Electrical Base Spec a peaking range of 0.01 dB to 3 dB is now defined. It is necessary to place a non-zero lower limit on the peaking, both to define a corner case as well as to maintain a common mathematical expression for the PLL transfer function in terms of ω_n and ζ .

Two sets of bandwidth and peaking are defined for 5.0 GT/s: 8-16 MHz with 3 dB of peaking and 5.0-16.0 MHz with 1 dB of peaking. This gives the designer the option of trading off between a low peaking PLL design vs. a low bandwidth design.

8.3.9.2 8.0 GT/s, 16.0 GT/s, 32.0 GT/s, and 64.0 GT/s Tx PLL Bandwidth and Peaking §

The Tx and Rx PLL bandwidth for 8.0 and 16.0 GT/s is 0.5-5 MHz with 1.0 dB of peaking or 0.5-4 MHz with 2.0 dB of peaking. The Tx and Rx PLL bandwidth for 32.0 GT/s is 0.5 to 1.8 MHz with 2.0 dB of peaking. The Tx and Rx PLL bandwidth for 64.0 GT/s is 0.5 to 1.0 MHz with 2.0 dB of peaking. The 8.0 GT/s PLL BW range is substantially lower than

the PLL bandwidths specified for 5.0 GT/s or 2.5 GT/s to reduce the amount of Refclk jitter at the sample latch of the Receiver. A non-zero value of 0.01 dB is given for the lower limit of the peaking to define all the peaking corners.

8.3.9.3 Series Capacitors §

PCI Express requires series capacitors to provide a DC block between Tx and Rx. The min/max capacitance spread has been decreased from that of the 2.5 and 5.0 GT/s standards, while the maximum value has been slightly increased. This change is necessary to minimize DC wander effects due to data scrambling implemented at 8.0 GT/s, 16.0 GT/s, 32.0 GT/s, and 64.0 GT/s. Note that 2.5 GT/s and 5.0 GT/s signaling must also propagate through these larger value capacitors, but the small increase in capacitor size has no adverse impact on either 2.5 GT/s or 5.0 GT/s signaling or low frequency in-band signaling such as Receiver detect.

8.3.10 Data Rate Independent Tx Parameters §

Table 8-7 Data Rate Independent Tx Parameters §

Symbol	Parameter Description	Value	Units	Notes
$V_{TX-DC-CM}$	Tx DC peak-peak common mode voltage	(min) 0 (max) 3.6	V	Total single-ended voltage a Tx can supply under any conditions with respect to ground. See also the $I_{TX-SHORT}$. See Note 1
$V_{TX-CM-DC-ACTIVE-IDLE-DELTA}$	Absolute delta of DC Common Mode Voltage during L0 and Electrical Idle	(min) 0 (max) 100	mV	$ V_{TX-CM-DC} \text{ [during L0]} - V_{TX-CM-Idle-DC} \text{ [during Electrical Idle]} \leq 100 \text{ mV}$ $V_{TX-CM-DC} = DC_{(avg)} \text{ of } V_{TX-D+} + V_{TX-D-} / 2$ $V_{TX-CM-Idle-DC} = DC_{(avg)} \text{ of } V_{TX-D+} + V_{TX-D-} / 2 \text{ [Electrical Idle]}$
$V_{TX-CM-DC-LINE-DELTA}$	Absolute Delta of DC Common Mode Voltage between D+ and D-	(min) 0 (max) 25	mV	$ V_{TX-CM-DC-D+} \text{ [during L0]} - V_{TX-CM-DC-D-} \text{ [during L0]} \leq 25 \text{ mV}$ $V_{TX-CM-DC-D+} = DC_{(avg)} \text{ of } V_{TX-D+} \text{ [during L0]} $ $V_{TX-CM-DC-D-} = DC_{(avg)} \text{ of } V_{TX-D-} \text{ [during L0]} $
$V_{TX-IDLE-DIFF-AC-P}$	Electrical Idle Differential Peak Output Voltage	(min) 0 (max) 20	mV	$V_{TX-IDLE-DIFF-AC-P} = V_{TX-Idle-D+} - V_{TX-Idle-D-} \leq 20 \text{ mV}$. Voltage must be band pass filtered to remove any DC component and HF noise. The bandpass is constructed from two first-order filters, the high pass and low pass 3 dB bandwidths are 10 kHz and 1.25 GHz, respectively.
$V_{TX-IDLE-DIFF-DC}$	DC Electrical Idle Differential Output Voltage	(min) 0 (max) 5	mV	$V_{TX-IDLE-DIFF-DC} = V_{TX-Idle-D+} - V_{TX-Idle-D-} \leq 5 \text{ mV}$. Voltage must be low pass filtered to remove any AC component. The low pass filter is first-order with a 3 dB bandwidth of 10 kHz.
$V_{TX-RCV-DETECT}$	The amount of voltage change allowed during Receiver Detection	(max) 600	mV	The total amount of voltage change in a positive direction that a Transmitter can apply to sense whether a low impedance Receiver is present. Note: Receivers display substantially different impedance for $V_{IN} < 0$ vs. $V_{IN} > 0$.

Symbol	Parameter Description	Value	Units	Notes
$T_{TX-IDLE-MIN}$	Minimum time spent in Electrical Idle	20 (min)	ns	The time a Tx must spend in Electrical Idle before transitioning to another state
$T_{TX-IDLE-SET-TO-IDLE}$	Maximum time to transition to a valid Electrical Idle after sending an EIOS	(max) 8	ns	After sending the required number of EIOSs, the Transmitter must meet all Electrical Idle specifications within this time. This is measured from the end of the last UI of the last EIOS to the Transmitter in Electrical Idle.
$T_{TX-IDLE-TO-DIFF-DATA}$	Maximum time to transition to valid diff signaling after leaving Electrical Idle	(max) 8	ns	Maximum time to transition to valid diff signaling after leaving Electrical Idle. This is considered a debounce time to the Tx.
$T_{CROSSLINK}$	Crosslink random timeout	(max) 1.0	ms	This random timeout helps resolve potential conflicts in the crosslink configuration.
C_{TX}	AC Coupling Capacitor	(min) 176 (max) 265	nF	All Transmitters shall be AC coupled. The AC coupling is required either within the media or within the transmitting component itself.
$Z_{TX-DIFF-DC}$	DC differential Tx impedance	(max) 120	Ω	Low impedance defined during signaling. The minimum value is bounded by $RL_{TX-DIFF}$.
$I_{TX-SHORT}$	Tx short circuit current	(max) 90	mA	Tx short circuit current. Note 1

Notes:

1. $I_{TX-SHORT}$ and $V_{TX-DC-CM}$ stipulate the maximum current/voltage levels that a Transmitter can generate and therefore define the worst case transients that a Receiver must tolerate.

8.4 Receiver Specifications §

8.4.1 Receiver Stressed Eye Specification §

All Receiver speeds are tested by means of a stressed eye applied over a calibration channel that approximates the near worst-case loss characteristics encountered in an actual channel. The recovered eye is defined at the input to the Receiver's latch. For 2.5 GT/s and 5.0 GT/s this point is equivalent to the Rx pins; for 8.0 GT/s, 16.0 GT/s, 32.0 GT/s, and 64.0 GT/s it is equivalent to the signal at the Rx die pad after behavioral Rx equalization has been applied.

8.4.1.1 Breakout and Replica Channels §

The closest practical measurement points to the Rx DUT are the coaxial connectors at the end of a breakout channel, while the Rx reference point of interest is the pin of the Rx. By constructing a replica channel that closely matches the electrical characteristics of the breakout channel it is possible to measure the signal as it would appear at the DUT's pin,

if the DUT were an ideal termination. Impedance targets for the Rx breakout and replica channels are 85 Ω differential and 42.5 Ω single-ended, and the impedance tolerance should be maintained within $\pm 5\%$ or better. Note that the impedance target for the Tx test breakout and replica channels is still 100 Ω differential and 50 Ω single-ended

In § Figure 8-28 the stressed eye is observed at TP2 with the signal sources connected to the calibration channel. A calibration channel will be required for each data rate. Once the stressed eye has been calibrated, the signal source is applied to the DUT. Note that TP1-TP2 encompasses all the components between the signal source and the equivalent of the DUT pin, thereby capturing all non-ideal characteristics in the overall insertion loss due to cabling and replica/breakout channel, excluding Rx package. The AC and DC loss from generator to TP1 are assumed to be zero or must be otherwise de-embedded. The $V_{RX-LAUNCH}$ (differential voltage swing) and Tx Equalization of the Signal Generator are calibrated at TP3 as shown in § Figure 8-28. Some Signal Generators do factory calibration with a cable and trying to de-embed to TP1 for these calibrations can cause inaccuracies. Only the loss from TP3 onward is counted in overall calibration channel loss.

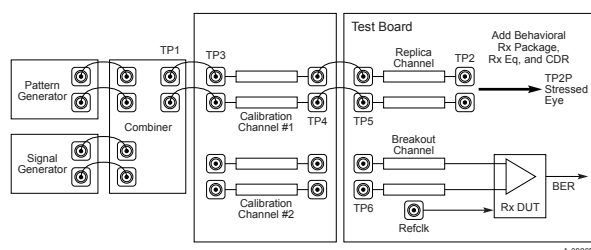


Figure 8-28 Rx Test board Topology for 16.0 and 32.0 GT/s §

8.4.1.2 Calibration Channel Insertion Loss Characteristics §

Calibration channels, each with a specified differential insertion loss at one of the PCIe data rates, provide the means of generating prescribed amounts of ISI that approximates a worst-case channel. For each data rate a single calibration channel loss mask is defined by means of two pairs of IL limits at a high and a low frequency. It is not acceptable to generate IL by means other than physical channel (PCB traces, cables, switches, small compensation delays, etc. are acceptable), such as specialized filters. The Calibration Channel needs to include all physical loss after TP3 as shown in § Figure 8-28 within the IL mask.

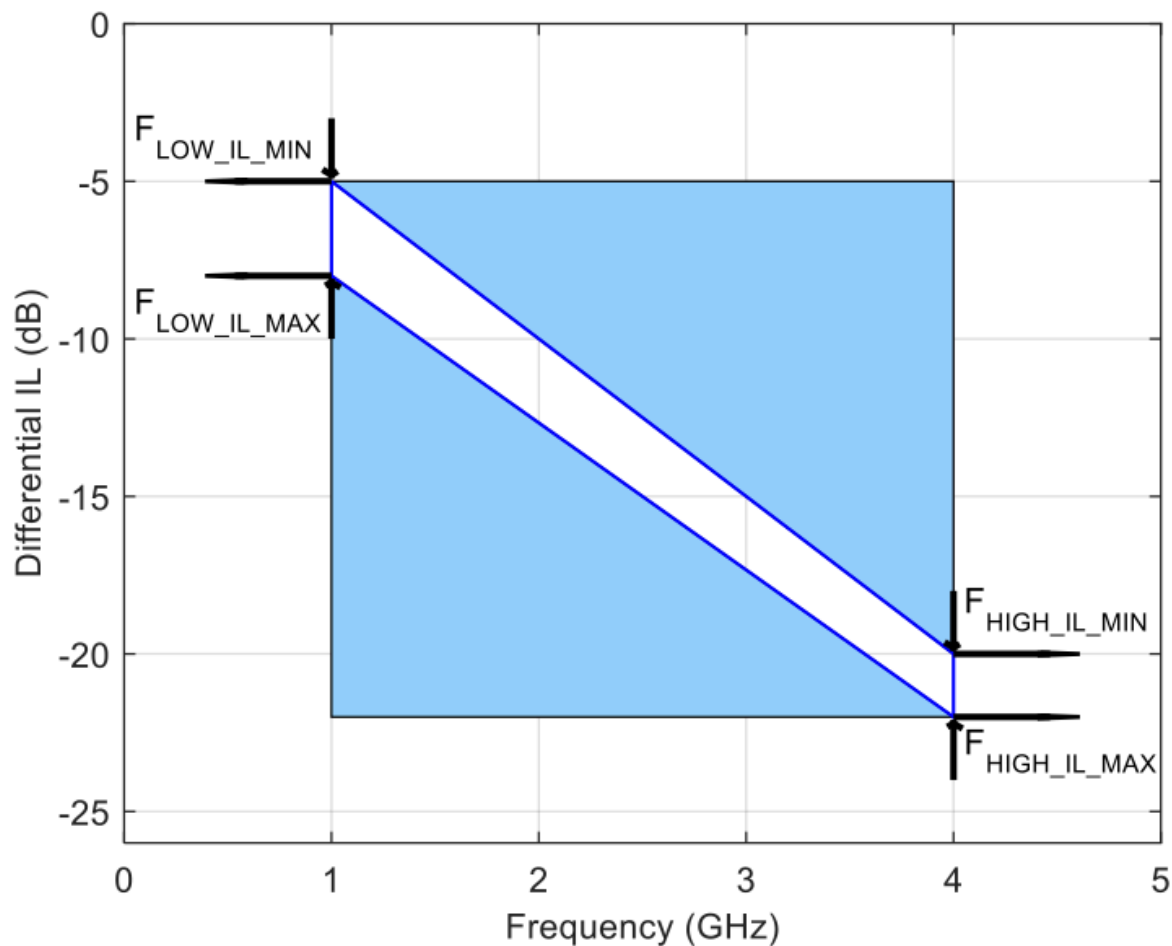


Figure 8-29 Example Calibration Channel IL Mask Excluding Rx Package for 8.0 GT/s §

The following table defines the calibration channel IL masks by means of four loss points at two frequency limits, thereby creating a quadrilateral shaped solution area. The calibration channel IL mask is provided as a guideline to minimize reflection so that the channel represents mostly ISI stress for the receiver test.

Table 8-8 Calibration Channel IL Limits §

Data Rate	$F_{LOW-IL-MIN}$	$F_{LOW-IL-MAX}$	$F_{HIGH-IL-MIN}$	$F_{HIGH-IL-MAX}$
2.5 GT/s	4.5 dB @ 1 GHz	5.0 dB @ 1 GHz	4.7 dB @ 1.25 GHz	5.2 dB @ 1.25 GHz
5.0 GT/s	4.5 dB @ 1 GHz	5.0 dB @ 1 GHz	10.0 dB @ 2.5 GHz	11.0 dB @ 2.5 GHz
8.0 GT/s	5 dB @ 1 GHz	8 dB @ 1 GHz	20 dB @ 4 GHz	22 dB @ 4 GHz
16.0 GT/s Root Port	4.2 dB @ 1 GHz	5.2 dB @ 1 GHz	22.5 dB @ 8 GHz	23.5 dB @ 8 GHz
16.0 GT/s Non-Root Port	4.2 dB @ 1 GHz	5.2 dB @ 1 GHz	24.5 dB @ 8 GHz	25.5 dB @ 8 GHz
32.0 GT/s Root Port	3.2 dB @ 1 GHz	4.2 dB @ 1 GHz	26.5 dB @ 16 GHz	27.5 dB @ 16 GHz
32.0 GT/s Non-Root Port	3.9 dB @ 1 GHz	4.9 dB @ 1 GHz	31.5 dB @ 16 GHz	32.5 dB @ 16 GHz

Data Rate	$F_{LOW-IL-MIN}$	$F_{LOW-IL-MAX}$	$F_{HIGH-IL-MIN}$	$F_{HIGH-IL-MAX}$
64.0 GT/s Root Port	3.0 dB @ 1 GHz	4.0 dB @ 1 GHz	23.5 dB @ 16 GHz	24.5 dB @ 16 GHz
64.0 GT/s Non-Root Port	3.6 dB @ 1 GHz	4.6 dB @ 1 GHz	27.5 dB @ 16 GHz	28.5 dB @ 16 GHz

Notes:

- Calibration channel plus Rx package is 28 dB nominally (informative) for 16.0 GT/s.
- Calibration channel plus Rx package is 36 dB nominally (informative) for 32.0 GT/s.
- Calibration channel plus Rx package is 32 dB nominally (informative) for 64.0 GT/s.
- Different reference packages are defined for devices containing Root Ports and all other device types at 16.0 GT/s, 32.0 GT/s, and 64.0 GT/s.
- It is recommended that some validation be done with shorter channels at 16.0 GT/s, 32.0 GT/s, and 64.0 GT/s.
- For 32.0 GT/s, a material at least as good as a Megtron-6 class material with loss of approximately 1.0 dB/inch at 16 GHz at typical room conditions must be used.
- For 64.0 GT/s, a material at least as good as a Megtron-6 class material with loss of approximately 1.0 dB/inch at 16 GHz under worst-case temperature and humidity conditions must be used to achieve system routing length of 13" for 1-connector server topologies.

The impedance targets for the Rx tolerancing interconnect environment are 100 Ω differential and 50 Ω single-ended for the 2.5, 5.0, and 8.0 GT/s channels and 85 Ω differential and 42.5 Ω single-ended for the 16.0 GT/s, 32.0 GT/s, and 64.0 GT/s channels; the impedance tolerance should be maintained within $\pm 5\%$ or better.

The calibration channel for 16.0 GT/s must meet the following return loss mask when measured from either end of the calibration channel:

- ≤ -12 dB for < 4 GHz
- ≤ -8 dB for ≥ 4 GHz and < 12 GHz
- ≤ -6 dB for ≥ 12 GHz and ≤ 16 GHz

The calibration channel for 32.0 GT/s and 64.0 GT/s must meet the following return loss mask when measured from either end of the calibration channel:

- ≤ -12 dB for < 4 GHz
- ≤ -10 dB for ≥ 4 GHz and < 16 GHz
- ≤ -6 dB for ≥ 16 GHz and ≤ 32 GHz

A calibration channel consists of a differential pair of PCB traces terminated at both ends by coaxial connectors. For 16.0 GT/s, 32.0 GT/s, and 64.0 GT/s the calibration channel includes a 4.0 (16.0 GT/s), 5.0 (32.0 GT/s), or 6.0 (64.0 GT/s) Card Electromechanical Specification compliant connector and edge finger placed at least 4 dB at Nyquist away from the coaxial connectors where the signal generator is connected. The calibration channel's electrical characteristics are defined in terms of differential insertion loss masks as shown in § Figure 8-29, where S_{DD21} is measured between TP3 (See § Figure 8-28) and TP2. Connections between TP4-TP5 represent cabling and are included in the S_{DD21} measurement. Loss before TP3 is effectively calibrated out by calibrating differential voltage swing and TX EQ at TP3 and is not included in the S_{DD21} measurement.

While the 8.0 GT/s, 16.0 GT/s, 32.0 GT/s, and 64.0 GT/s S-parameter masks do not extend below 1.0 GHz, all calibration channels must be well behaved below 1.0 GHz and must not have a DC resistance in excess of 7.5 ohms, as measured by the sum of the resistances of the D+ and D- traces. This limitation on DC resistance guarantees that the calibration channel low frequency characteristic is consistent with the extrapolations of the S_{DD21} masks to DC. The calibration loss targets for devices containing Root Ports and other devices are different because the reference package models have

different losses. For 16.0 GT/s, 32.0 GT/s, and 64.0 GT/s the insertion loss range $F_{\text{HIGH-IL-MIN}}$ to $F_{\text{HIGH-IL-MAX}}$ is the nominal loss. The calibration channel must have a series of loss options covering a range from at least 2 dB below $F_{\text{HIGH-IL-MIN}}$ to 3 dB above $F_{\text{HIGH-IL-MAX}}$ (for example for the non-root case this means a loss range from -22.5 to -28.5 dB) with loss delta between consecutive options of 0.5 dB or less.

IMPLEMENTATION NOTE:

16.0 GT/S CALIBRATION CHANNEL REFERENCE DESIGN §

This section gives an example of a 16.0 GT/s calibration channel that was built and tested to meet the requirements in this specification. A high-level block diagram of the calibration channel is shown in § Figure 8-30. Note that this example fixture covers a wider loss range than required by the specification and can cover both root and non-root cases. The test fixture includes four PCBs:

16.0 GT/s Rx Calibration Base Boards

Sixteen differential pairs (85 Ohm Nominal Impedance) routed from SMA connectors to a CEM through-hole connector. There are three different base boards to achieve the following insertion loss ranges – The insertion loss of the differential pairs for the base board is varied as follows @ 8.0 GHz in 0.5 dB steps.

Low-Loss Base Board: 4-11.5 dB

Mid-Loss Base Board: 12-19.5 dB

High-Loss Base Board: 20-27.5 dB

All traces are routed as microstrip on the bottom layer. The SMA connectors and CEM connectors are optimized with layout techniques at 8.0 GHz.

16.0 GT/s Rx Calibration Riser Board

Sixteen differential pairs (85 Ohm Nominal Impedance) routed from SMA connectors to Gold Edge Fingers. The insertion loss of the differential pairs is fixed at 4 dB nominal @ 8.0 GHz for all sixteen pairs. All traces are routed as microstrip on the bottom layer. The SMA connectors and CEM connectors are optimized with layout techniques at 8.0 GHz.

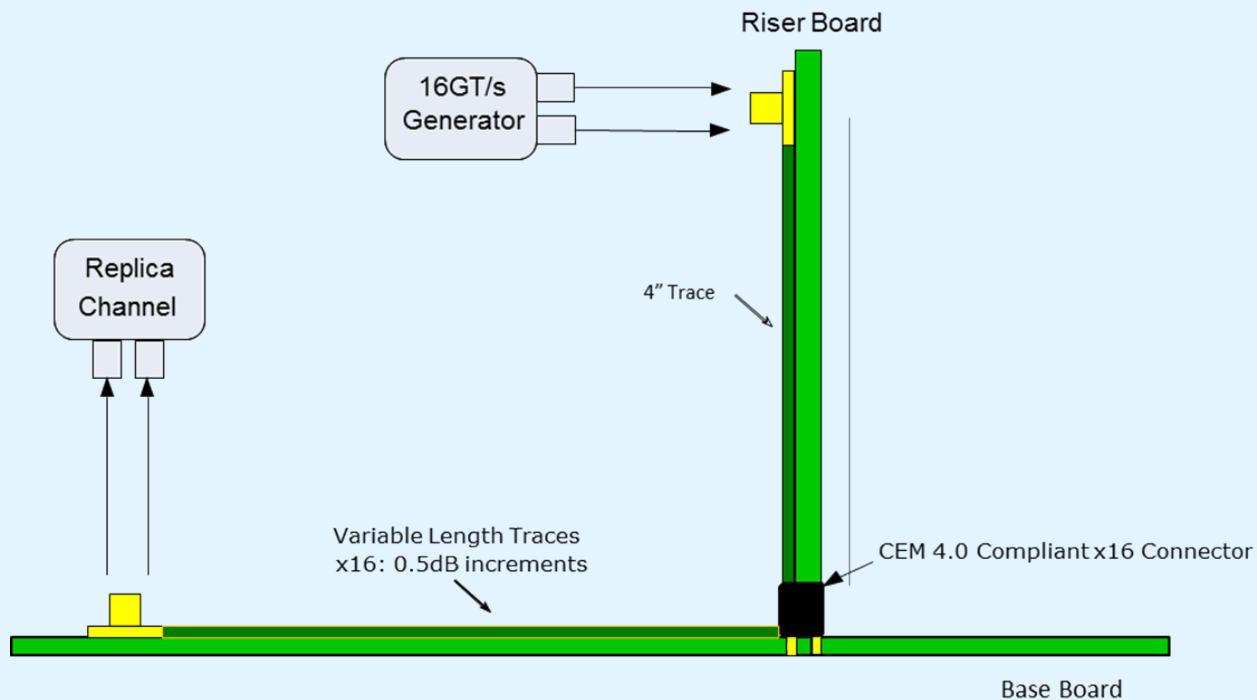


Figure 8-30 Example 16.0 GT/s Calibration Channel §

The stackup for both boards is shown in § Figure 8-31 where 65% is the estimated copper fill percentage. § Figure 8-31 includes stackups for both nominal 85 Ω and 100 Ω stackups - the 85 Ω stackup is used for the calibration channel example.

Material	Layer	Dielectric	Copper Fill / DK	Starting Copper oz	Finished Copper oz	Copper Thickness	Single Ended Impedance	Finished Trace Single Ended	Calculated Impedance	SF Ref Layers	Differential Impedance	Finished Trace/Gap Differential	Calculated Impedance	Diff Ref Layers
1-1080,1-2113	1	.0062	Top	.5	1.5	0.00210	42.50 $\pm$ 1-5% 50 Ω $\pm$ 1-5%	.0135 .010	42.77 50.11	2	85 Ω $\pm$ 1-5% 100 Ω $\pm$ 1-5%	.0075 / .005 .0052 / .006	84.95 100.38	2
	2	.0062	Plane	65% DK=4.0	1	0.00130				2				
FILLER	3	.043	Plane	65% DK=4.0	1	0.00130								
1-1080,1-2113	4	.0062	Bot	.5	1.5	0.00210	42.50 $\pm$ 1-5% 50 Ω $\pm$ 1-5%	.0135 .010	42.77 50.11	3 3	85 Ω $\pm$ 1-5% 100 Ω $\pm$ 1-5%	.0075 / .005 .0052 / .006	84.95 100.38	3 3
		.0622	Final Thickness (After Plating)											

Figure 8-31 Stackup for Example 16.0 GT/s Calibration Channel §

The pad stack for the CEM connector drill holes is shown in § Figure 8-32 and the pad stack for the SMA drill holes is shown in § Figure 8-33.

Padstack: C40P28P3M3-A59 Type: through Inner pads: Optional								
Layer	Pad Type	Geometry	Width	Height	Offset X	Offset Y	Flash Name	Shape Name
TOP	ANTI	CIRCLE	59.00	59.00	0.00	0.00		
TOP	THERMAL	RECTANGLE	80.00	80.00	0.00	0.00	TH80X60X15_4X45	
TOP	REGULAR	CIRCLE	40.00	40.00	0.00	0.00		
GND-2	ANTI	CIRCLE	59.00	59.00	0.00	0.00		
GND-2	THERMAL	RECTANGLE	80.00	80.00	0.00	0.00	TH80X60X15_4X45	
GND-2	REGULAR	CIRCLE	40.00	40.00	0.00	0.00		
GND-3	ANTI	CIRCLE	59.00	59.00	0.00	0.00		
GND-3	THERMAL	RECTANGLE	80.00	80.00	0.00	0.00	TH80X60X15_4X45	
GND-3	REGULAR	CIRCLE	40.00	40.00	0.00	0.00		
BOTTOM	ANTI	CIRCLE	59.00	59.00	0.00	0.00		
BOTTOM	THERMAL	RECTANGLE	80.00	80.00	0.00	0.00	TH80X60X15_4X45	
BOTTOM	REGULAR	CIRCLE	40.00	40.00	0.00	0.00		
internal_pad_def	ANTI	CIRCLE	59.00	59.00	0.00	0.00		
internal_pad_def	THERMAL	RECTANGLE	80.00	80.00	0.00	0.00	TH80X60X15_4X45	
internal_pad_def	REGULAR	CIRCLE	40.00	40.00	0.00	0.00		
SOLDERMASK_TOP	REGULAR	CIRCLE	40.00	40.00	0.00	0.00		
SOLDERMASK_BOTTOM	REGULAR	CIRCLE	40.00	40.00	0.00	0.00		
PASTEMASK_TOP	REGULAR	NULL	0.00	0.00	0.00	0.00		
PASTEMASK_BOTTOM	REGULAR	NULL	0.00	0.00	0.00	0.00		
FILMMASKTOP	REGULAR	NULL	0.00	0.00	0.00	0.00		
FILMMASKBOTTOM	REGULAR	NULL	0.00	0.00	0.00	0.00		

Drill Data for C40P28P3M3-A59

Hole Type	Drill Dia	Plating	Figure	Characters	Width	Height	Offset X	Offset Y	Pos Tolerance	Neg Tolerance	Non-Standard
CIRCLE DRILL	28.00	PLATED	DIAMOND		50.00	50.00	0.00	0.00	3.00	3.00	

Figure 8-32 CEM Connector Drill Hole Pad Stack [§]

Padstack: C60_BOT75P20_SSM40P3M3 Type: through Inner pads: Optional

Layer	Pad Type	Geometry	Width	Height	Offset X	Offset Y	Flash Name	Shape Name
TOP	ANTI	CIRCLE	45.00	45.00	0.00	0.00		
TOP	THERMAL	RECTANGLE	5.00	5.00	0.00	0.00	5MIL_PAD	
TOP	REGULAR	CIRCLE	60.00	60.00	0.00	0.00		
GND-2	ANTI	CIRCLE	45.00	45.00	0.00	0.00		
GND-2	THERMAL	RECTANGLE	5.00	5.00	0.00	0.00	5MIL_PAD	
GND-2	REGULAR	CIRCLE	60.00	60.00	0.00	0.00		
GND-3	ANTI	CIRCLE	45.00	45.00	0.00	0.00		
GND-3	THERMAL	RECTANGLE	5.00	5.00	0.00	0.00	5MIL_PAD	
GND-3	REGULAR	CIRCLE	60.00	60.00	0.00	0.00		
BOTTOM	ANTI	CIRCLE	45.00	45.00	0.00	0.00		
BOTTOM	THERMAL	RECTANGLE	5.00	5.00	0.00	0.00	5MIL_PAD	
BOTTOM	REGULAR	CIRCLE	75.00	75.00	0.00	0.00		
internal_pad_def	ANTI	CIRCLE	45.00	45.00	0.00	0.00		
internal_pad_def	THERMAL	RECTANGLE	5.00	5.00	0.00	0.00	5MIL_PAD	
internal_pad_def	REGULAR	CIRCLE	60.00	60.00	0.00	0.00		
SOLDERMASK_TOP	REGULAR	CIRCLE	60.00	60.00	0.00	0.00		
SOLDERMASK_BOTTOM	REGULAR	CIRCLE	40.00	40.00	0.00	0.00		
PASTEMASK_TOP	REGULAR	NULL	0.00	0.00	0.00	0.00		
PASTEMASK_BOTTOM	REGULAR	NULL	0.00	0.00	0.00	0.00		
FILMMASKTOP	REGULAR	NULL	0.00	0.00	0.00	0.00		
FILMMASKBOTTOM	REGULAR	NULL	0.00	0.00	0.00	0.00		

Drill Data for C60_BOT75P20_SSM40P3M3

Hole Type	Drill Dia	Plating	Figure	Characters	Width	Height	Offset X	Offset Y	Pos Tolerance	Neg Tolerance	Non-Standard
CIRCLE DRILL	20.00	PLATED	CIRCLE	E	60.00	60.00	0.00	0.00	3.00	3.00	

Figure 8-33 Pad Stack for SMA Drill Holes §

IMPLEMENTATION NOTE:

32.0 GT/S CALIBRATION CHANNEL REFERENCE DESIGN §

This section gives an example of a 32.0 GT/s calibration channel that was built and tested to meet the requirements in this specification. A high-level block diagram of the calibration channel is shown in § Figure 8-34. Note this example fixture covers a wider loss range than required by the specification and can cover both root and non-root cases. The test fixture includes two PCBs:

32.0 GT/s Rx Calibration Base Boards

Sixteen differential pairs (85 Ohm Nominal Impedance) routed on a Megtron-6 PCB from MMPX connectors to a CEM surface mount connector. There are three different base boards to achieve the following insertion loss ranges - The insertion loss of the differential pairs for the base board is varied as follows @ 16.0 GHz in 0.5 dB steps.

Low-Loss Base Board: 4.0-11.5 dB

Mid-Loss Base Board: 12.0-19.5 dB

High-Loss Base Board: 20.0-27.5 dB

All traces are routed as microstrip on the top layer. The MMPX connectors and CEM connectors are optimized with layout techniques at 16.0 GHz

For information on MMPX connectors refer to Huber+Suhner microminiature connectors.

32.0 GT/s Rx Calibration Riser Board

Sixteen differential pairs (85 Ohm Nominal Impedance) routed on a Megtron-6 PCB from MMPX connectors to Gold Edge Fingers. The insertion loss of the differential pairs is fixed at 8 dB nominal @ 16.0 GHz for all sixteen pairs. All traces are routed as microstrip on the top layer. The MMPX connectors and CEM connectors are optimized with layout techniques at 16.0 GHz.

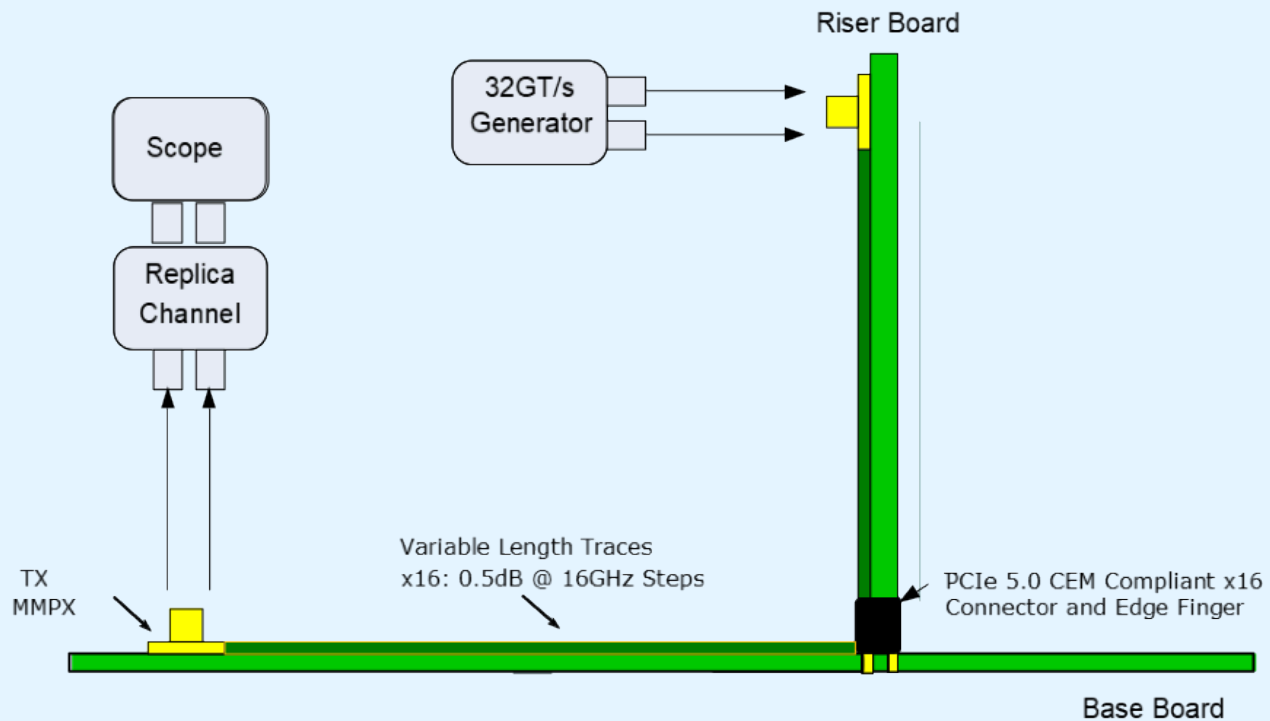


Figure 8-34 Example 32.0 GT/s Calibration Channel §

The stack-up for both 85 Ohm boards is shown in § Figure 8-35 where 65% is the estimated copper fill percentage.

Material	Layer	Starting Cu oz	Finished Cu oz	Dielectric	Single Ended Lyr	Single Ended Impedance	Designed Trace Width	Finished Trace Width	Calculated Impedance	Ref Lyr A	Ref Lyr B	Differential Impedance	Designed Trace/Space	Finished Trace/Space	Calculated Impedance	Ref Lyr A	Ref Lyr B
1-3313(54%), 1-1078(75%)	1	Signal	.5	1.5		42.5 Ω	.00945	.0155	42.18 Ω			85 Ω	.00675/.00525	.0115/.0095	85.94 Ω	2	
						50 Ω	.008	.0115	49.7 Ω			100 Ω	.005/.007	.006/.006	100.45 Ω	2	
Filler Material	2	Plane	1	1													
1-3313(54%), 1-1078(75%)	3	Plane	1	1													
	4	Signal	.5	1.5		50 Ω	.008	.0115	49.7 Ω			100 Ω	.005/.007	.006/.006	100.45 Ω	2	
						42.5 Ω	.00945	.0155	42.18 Ω			85 Ω	.00675/.00525	.0115/.0095	85.94 Ω	2	

Final Thickness (After Plating): 0.062

Figure 8-35 Stack-up for Example 32.0 GT/s Calibration Channel §

64.0 GT/s Calibration channel reference design may be added in Rev 0.9.

8.4.1.3 Post Processing Procedures §

The Receiver test requires that the stressed eye characteristics be measured at TP2 (which is accessible) and then post-processed to yield a signal as it would appear at test point two post-processed (TP2P) (which is not accessible) for 8.0, 16.0, 32.0, and 64.0 GT/s. TP2P defines a reference point that comprehends the effects of the behavioral Rx package plus Rx equalization and represents the only location where a meaningful EH and EW limits can be defined.

8.4.1.4 Behavioral Rx Package Models §

Behavioral Rx package models are included as part of the post processing to allow the calibrated eye to comprehend package insertion loss. A separate pair of package models is defined for 8.0, 16.0, 32.0, and 64.0 GT/s eye calibration. At 8.0 GT/s, separate package models are defined for TX and RX ports to reflect the smaller CPAD capacitance typical in most receiver implementations. At 16.0, 32.0, and 64.0 GT/s, separate package models are defined for devices containing Root Ports and all other devices. This is necessary to allow a reasonable channel solution space and assumes that devices containing Root Complexes are usually large and socketed, while all other devices tend to be unsocketed and smaller. The 16.0 GT/s Root and Non-Root behavioral Rx package models have been constructed to represent respective package loss characteristics for high loss, but not worst-case loss, packages. The 32.0 and 64.0 GT/s Root and Non-Root behavioral Rx package models have been constructed to represent package loss characteristics for worst case packages.

The 8.0, 32.0, and 64.0 GT/s stressed eye test for all devices and the 16.0 GT/s stressed eye test for Non-Root Package devices that support captive channels are required to use the appropriate behavioral package (see § Section 8.3.3.11). For all other device types, if the actual Rx package performance is worse than that of the behavioral package, then the actual package models are permitted to be used. If the actual package models are used, the calibration channel must be adjusted such that the total channel loss including the embedded actual package remains at 28 dB nominal. Note that form factor overall requirements still need to be met. The Rx package performance is assessed using the methodology defined in § Section 8.5.1.2 .

Details of the behavioral Rx packages are provided in § Section 8.5.1.1 of the Channel Tolerancing section. S-parameter models for the behavioral Rx package models are available as design collateral. The reference impedance at the pad side of the packages model is assumed to be $2 \times 50 \Omega$.

8.4.1.5 Behavioral CDR Model §

Post processing shall include a behavioral CDR model with a data rate dependent transfer function. A first order CDR transfer function is utilized for Receivers operating with a CC Refclk architecture except for 32.0 GT/s and 64.0 GT/s. For Receivers operating in IR Refclk mode an alternate CDR transfer function is required. For a given data rate the behavioral CDR used for Rx testing is the same as the corresponding CDR used for Tx testing. For details on behavioral CDR functions refer to § Section 8.3.5.5 .

8.4.1.6 No Behavioral Rx Equalization for 2.5 and 5.0 GT/s §

The combination of worst-case channel, behavioral Rx package, and Tx jitter at 2.5 and 5.0 GT/s will yield open eyes, when the appropriate Tx presets are set. Therefore, there is no need to define a behavioral Rx equalization or to adjust the Tx equalization setting. Actual implementations of 2.5 and 5.0 GT/s receivers may, of course, include equalization.

8.4.1.7 Behavioral Rx Equalization for 8.0, 16.0, 32.0, and 64.0 GT/s §

As measured at TP2, stressed eyes at 8.0, 16.0, 32.0, and 64.0 GT/s will usually be closed, making direct measurement of the stressed eye jitter parameters unfeasible. This problem is overcome by employing a behavioral Receiver equalizer that implements both CTLE and a 1-tap DFE (8.0 GT/s) or a 2-tap DFE (16.0 GT/s) or a 3-tap DFE (32.0 GT/s) or a 16-tap DFE (64.0 GT/s).

Rx equalization algorithms of CTLE and DFE are only intended to be a means for obtaining an open eye in the presence of calibration channel ISI plus the other signal impairment terms and for channel compliance. The behavioral Rx equalization algorithms are not intended to serve as a guideline for implementing actual Receiver equalization. For example, additional DFE taps can have significant benefit in actual implementations where the CTLE may differ from the

behavioral equalizer and/or CTLE selection may not always be optimal. Channel loss characteristics can vary significantly with temperature and humidity and a real Receiver must be able to continue to function at the target BER through such variations.

8.4.1.8 Behavioral CTLE (8.0 and 16.0 GT/s) §

8.0 and 16.0 GT/s behavioral Rx equalization defines a 1st order CTLE with fixed LF and HF poles, and an adjustable DC gain (A_{DC}) specified according to the family of curves shown in § Figure 8-37. For the 8.0 GT/s rates A_{DC} is adjustable over a minimum range of -6 to -12 dB in steps of 1.0 dB.

$$H(s) = \omega_{P2} \times \frac{s + \omega_{P1} \times A_{DC}}{(s + \omega_{P1}) \times (s + \omega_{P2})}$$

$$\omega_{P1} = \text{pole 1} = 2\pi \times 2 \text{ GHz}$$

$$\omega_{P2} = \text{pole 2} = 2\pi \times 8 \text{ GHz}$$

$$A_{DC} = \text{dc gain}$$

Figure 8-36 Transfer Function for 8.0 GT/s Behavioral CTLE §

The following diagram illustrates the gain vs. frequency behavior of the CTLE as A_{DC} is varied over its minimum to maximum range in 1.0 dB steps.

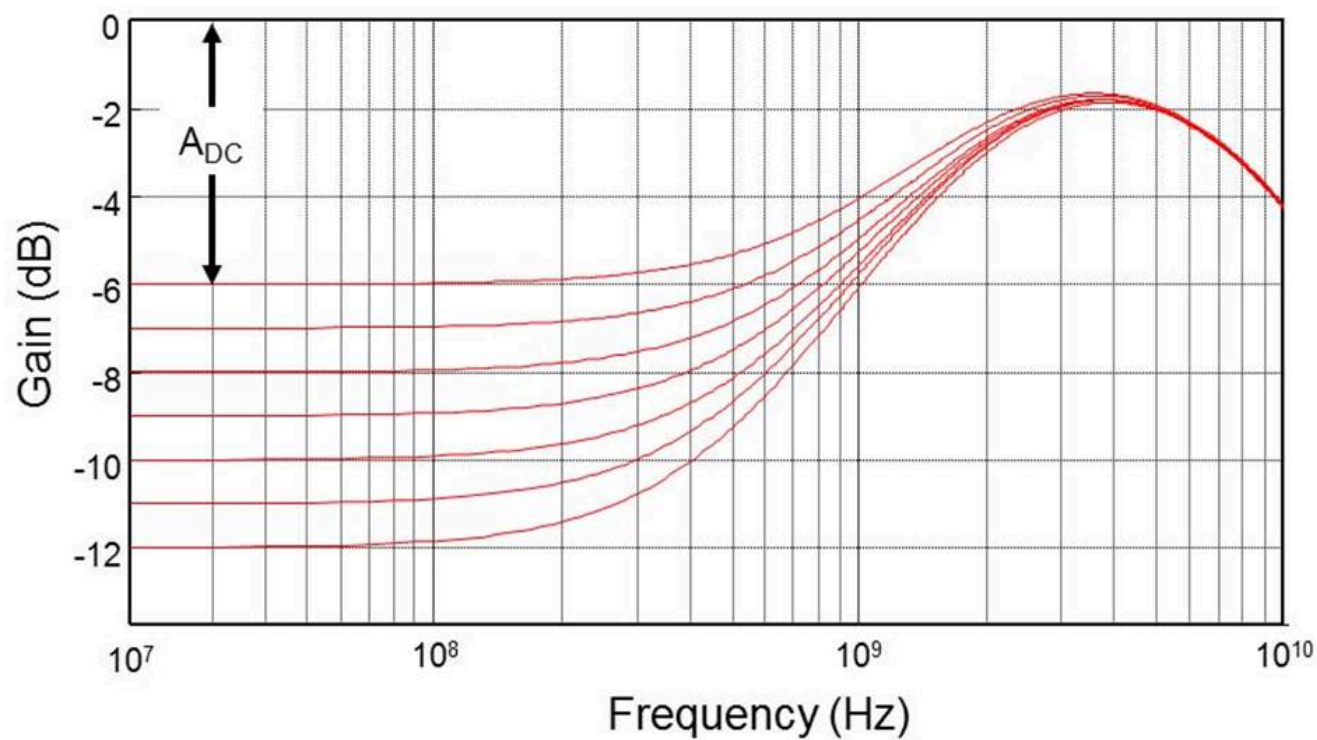


Figure 8-37 Loss Curves for 8.0 GT/s Behavioral CTLE §

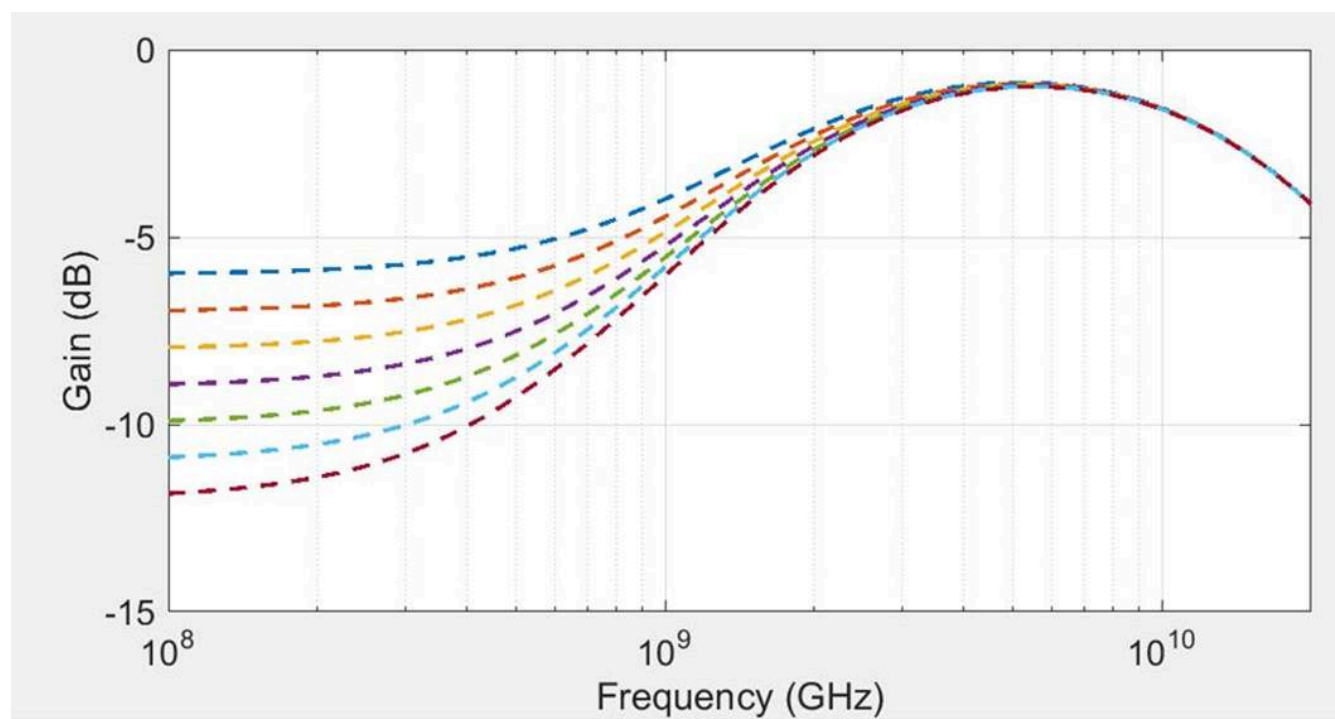


Figure 8-38 Loss Curves for 16.0 GT/s Behavioral CTLE §

A Receiver operating at 16.0 GT/s utilizes a similar set of CTLE curves with different pole locations. The difference is that $\omega_{p1} = \text{pole 1} = 2\pi \times 2 \text{ GHz}$ and $\omega_{p2} = \text{pole 2} = 2\pi \times 16.0 \text{ GHz}$. The range for A_{DC} remains the same as that for 8.0 GT/s.

8.4.1.9 Behavioral CTLE (32.0 and 64.0 GT/s) §

32.0 GT/s behavioral Rx equalization defines a 2nd order CTLE with fixed poles, and an adjustable DC gain (A_{DC}) specified according to the family of curves shown in § Figure 8-39. The A_{DC} is adjustable over a range of -5 to -15 dB in steps of 1.0 dB.

$$H(s) = \frac{\omega_{p1} \times \omega_{p3} \times \omega_{p4}}{\omega_{z1}} \times \frac{(s + \omega_{z1})(s + \omega_{p2} \times A_{DC})}{(s + \omega_{p1})(s + \omega_{p2})(s + \omega_{p3})(s + \omega_{p4})}$$

$$\omega_x = 2\pi \times F_x$$

$$F_{p1} = 1.65 \times F_{z1}$$

$$F_{p2} = 9.5 \text{ GHz}$$

$$F_{p3} = 28 \text{ GHz}$$

$$F_{p4} = 28 \text{ GHz}$$

$$F_{z1} = 450 \text{ MHz}$$

$$F_{z2} = \text{mag}(\text{DC gain}) \times F_{p2}$$

Equation 8-26 Behavioral CTLE at 32.0 GT/s §

§ Figure 8-39 illustrates the gain vs. frequency behavior of the CTLE as A_{DC} is varied over its minimum to maximum range in 1.0 dB steps. Note that the maximum frequency of the CTLE curves is 200 GHz which ensures accuracy in the time-domain post-processing simulation tools.

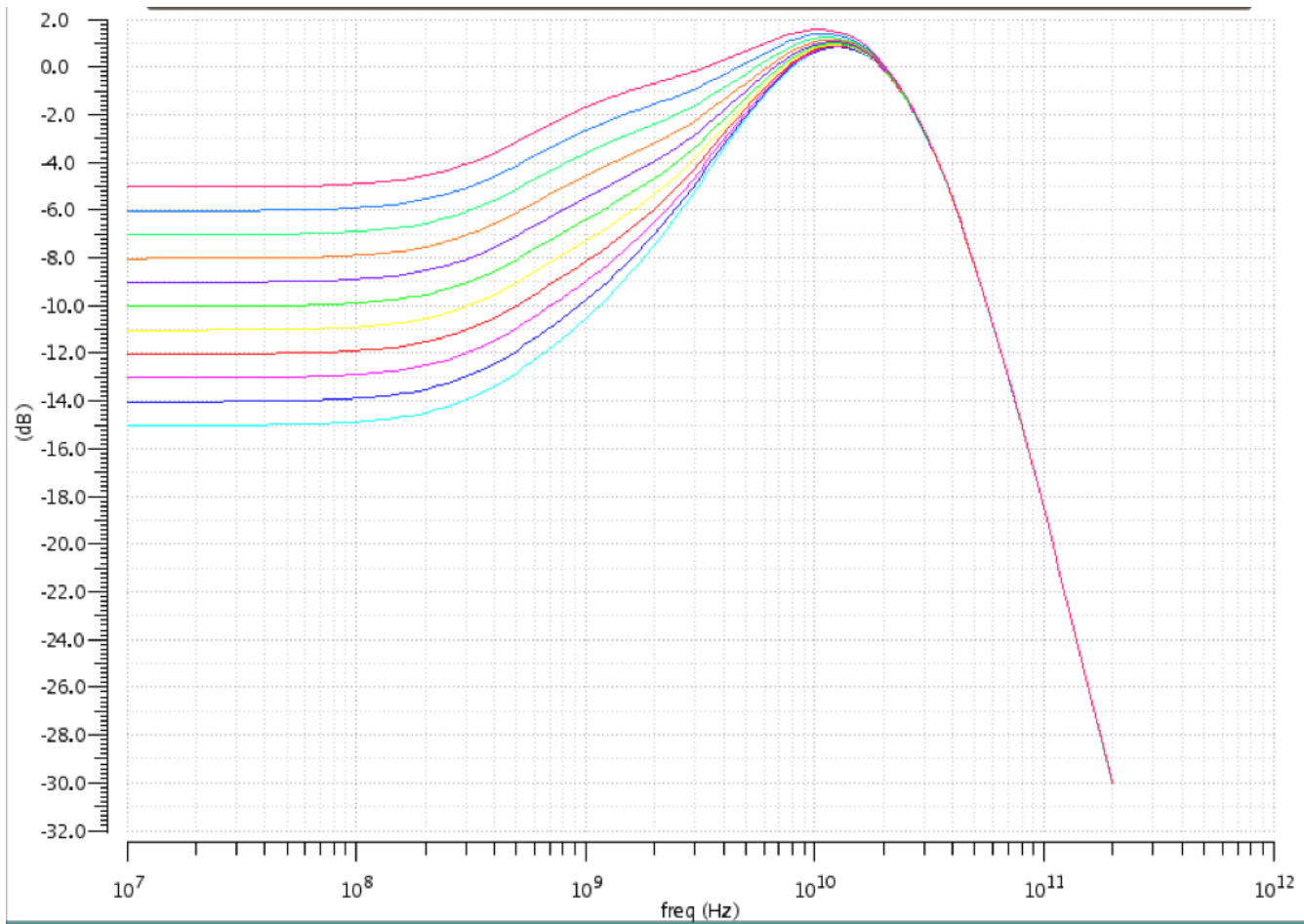


Figure 8-39 Loss Curves for 32.0 GT/s Behavioral CTLE §

64.0 GT/s behavioral Rx equalization defines a CTLE with six poles and three zeros, and an adjustable DC gain (ADC) specified according to the family of curves shown in § Figure 8-40. The ADC is adjustable over a range of -5 to -15 dB in steps of 1.0 dB. The maximum frequency of the CTLE curves is 250 GHz.

$$H(s) = \frac{\omega_{p1} \times \omega_{p3} \times \omega_{p4} \times \omega_{p5} \times \omega_{p6}}{\omega_{z1} \times \omega_{z3}} \times \frac{(s + \omega_{z1}) \times (s + \omega_{p2} \times A_{DC}) \times (s + \omega_{z3})}{(s + \omega_{p1}) \times (s + \omega_{p2}) \times (s + \omega_{p3}) \times (s + \omega_{p4}) \times (s + \omega_{p5}) \times (s + \omega_{p6})}$$

$$\omega_x = 2\pi \times F_x$$

$$F_{P1} = 1.30 \times F_{Z1}$$

$$F_{P2} = 7.7 \text{ GHz}$$

$$F_{P3} = 22.0 \text{ GHz}$$

$$F_{P4} = 28.0 \text{ GHz}$$

$$F_{P5} = 32.0 \text{ GHz}$$

$$F_{P6} = 32.0 \text{ GHz}$$

$$F_{Z1} = 250 \text{ MHz}$$

$$F_{Z2} = \text{mag}(\text{DC gain}) \times F_{P2}$$

$$F_{Z3} = 7.7 \text{ GHz}$$

Equation 8-27 Behavioral CTLE at 64.0 GT/s §

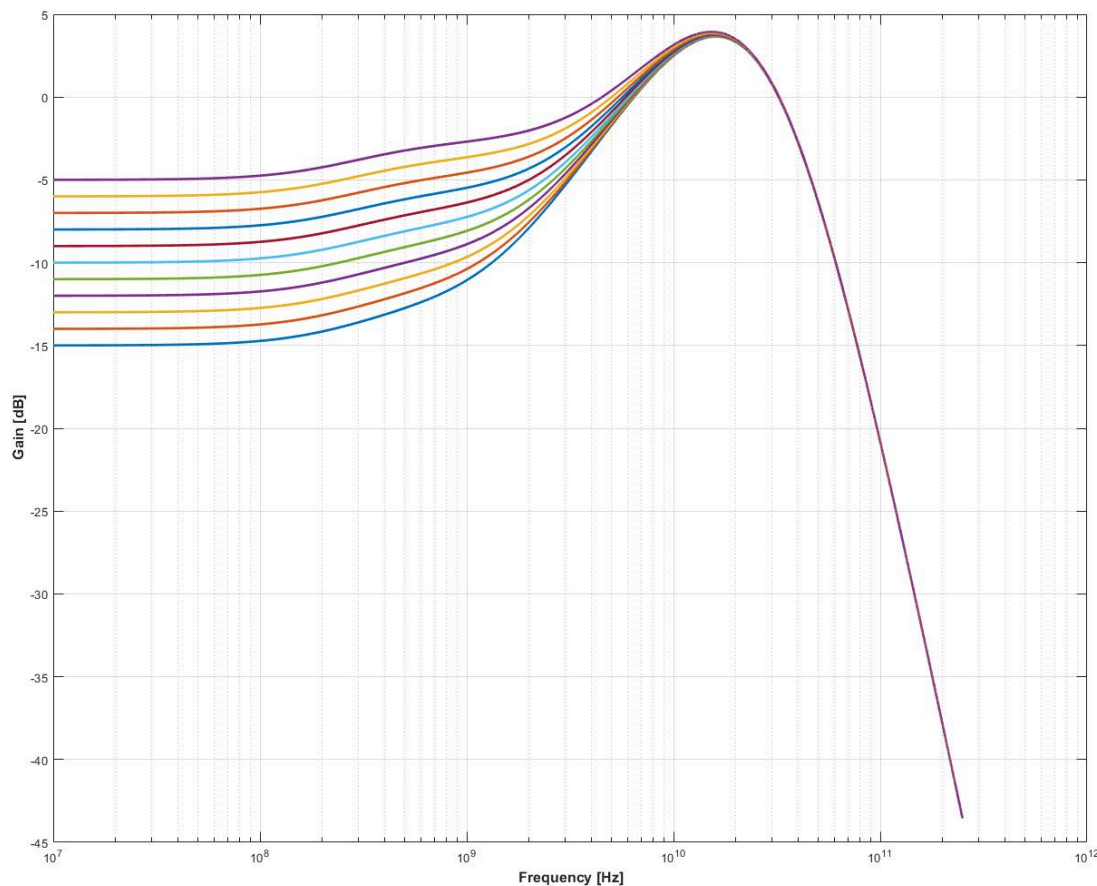


Figure 8-40 Loss Curves for 64.0 GT/s Behavioral CTLE §

8.4.1.10 Behavioral DFE (8.0, 16.0, 32.0, and 64.0 GT/s Only) §

At 8.0 GT/s the combination of a 1st order CTLE and a one-tap DFE algorithm is required for calibrating the stressed eye when employing the max length calibration channel. The DFE may be represented by the following equation and flow diagram. For 8.0 GT/s and 16.0 GT/s the limits on d_1 are ± 30 mV. For 32.0 GT/s the limit on d_1 is defined as a ratio of the tap magnitude (h_1) to the cursor strength (h_0). The h_1/h_0 ratio must be less than or equal to 0.8. Note that the h_1/h_0 limit of 0.8 is only to bound the behavior of the reference receiver and does not indicate that real implementations will be safe from error bursts due to large h_1/h_0 causing undetected data errors if the h_1/h_0 ratio is below 0.8. Implementers must do their own analysis for their specific designs on the largest safe h_1/h_0 ratio. Note that an optional precoding mechanism is provided at 32.0 GT/s that receivers can optionally enable to reduce the risk of DFE related error bursts in high transition data patterns causing silent data corruption. For 16.0 GT/s the limits on d_2 are ± 20 mV. For 32.0 GT/s the limits on d_2 and d_3 are ± 20 mV.

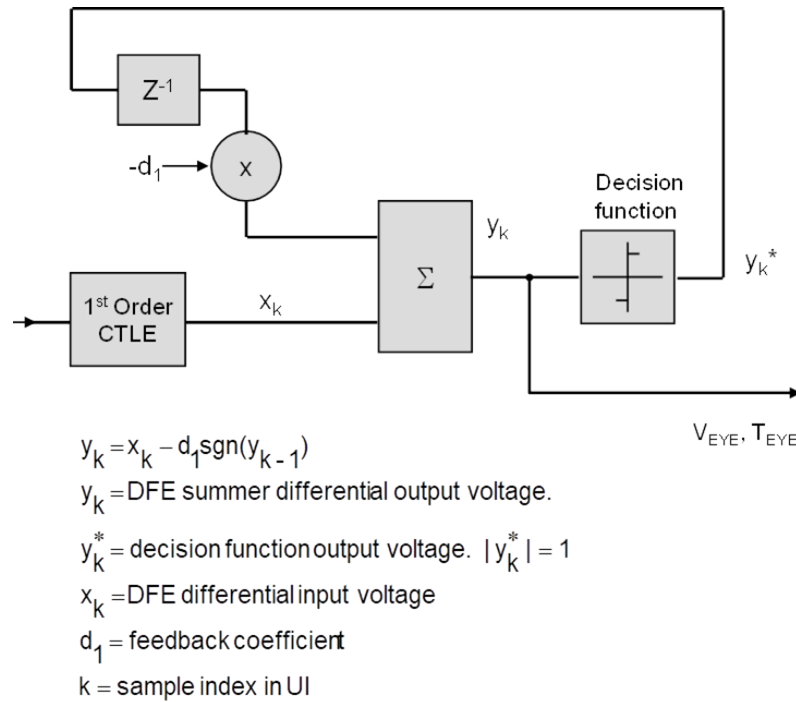


Figure 8-41 Variables Definition and Diagram for 1-tap DFE §

16.0 GT/s Receiver tolerancing utilizes a CTLE and a 2-tap behavioral DFE as illustrated below. Other than the inclusion of the second tap, it is identical to the 1-tap DFE shown above. The 32.0 GT/s Receiver tolerancing utilizes a CTLE and a 3-tap behavior DFE.

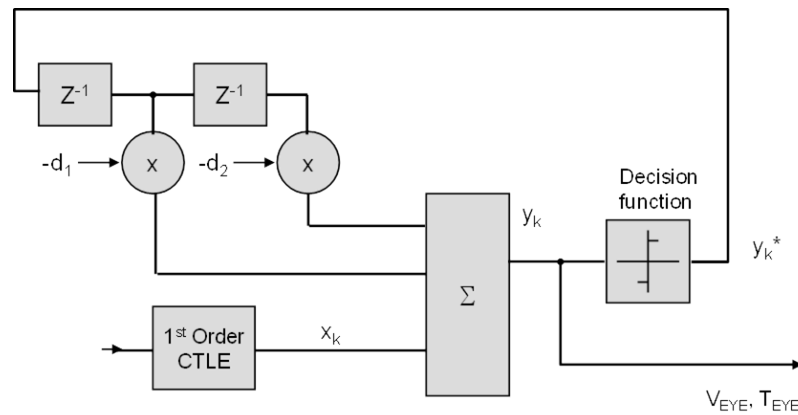


Figure 8-42 Diagram for 2-tap DFE §

For 64.0 GT/s, the feedback signal y_k^* can take values of -1, -1/3, +1/3 and +1. In this case, the limit on d_1 is defined as a ratio of the tap magnitude ($|d_1|$) to the cursor magnitude at the input of the DFE (h_0). To constrain DFE burst errors, the $|d_1/h_0|$ ratio must be less than 0.55 and the weighted-sum of the tap magnitudes defined as $(|d_1| + |d_2| + 0.85*|d_3| + 0.60*|d_4| + 0.25*|d_5| + 0.10*|d_6| + 0.05*|d_7| + 0.05*|d_8| + 0.05*|d_9| + 0.05*|d_{10}| + 0.05*|d_{11}| + 0.05*|d_{12}| + 0.05*|d_{13}| + 0.05*|d_{14}| + 0.05*|d_{15}| + 0.05*|d_{16}|)/h_0$ must be less than 0.85. The limits on DFE tap magnitudes are only to bound the behavior of the reference receiver and do not indicate that real implementations will be safe from error bursts by

ensuring the DFE tap magnitude limits specified for the reference receiver. Implementers must do their own analysis for their specific designs on the largest safe DFE tap magnitudes.

8.4.2 Stressed Eye Test §

Rx testing at 16.0, 32.0, and 64.0 GT/s requires only a single stressed voltage/stressed jitter test per data rate.

When testing a Receiver, it is required to have other PCI Express Lanes on the DUT sending or receiving data. Similarly, if the device supports other I/O, it should also be sending or receiving on these interfaces. The goal is to have the Rx test environment replicate the noise environment found in a real system as closely as possible.

8.4.2.1 Procedure for Calibrating a Stressed EH/EW Eye §

The goal of calibrating a stressed voltage/jitter eye is to present the Receiver under test with simultaneously worst case margins whose distortion characteristics are like an eye produced by a real channel. Much of the distortion consists of the ISI produced by the calibration channel. Incremental changes of R_j and differential voltage are allowed to adjust the EW and EH, respectively at 8.0 GT/s. Incremental changes of S_j , $V_{RX-DIFF-INT}$, and differential voltage swing from nominal values may be used to adjust the EW and EH at 16.0, 32.0, and 64.0 GT/s. Refer to § Table 8-11 for initial values of various stress parameters for all data rates.

The reference point where EH/EW is defined corresponds to input to the Receiver latch at 8.0, 16.0, 32.0, and 64.0 GT/s. Since this point is not physically accessible it is necessary to construct its equivalent by means of a post-processing procedure. A two million unit interval data record of compliance pattern or a step that has been averaged 1024 times at TP2 is first post processed to mathematically include the additional signal distortion caused by the behavioral Receiver package. If a compliance pattern waveform is used then all stresses except $V_{RX-CM-INT}$ are turned on if a step is used then all stresses are turned off. Then the resulting signal is recovered by means of Rx equalization, and a behavioral CDR function, resulting in an equivalent eye. The requirements for the waveform post processing tool used for the EH/EW calibration are described further in § Section 8.4.2.1.1. If the receiver calibration eye margin simulation tool uses a step response, the R_j , S_j , and $V_{RX-DIFF-INT}$ are input parameters to the simulation tool. If the receiver calibration eye margin simulation tool uses compliance pattern waveform, the R_j , S_j , and $V_{RX-DIFF-INT}$ are inputs to the waveform. In either case, the stress parameters must be calibrated.

As the calibration procedure of the signal generator output contains steps where the generator is connected directly to measurement instrumentation, the transition time of the output waveform can be very fast. Therefore, it is important that the bandwidth of instrumentation used to calibrate the generator be matched appropriately to the edge rate of the generator output. This specification requires the use of a generator for 16.0 GT/s testing whose outputs have a rise time of 14 ps-19 ps (20% / 80%) which also requires a minimum oscilloscope bandwidth of 25 GHz. This oscilloscope bandwidth is also the minimum required bandwidth for transmitter measurements at 16.0 GT/s. For 32.0 and 64.0 GT/s testing the specification requires the use of a generator whose outputs have a rise time of 7.5 - 15.0 ps (20%/80% measured with P4) which requires a minimum oscilloscope bandwidth of 50 GHz. This oscilloscope bandwidth is also the minimum required for transmitter measurements at 32.0 and 64.0 GT/s. A minimum oscilloscope sampling rate that captures at least 4 samples per unit interval is required for all data rates.

For the eye calibration process, the Tx equalization is fixed to the preset that gives the optimal eye area with the post processing tool being used for calibration. Once the testing procedure is under way the Tx preset may be adjusted to yield the best eye margins with the DUT. During EH/EW calibration S_j is initially set to 100 MHz with a nominal amplitude of 0.1 UI for 8.0, 16.0, and 32.0 GT/s, with a nominal amplitude of 0.05 UI for 64.0 GT/s. The 100 MHz S_j amplitude will be swept during the stressed eye calibration. Tx EQ and differential voltage swing calibration are done at TP3 as shown in § Figure 8-28. The coaxial cable from TP1 to TP3 is considered part of the generator and not included in the channel insertion loss measurements for 32.0 GT/s and 64.0 GT/s stressed eye calibration, but the coaxial cable is included in the total channel insertion loss measurement at 16.0 GT/s to keep consistency with the 16.0 GT/s measurement methodology adopted in PCIe 4.0. For calibration at 16.0 GT/s the following process is used to calibrate the eye:

1. Calibrate the stress values to the nominal values in § Table 8-11.
2. Select an initial test channel length that gives a loss at TP2P at 8 GHz of 27 dB±0.5 dB.
3. Measure the eye diagram for each TX EQ preset using the nominal TX Eq for the preset +/- 0.1 dB and select the TX EQ preset that gives the largest eye area.

For all EH, EW and eye area measurements performed in receiver calibration the A_{DC} in the reference receiver CTLE is varied over its minimum to maximum range in 0.25 dB steps. This is done to improve repeatability and accuracy in automated Rx calibration software and is only done for stressed eye calibration (not for channel compliance, etc.)

4. Increase the calibration channel loss to the next available length/loss and measure the new eye diagram at the selected preset. Continue to increase the length/loss until either the height or width have fallen below the targets in § Table 8-11 then the previous calibration channel length/loss is selected. If neither the height or width have fallen below the targets and the TP3 (§ Figure 8-28) to TP2P loss at 8 GHz has reached 30.0 dB then advance to the next step.
5. For the selected calibration channel length/loss, measure the eye diagram for each TX EQ preset and select the preset that gives the largest eye area. Note that this may be a different preset than step 3 due to the length/loss change.
6. Adjust S_j , $V_{RX-DIFF-INT}$, and Voltage Swing to make final adjustments to the eye by sweeping them through the following ranges:
 - a. S_j 5 to 10 ps PP.
 - b. $V_{RX-DIFF-INT}$ 10 to 25 mV at TP2.
 - c. Differential Voltage Swing 720 to 800 mV PP at TP1.
7. If the final S_j value is less than 0.1 UI then the R_j level is reduced so the eye width meets the target eye width with 0.1 UI of 100 MHz S_j .
8. If there are multiple combinations of S_j , $V_{RX-DIFF-INT}$, and Voltage Swing that give valid solutions first pick the combination that is closest to the target eye width (18.75 ps). If there are multiple S_j , $V_{RX-DIFF-INT}$, and Voltage Swing combinations that are equally close to the target eye width then pick the one with S_j closest to nominal. The selected values must give a mean eye height and width (over at least 5 measurements exact number of measurements needed for stable values will depend on lab set-up and tools) within the following ranges at BER E-12:
 - a. Eye height 15 mV +/- 1.5 mV
 - b. Eye width 18.75 ps +/- 0.5 ps

For calibration at 32.0 GT/s the following process is used to calibrate the eye:

1. Measure the eye diagram for each TX EQ preset using the nominal TX Eq for the preset +/- 0.3 dB and select the TX EQ preset that gives the largest eye area.

For all EH, EW and eye area measurements performed in receiver calibration the A_{DC} in the reference receiver CTLE is varied over its minimum to maximum range in 1.0 dB steps. Measure the eye diagram varying the following parameters with the maximum indicated step size for each variable parameter:

- a. Calibrate the stress values to the initial values in § Table 8-11. The R_j stress is kept constant.
- b. Channel loss from TP3 to TP2P varied from 34.0 to 37.0 dB with a maximum 0.5 dB step size.

Note that if your actual channel loss comes out to slightly above 37 or slightly below 34 that these cases are excluded (loss must be between 34.0 and 37.0 dB)

- c. S_j varied from 1 to 5 ps PP with a maximum 0.25 ps step size with S_j measured at TP3

- d. $V_{RX-DIFF-INT}$ 5 to 30 mV at TP2 with a maximum 2.5 mV step size
2. If the final S_j value is less than 0.1 UI then the R_j level is reduced so the eye width meets the target eye width with 0.1 UI of 100 MHz S_j .
3. If there are multiple combinations of S_j , $V_{RX-DIFF-INT}$, and channel loss that give valid solutions first pick the combination with the highest channel loss. If there are multiple combinations that work with the highest channel loss, then select the combination that is closest to the target eye height (15.0 mV). The selected values must give a mean eye height and width (over at least 5 measurements exact number of measurements needed for stable values will depend on lab set-up and tools) within the following ranges at BER E-12. A specific method for finding the combination of stress values that meet these criteria is outside the scope of this specification. If and only if no stress combinations can be found, then the voltage swing may also be varied from 720 to 800 mV:

Note that because the first tiebreaker is highest loss - most approaches will start with the highest allowed channel loss.

- a. Eye height 15 mV +/- 1.5 mV
- b. Eye width 9.375 ps +/- 0.5 ps

For calibration at 64.0 GT/s the following process is used to calibrate the eye:

1. Measure the PAM4 eye diagrams for each TX EQ preset of Q0-Q9 using the nominal TX Eq for the preset +/- 0.3 dB and select the TX EQ preset that gives the largest top eye area.

For all EH, EW and eye area measurements performed in receiver calibration the ADC in the reference receiver CTLE is varied over its minimum to maximum range in 1.0 dB steps. Measure the eye diagrams varying the following parameters with the maximum indicated step size for each variable parameter:

- a. Calibrate the stress values to the initial values in § Table 8-11. The R_j stress is kept constant.
- b. Channel loss from TP3 to TP2P varied from 30.0 to 33.0 dB with a maximum 0.5 dB step size.

Note that if your actual channel loss comes out to slightly above 33 or slightly below 30 that these cases are excluded (loss must be between 30.0 and 33.0 dB)

- c. S_j varied from 1 to 3 ps PP with a maximum 0.25 ps step size with S_j measured at TP3
- d. $V_{RX-DIFF-INT}$ 5 to 25 mV at TP2 with a maximum 2.0 mV step size
2. If the final S_j value is less than 0.05 UI then the R_j level is reduced so the eye width meets the target eye width with 0.05 UI of 100 MHz S_j .
3. If there are multiple combinations of S_j , $V_{RX-DIFF-INT}$, and channel loss that give valid solutions first pick the combination with the highest channel loss. If there are multiple combinations that work with the highest channel loss, then select the combination that is closest to the target top eye height (6.0 mV). The selected values must give a mean eye height and width (over at least 5 measurements exact number of measurements needed for stable values will depend on lab set-up and tools) within the following ranges at BER of 10⁻⁶. A specific method for finding the combination of stress values that meet these criteria is outside the scope of this specification. If and only if no stress combinations can be found, then the voltage swing between PAM4 voltage level 0 and level 3 may also be varied from 720 to 800 mV:

Note that because the first tiebreaker is highest loss - most approaches will start with the highest allowed channel loss.

- a. Top eye height 6 mV +/- 0.5 mV
- b. Top eye width 3.125 ps +/- 0.3 ps

For calibration at 64.0 GT/s, if the Differential Noise and/or Common Mode Noise are generated using an external source, any broadband differential noise introduced by the noise source should be characterized and added to the step response based calibration procedure through inclusion of SNDR in the modeling of the BERT transmitter.

Based upon the data rate at which the Rx is being tested, Rj or Sj and differential interference sources are adjusted to fall within the V_{RX-ST} and T_{RX-ST} limits. The EH and EW ranges are designed to account for post processing or measurement errors. § Figure 8-43 shows the process for calibrating the stressed jitter eye at 8.0 GT/s.

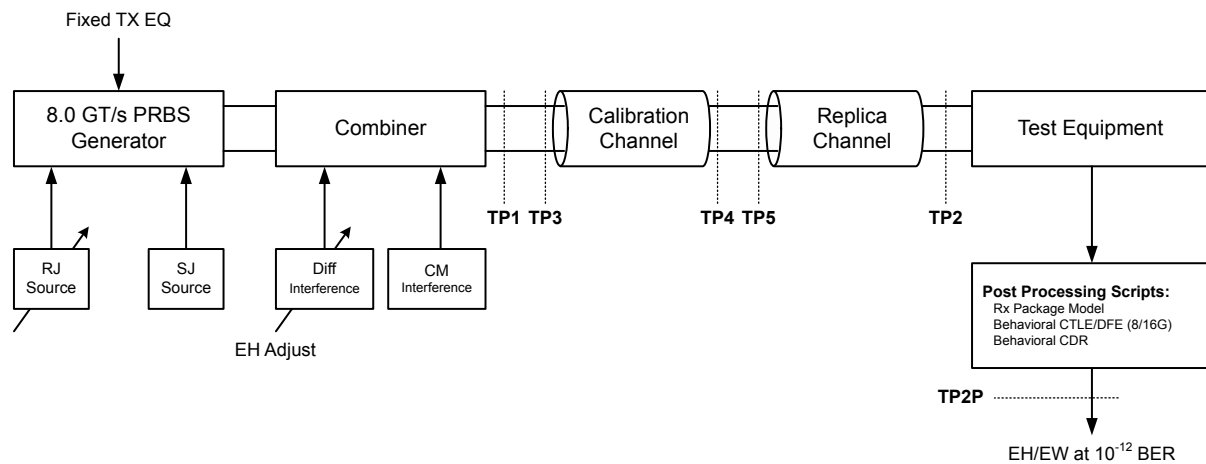


Figure 8-43 Layout for Calibrating the Stressed Jitter Eye at 8.0 GT/s §

§ Figure 8-44 shows the process for calibrating the stressed jitter eye common for 16.0, 32.0, and 64.0 GT/s data rates. The PRBS Generator provides the required data rate and the eye is calibrated to the data rate specific target EH and EW at the corresponding BER.

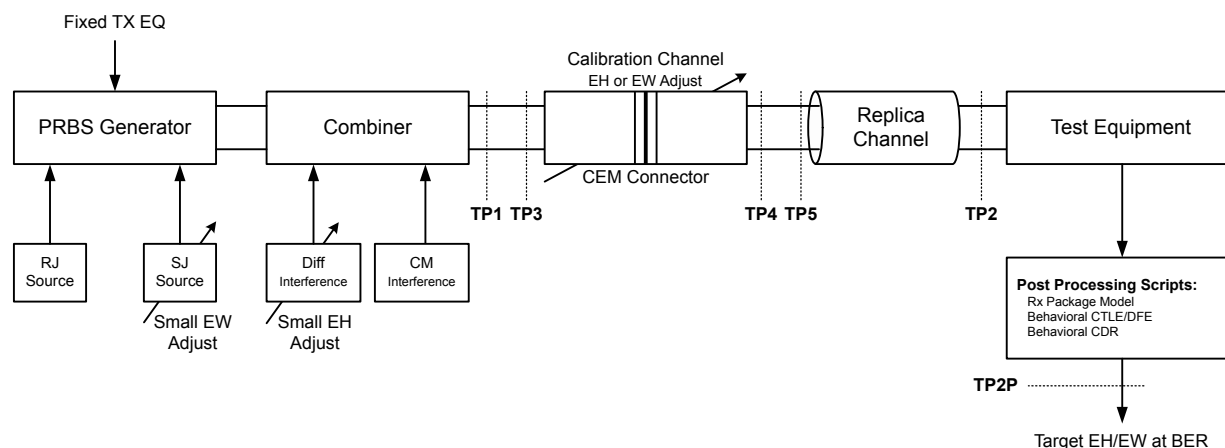


Figure 8-44 Layout for Calibrating the Stressed Jitter Eye at 16.0, 32.0, and 64.0 GT/s §

Table 8-11 Stressed Jitter Eye Parameters §

Symbol	Parameter	2.5 GT/s	5.0 GT/s	8.0 GT/s	16.0 GT/s	32.0 GT/s	64.0 GT/s	Units	Details
$V_{RX-LAUNCH}$	Generator launch voltage	800 to 1200	800 to 1200	800 to 1200	800	800	800	mV PP	Note 1
T_{RX-UI}	Unit Interval	400	200	125	62.5	31.25	31.25	ps	
T_{RX-ST}	Target Eye width (Top Eye for PAM4)	0.4	0.32	0.30	0.30	0.30	0.10	UI	Note 3, 4, 8, 10
V_{RX-ST}	Target Eye height (Top Eye for PAM4)	175	100	25	15	15	6	mV PP	Note 2, 4, 8, 9
$T_{RX-ST-SJ}$	Swept Sj	N/A	75 ps (max) See Note 11	See § Section 8.4.2.2.1				ps	Note 5
$T_{RX-ST-RJ}$	Random Jitter	N/A	3.4	(max) 3.0	1.0	0.5	0.25	ps RMS	Note 6, 7
$V_{RX-DIFF-INT}$	Differential noise	N/A	N/A	14	14	20	15	mV PP	Note 7, 12 Adjust to set EH. Frequency = 2.1 GHz
$V_{RX-CM-INT}$	Common mode noise	150	150	150	150	150	75	mV PP	Note 8
$V_{SSC-RES}$	SSC Residual	N/A	75	N/A	500	N/A	N/A	ps	Note 11, 13

Notes:

Symbol	Parameter	2.5 GT/s	5.0 GT/s	8.0 GT/s	16.0 GT/s	32.0 GT/s	64.0 GT/s	Units	Details
$V_{RX-LAUNCH}$ may be adjusted to meet V_{RX-ST} as long as the outside eye voltage at TP2 does not exceed 1300 mVPP for calibration at 2.5, 5.0, and 8.0 GT/s. $V_{RX-LAUNCH}$ is adjusted from 800 to 720 mV for 16.0, 32.0, and 64.0 GT/s calibration with 800 mV as the nominal value. - Voltages shown for 2.5 GT/s and 5.0 GT/s are at the Rx pins. - Eye widths shown for 2.5 GT/s and 5.0 GT/s are at the Rx pins. V_{RX-ST} and T_{RX-ST} are referenced to TP2P for 8.0, 16.0, 32.0, and 64.0 GT/s and TP2 for 2.5 and 5.0 GT/s. For 8.0, 16.0, 32.0, and 64.0 GT/s behavioral equalization are applied to the data at TP2. At 64.0 GT/s, V_{RX-ST} and T_{RX-ST} correspond to top eye height and eye width at 10^{-6} BER. $T_{RX-ST-SJ}$ may be measured at either TP1 or TP2. Only 8.0, 16.0, 32.0 and 64.0 GT/s receivers are tested with Sj mask. $T_{RX-ST-RJ}$ may be adjusted to meet the target value for T_{RX-ST} at 8.0 GT/s. Rj is measured at TP1 to prevent data-channel interaction from adversely affecting the accuracy of the Rj calibration. Rj is applied over the following range: The low frequency limit may be between 1.5 and 10 MHz, and the upper limit is 1.0 GHz. - Both $T_{RX-ST-RJ}$ and $V_{RX-DIFF-INT}$ are limited to prevent the stressed eye from containing excessive amounts of jitter or noise distortion that are unrepresentative of a real channel. Too many of these distortion components produces a signal that cannot be equalized by an actual Receiver. - Defined as a single tone at 120 MHz. Measurement made at TP2 without post-processing. Common mode is turned off during T_{RX-ST} and V_{RX-ST} calibration and then turned on for the stressed eye jitter test. - For 2.5 GT/s and 5.0 GT/s Rx calibration variable channel loss is used to achieve the target eye height. - For 2.5 GT/s Rx calibration 100 MHz Sj is used to achieve the target eye width. - For 33 kHz SSC residual for common clock architecture testing only when testing at 5 GT/s. - Frequency for $V_{RX-DIFF-INT}$ is chosen to be slightly above the first pole of the reference CTLE. - Applied for CC testing only as a triangular phase modulation with a frequency between 30 kHz to 33 kHz when testing at 16.0 GT/s with no 32.0 GT/s and no 64.0 GT/s support and when the Sj mask of § Figure 8-52 and a first order CDR transfer function are used.									

8.4.2.1.1 Post Processing Tool Requirements §

A waveform post processing tool or a channel compliance methodology tool may be used for Rx stressed eye calibration at 16.0, 32.0, or 64.0 GT/s. If a waveform post processing tool is used to calibrate the EH/EW for the RX stressed eye testing, the tool must be consistent with the channel compliance methodology tool based on a consistency test defined as follows:

- The test channel is the long Rx calibration channel with the Root reference package applied in post-processing to give a total loss of 28.0 dB at 8 GHz (16.0 GT/s), 36.0 dB at 16 GHz (32.0 GT/s), or 32.0 dB at 16 GHz (64.0 GT/s). This means that the physical channel loss is 23.0 dB (16.0 GT/s), 27.0 dB (32.0 GT/s), or 24.0 dB (64.0 GT/s).
- All measurements are done at TP2.
- A step pattern with 512 ones and zeros is captured through the test channel by averaging 1024 times on a real time oscilloscope. The step is saved with an x-axis resolution of 1 ps or less to be used as the transmit waveform for the channel compliance methodology. The step pattern is captured for each preset using the nominal Tx EQ for each preset.
- The channel compliance methodology is run with no Tx EQ applied in simulation using the nominal stress values for Rx stressed eye calibration for each of the captured steps. The Tx EQ preset that produced the largest eye area is selected for exact eye height and width calibration.

- The channel compliance methodology is used with the selected Tx EQ preset to produce an EH/EW of
 - 15 mV and 0.3 UI @ 10^{-12} BER (16.0 GT/s),
 - 15 mV and 0.3 UI @ 10^{-12} BER (32.0 GT/s), or
 - top eye height of 6.0 mV and top eye width of 0.1 UI @ 10^{-6} BER (64.0 GT/s)

by adjusting the S_j , $V_{RX-DIFF-INT}$ and voltage swing at the Transmitter output.

- A pattern generator is calibrated to have the same jitter stress levels and Tx Swing as those used in the channel compliance simulations that produced the target eye height and eye width.
- 2 million unit interval waveforms with compliance pattern are captured at each Tx EQ at the end of the channel. The Tx EQ is calibrated to the nominal values for each preset at the pattern generator output before doing the captures.
- For the preset that gives the largest eye area with the waveform post processing tool the EH and EW (@ 10^{-12} BER for 8.0, 16.0, and 32.0 GT/s and @ 10^{-6} BER for 64.0 GT/s) & must match the target EH and EW from the channel compliance methodology within +/- 15%.

8.4.2.2 Procedure for Testing Rx DUT §

8.4.2.2.1 S_j Mask §

Once a calibrated EH and EW have been obtained, the cables are moved to connect the Rx DUT to the far end of calibration channel. For the testing of the Rx DUT, the BERT Tx must transmit Modified Compliance Pattern. The Tx equalization may then be optimized with the assumption that the DUT Rx will also optimize its equalization. S_j is set to an initial value of 0.1 UI at 100 MHz and the Receiver CDR must achieve lock. For 64.0 GT/s, S_j is set to an initial value of 0.05 UI at 100 MHz and the Receiver CDR must achieve lock. At 8.0, 16.0, 32.0, and 64.0 GT/s the 100 MHz S_j initial tone is removed and then the appropriate swept S_j profile is tested. At 16.0, 32.0, and 64.0 GT/s an additional S_j tone at 210 MHz is present for all testing. At 16.0 and 32.0 GT/s, the amplitude of this additional tone is equal to the amplitude of the 100 MHz S_j required to achieve the target eye width minus 0.1 UI. If the calibration S_j level was less than 0.1 UI then no additional tone at 210 MHz is used. At 64.0 GT/s, the amplitude of this additional tone is equal to the amplitude of the 100 MHz S_j required to achieve the target eye width minus 0.05 UI. If the calibration S_j level was less than 0.05 UI then no additional tone at 210 MHz is used. Different S_j profiles are used, depending on data rate and whether the Rx under test operates in the CC mode or the IR Refclk mode. See § Table 8-11.

The S_j (pp value) profiles shown in § Figure 8-45, § Figure 8-46, § Figure 8-47, § Figure 8-48, § Figure 8-49, § Figure 8-50, and § Figure 8-51 consist of swept tones at 33 kHz and from 400 kHz to 100 MHz, representing the swept S_j frequency range. For 8.0 and 16.0 GT/s SRIS mode with 5000 ppm SSC, the magnitude of the 33 kHz spur is 25 ns pp. For 32.0 and 64.0 GT/s SRIS mode with 3000 ppm SSC, the magnitude of the 33 kHz spur is 15 ns pp. A Receiver must meet the target BER (10^{-6} for 64.0 GT/s and 10^{-12} for all other data rates) over the entire swept S_j frequency range. It is not necessary to test a Receiver over the entire S_j frequency range, but a sufficient number of frequency points should be tested to guarantee that the Rx does not fail the target BER at some resonance frequency. The 33 kHz frequency point must be tested and treated as another frequency point. The 33 kHz frequency will not be kept on during other frequency measurements. Note that no SSC is applied at the source. Swept S_j is required only for testing Receivers at 8.0, 16.0, 32.0, and 64.0 GT/s. Receiver operation at 2.5 GT/s and 5.0 GT/s is tested using a single 33 kHz S_j tone.

Receivers operating at 8.0 GT/s in the IR mode use the S_j mask profile shown in § Figure 8-45. The magnitude of the 33 kHz spur is 25 ns pp, or 200 Upp. The equation of the swept S_j curve is shown on § Figure 8-45.

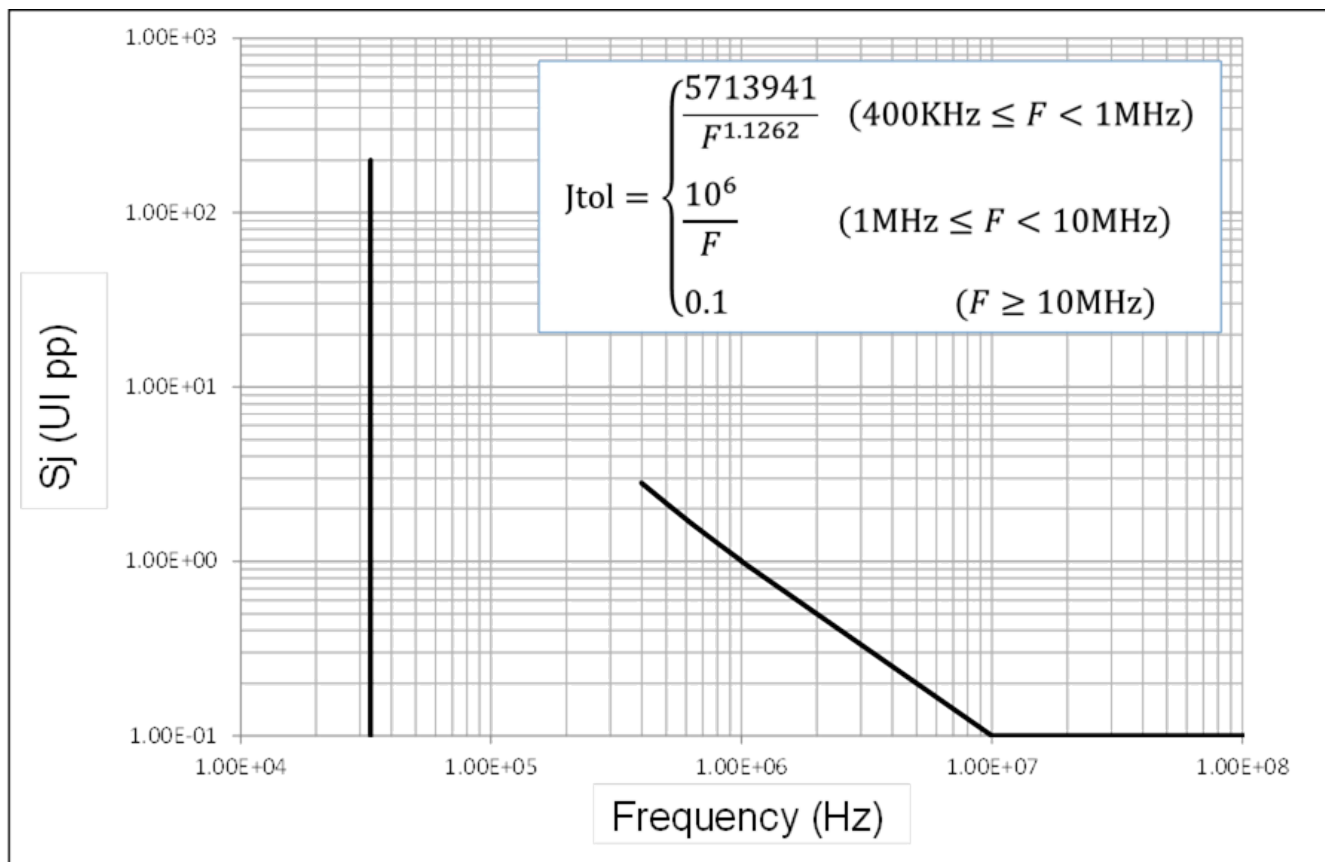


Figure 8-45 Sj Mask for Receivers Operating in IR mode at 8.0 GT/s §

Receivers operating at 16.0 GT/s in the Independent Refclk (IR) mode use the Sj mask profile shown in § Figure 8-46. The magnitude of the 33 kHz spur is 25 ns pp, or 400 UIpp. The equation of the swept Sj curve is shown on the § Figure 8-46.

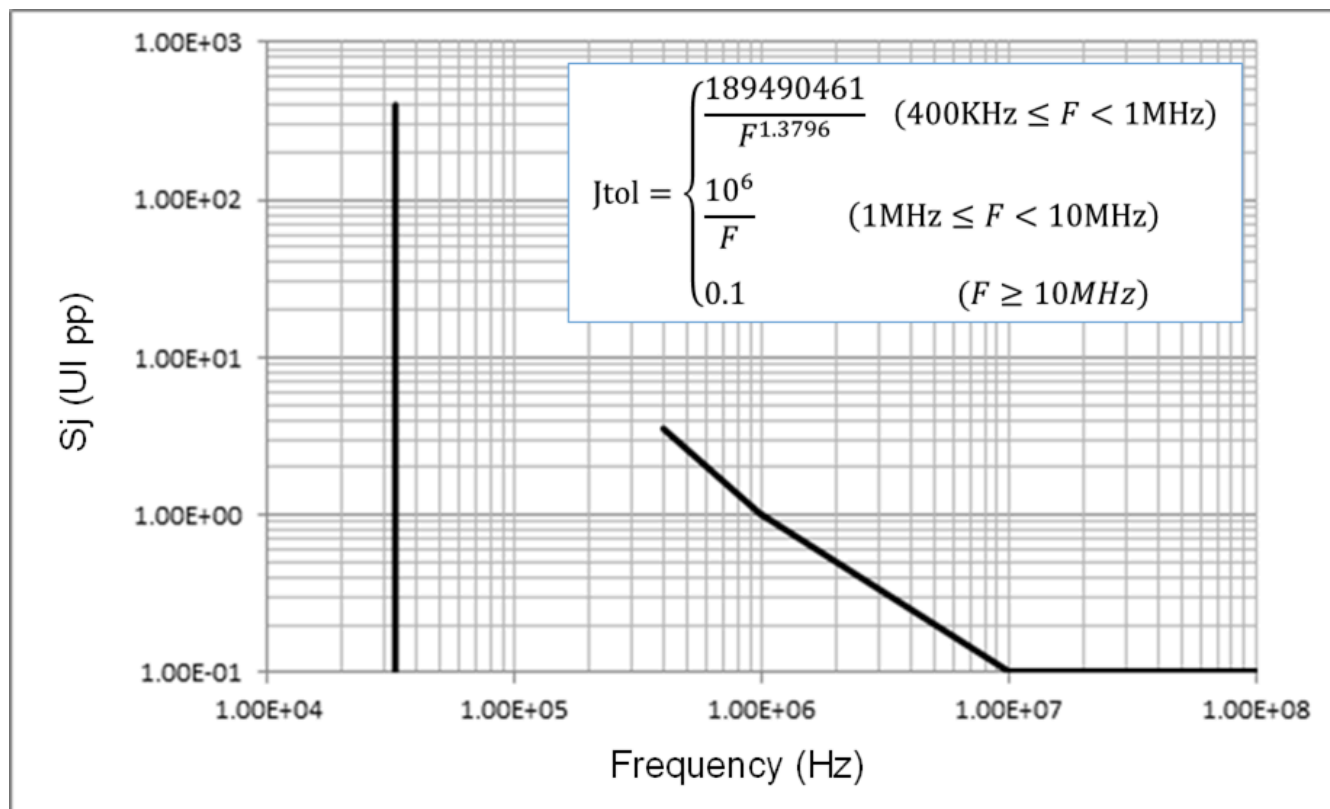


Figure 8-46 Sj Mask for Receivers Operating in SRIS mode at 16.0 GT/s §

Receivers operating at 16.0 GT/s in the Common Clock (CC) Refclk mode use the Sj mask profile shown in § Figure 8-47. The magnitude of the 33 kHz spur is 1 ns pp, or 16 UIpp. The equation of the swept Sj curve is shown on the § Figure 8-47. Devices that do not support 32.0 GT/s have the option to use the Sj mask defined in § Figure 8-52. In this case, a 500 ps-pp triangular SSC modulation has to be applied at the source for both channel calibration and RX compliance, as specified in § Table 8-11.

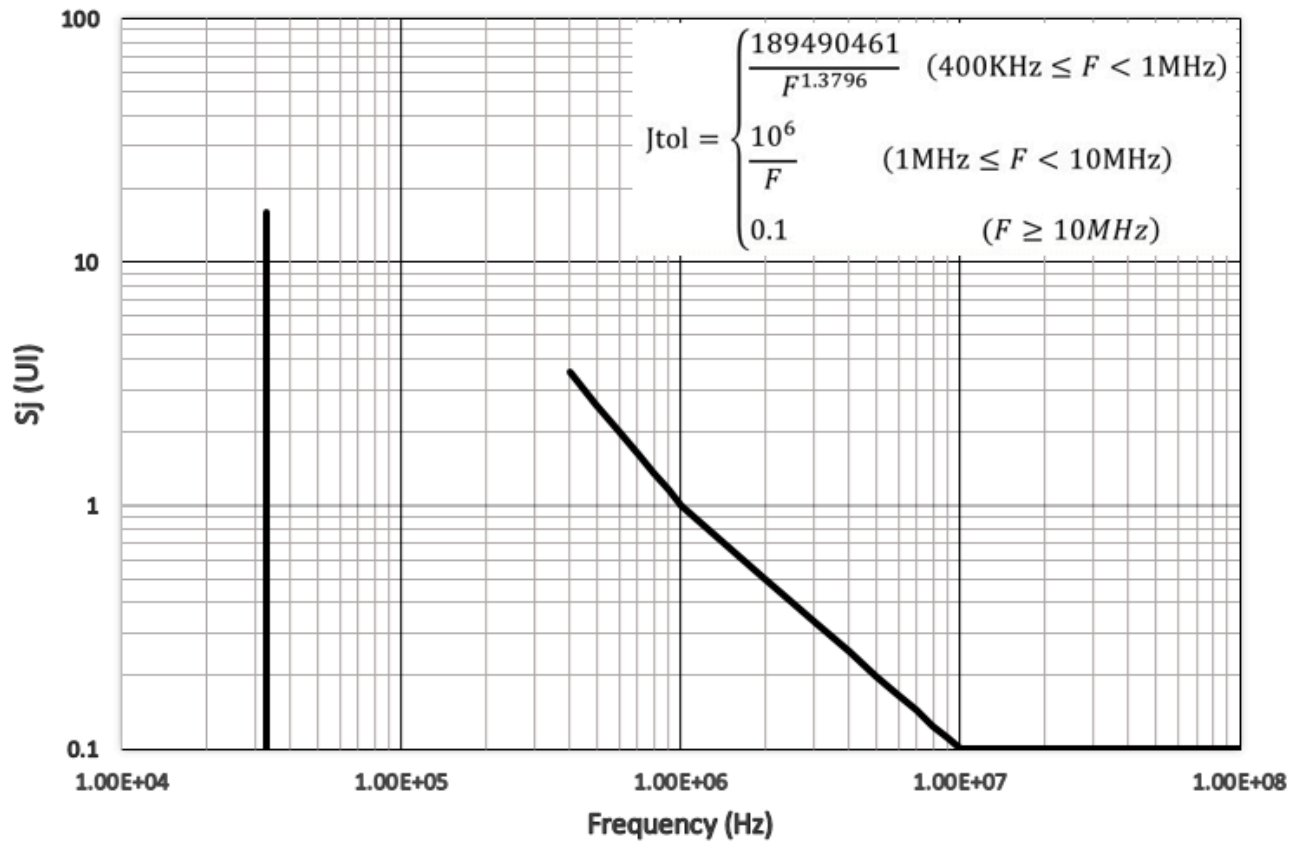


Figure 8-47 Sj Mask for Receivers Operating in CC mode at 16.0 GT/s §

Receivers operating at 32.0 GT/s in the IR mode use the Sj mask profile shown in § Figure 8-48. The magnitude of the 33 kHz spur is 15 ns pp, or 480 UIpp. The equation of the swept Sj curve is shown on the § Figure 8-48.

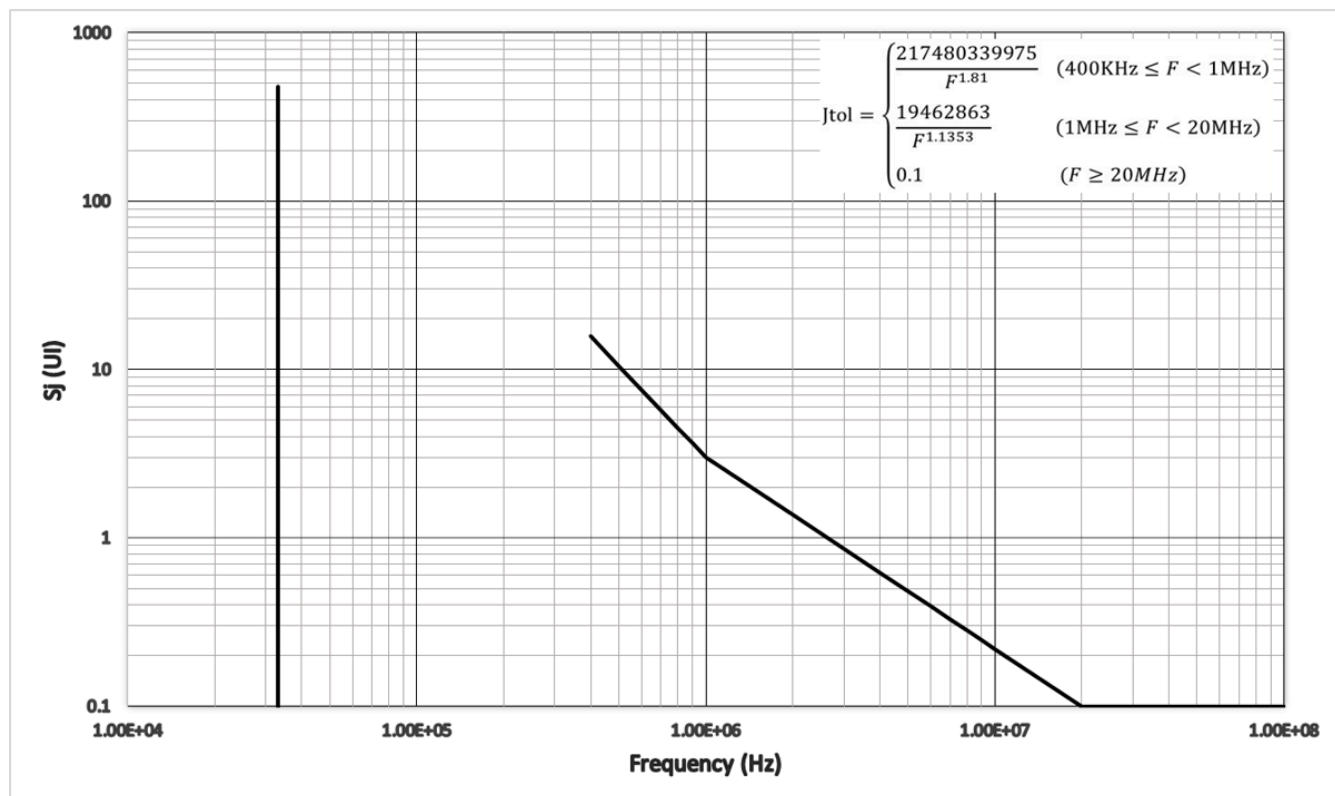


Figure 8-48 Sj Mask for Receivers Operating in SRIS mode at 32.0 GT/s §

Receivers operating at 32.0 GT/s in the CC Refclk mode use the Sj mask profile shown in § Figure 8-49. The magnitude of the 33 kHz spur is 1 ns pp, or 32 U_{lpp}. The equation of the swept Sj curve is shown on the § Figure 8-49.

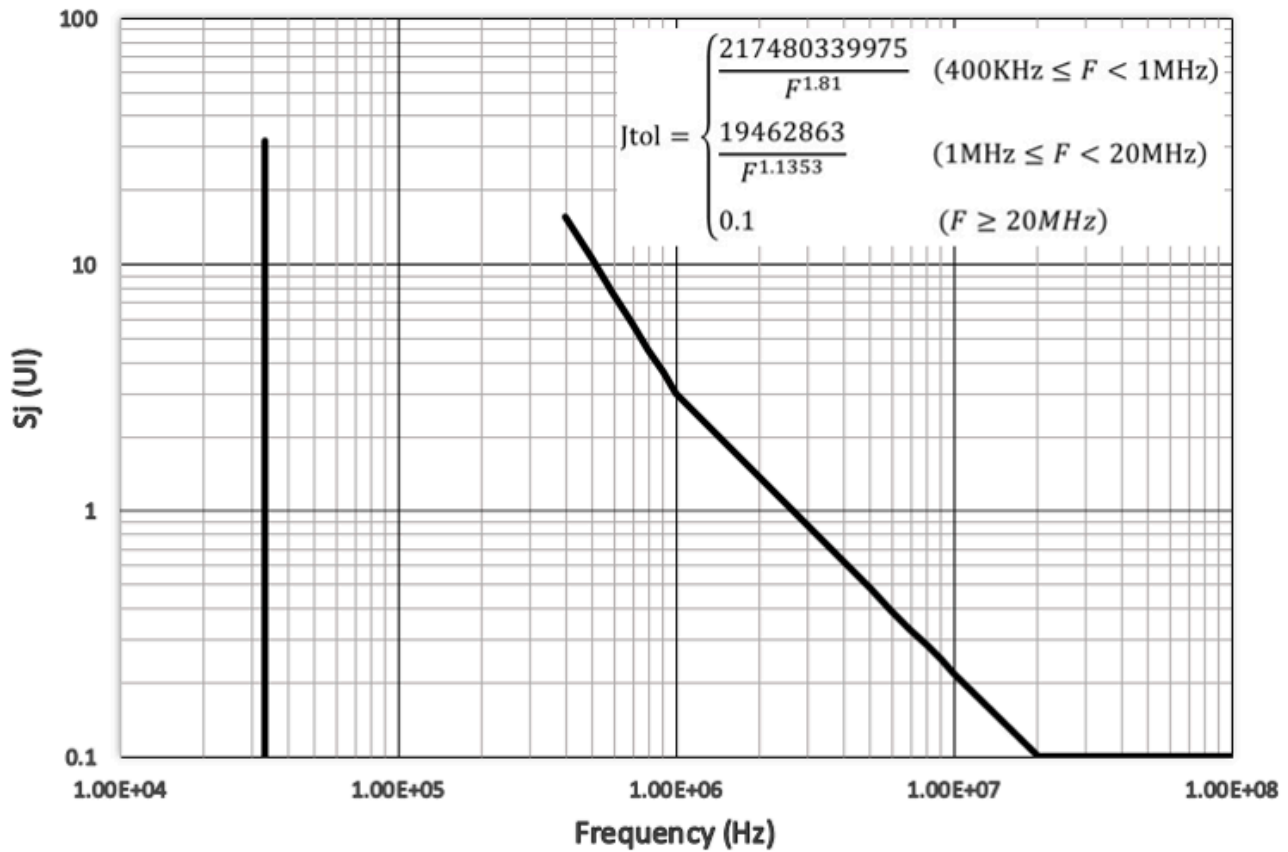


Figure 8-49 Sj Mask for Receivers Operating in CC mode at 32.0 GT/s §

Receivers operating at 64.0 GT/s in the IR mode use the Sj mask profile shown in § Figure 8-50. The magnitude of the 33 kHz spur is 15 ns pp, or 480 UI pp. The equation of the swept Sj curve is shown on the § Figure 8-50.

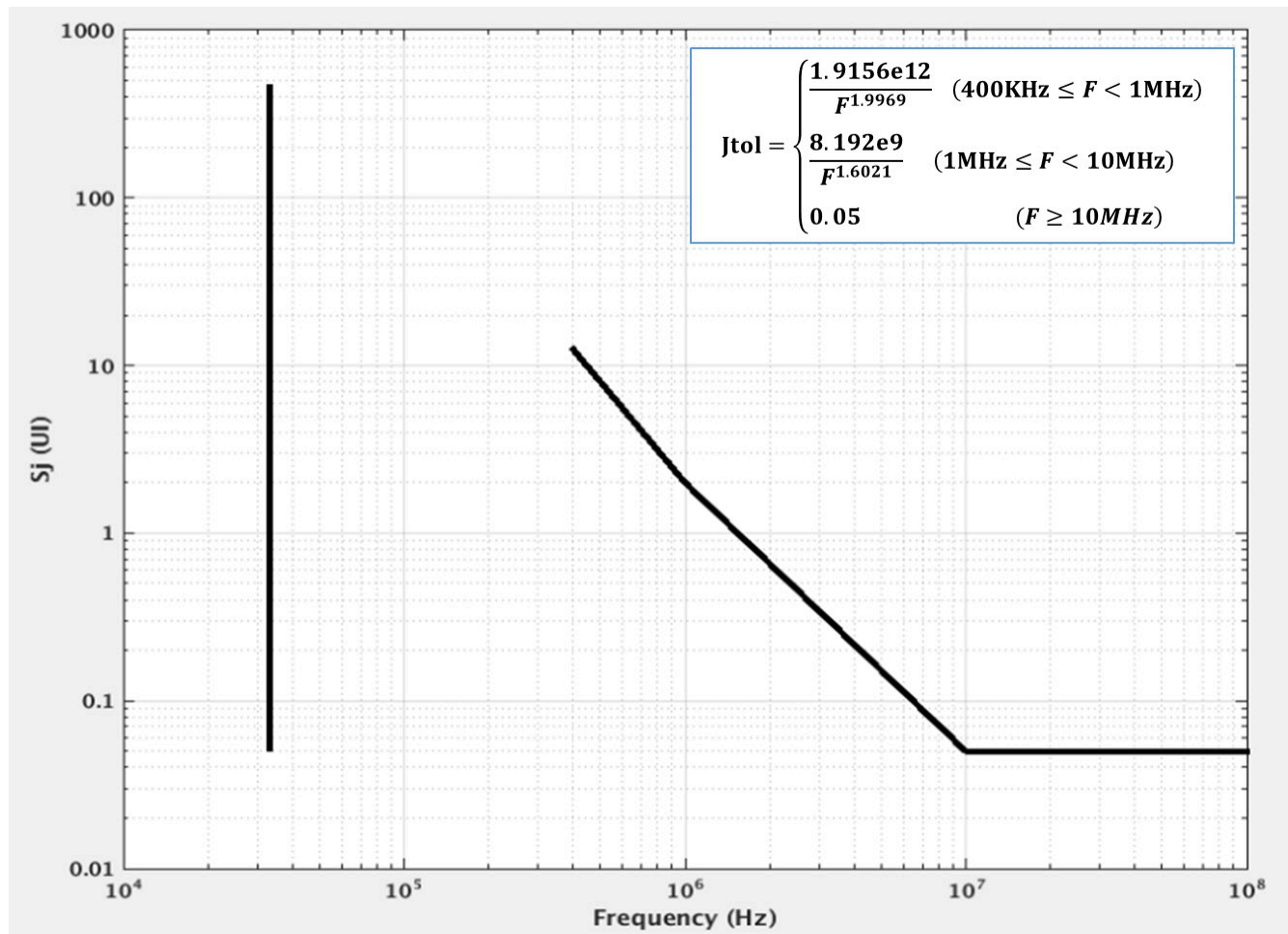


Figure 8-50 Sj Mask for Receivers Operating in SRIS mode at 64.0 GT/s §

Receivers operating at 64.0 GT/s in the CC Refclk mode use the Sj mask profile shown in § Figure 8-51. The magnitude of the 33 kHz spur is 1 ns pp, or 32 UIpp. The equation of the swept Sj curve is shown on the § Figure 8-51.

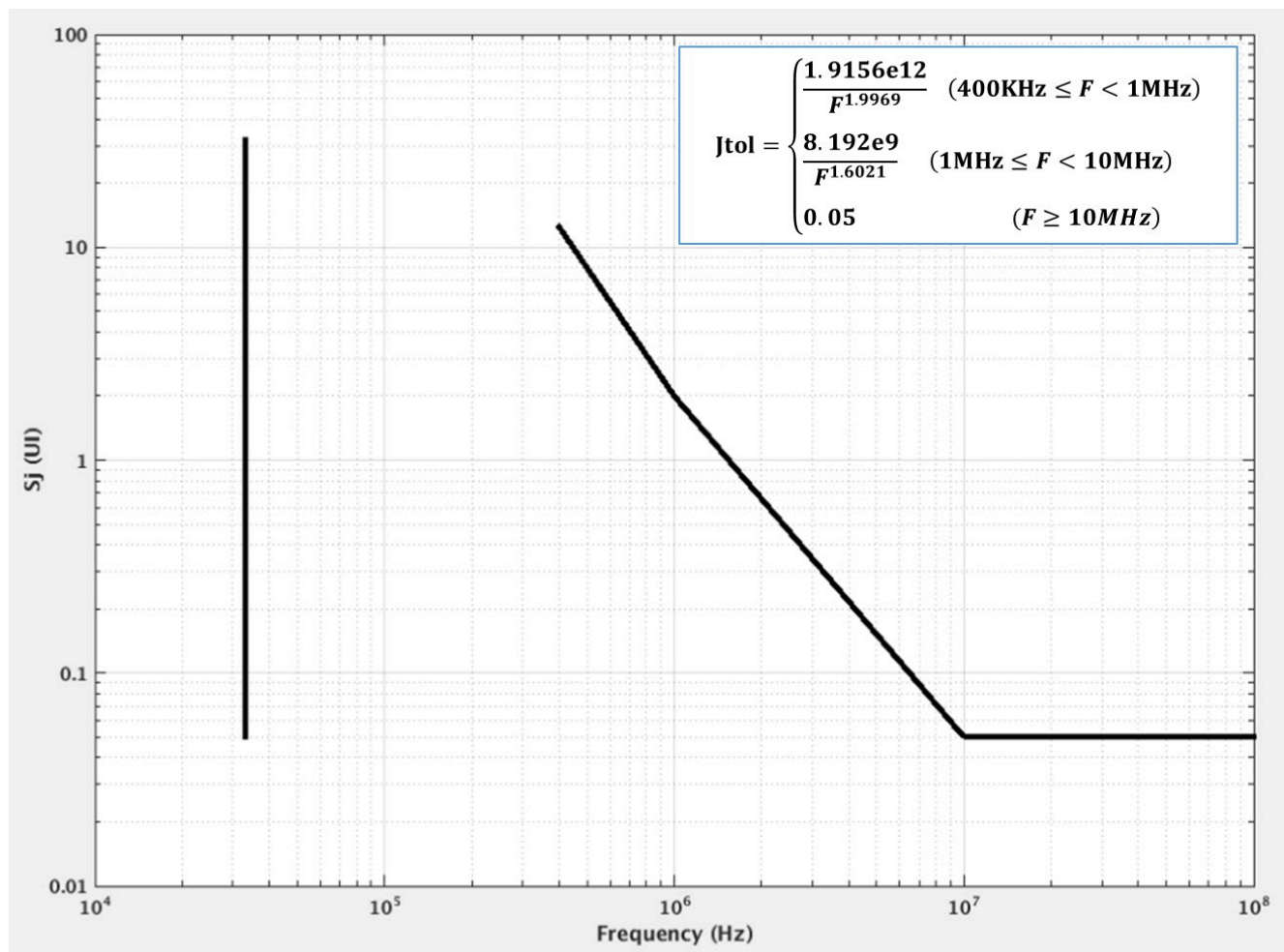


Figure 8-51 Sj Mask for Receivers Operating in CC mode at 64.0 GT/s §

Receivers operating in the CC Refclk mode at 8.0 GT/s shall utilize the Sj profile shown in § Figure 8-52. The testing procedure is identical to that used for the IR mode, except that the clock topology differs. See § Section 8.4.2.3 for details. Receivers operating at 16.0 GT/s in the CC Refclk mode in devices that do not support 32.0 GT/s also have the option to use the Sj mask profile shown in § Figure 8-52, with additional residual SSC applied per § Table 8-11.

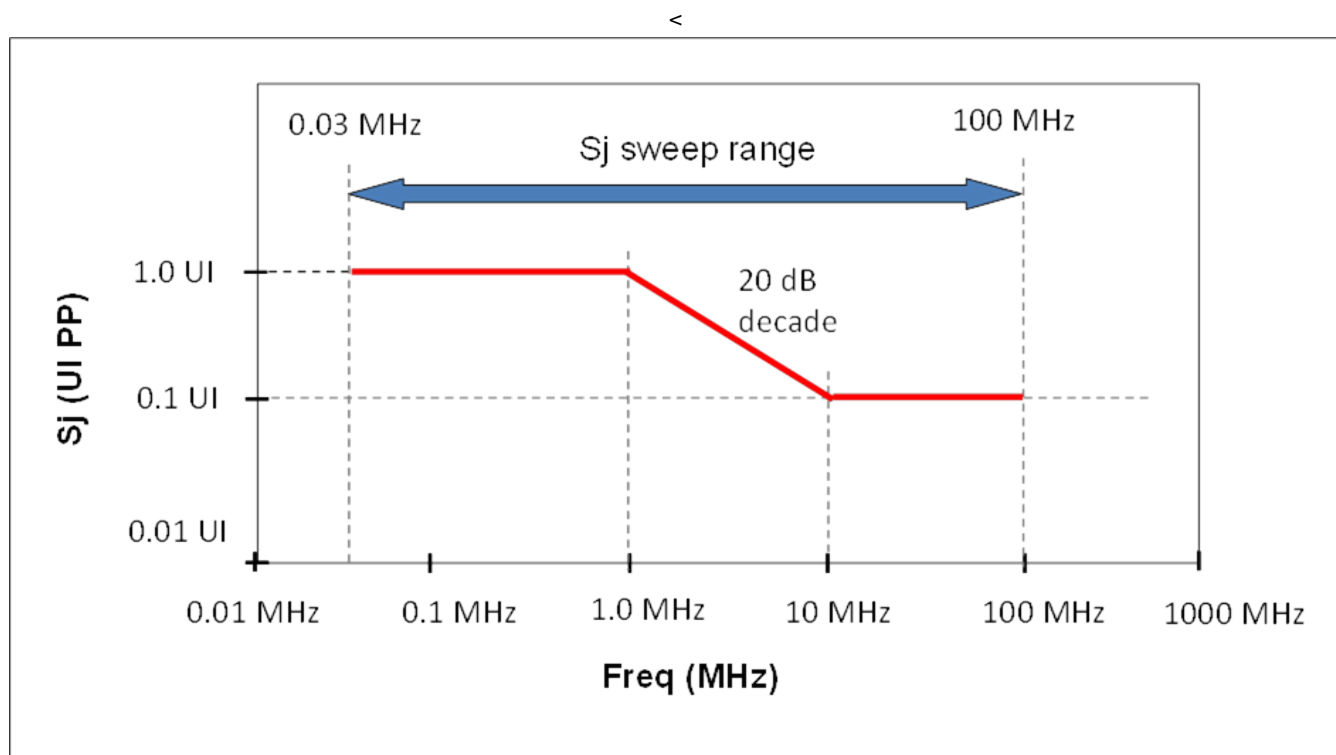


Figure 8-52 S_j Masks for Receivers Operating in CC Mode at 8.0 GT/s §

8.4.2.3 Receiver Refclk Modes §

A Rx is permitted to support one or both of two clock modes: CC and IR although only one clock mode may be operational at a given time. Receivers can support more than one Refclk mode by selecting a mode at power-up or by means of strapping pins, etc.

8.4.2.3.1 Common Refclk Mode §

§ Figure 8-53 shows the Refclk connection for a receiver in the Common Clock Refclk mode. A single Refclk source drives both the Generator and the DUT. This test utilizes the S_j mask specified in § Section 8.4.2.2.1 .

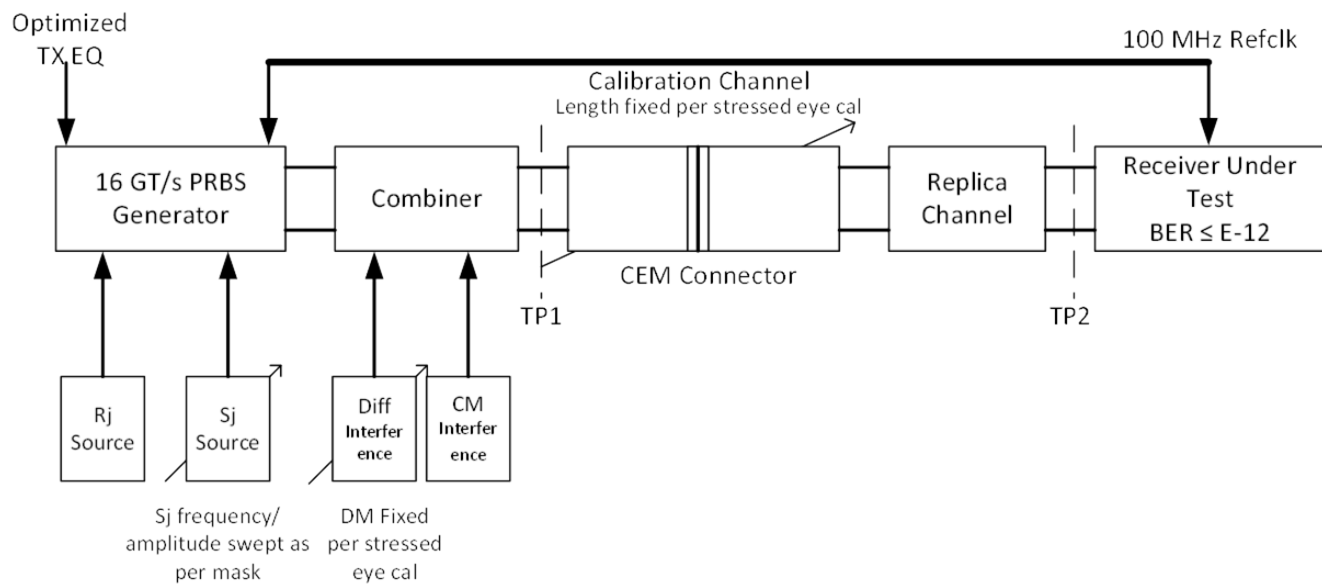


Figure 8-53 Layout for Jitter Testing Common Refclk Rx at 16.0 GT/s §

8.4.2.3.2 Independent Refclk Mode §

§ Figure 8-54 illustrates the configuration for testing a Receiver in the IR Refclk mode. A Refclk source with SSC is required for the DUT. The test utilizes the Sj mask specified in § Section 8.4.2.2.1. The generator must be able to produce a large 33 KHz Sj tone while Sj is swept as shown in § Section 8.4.2.2.1.

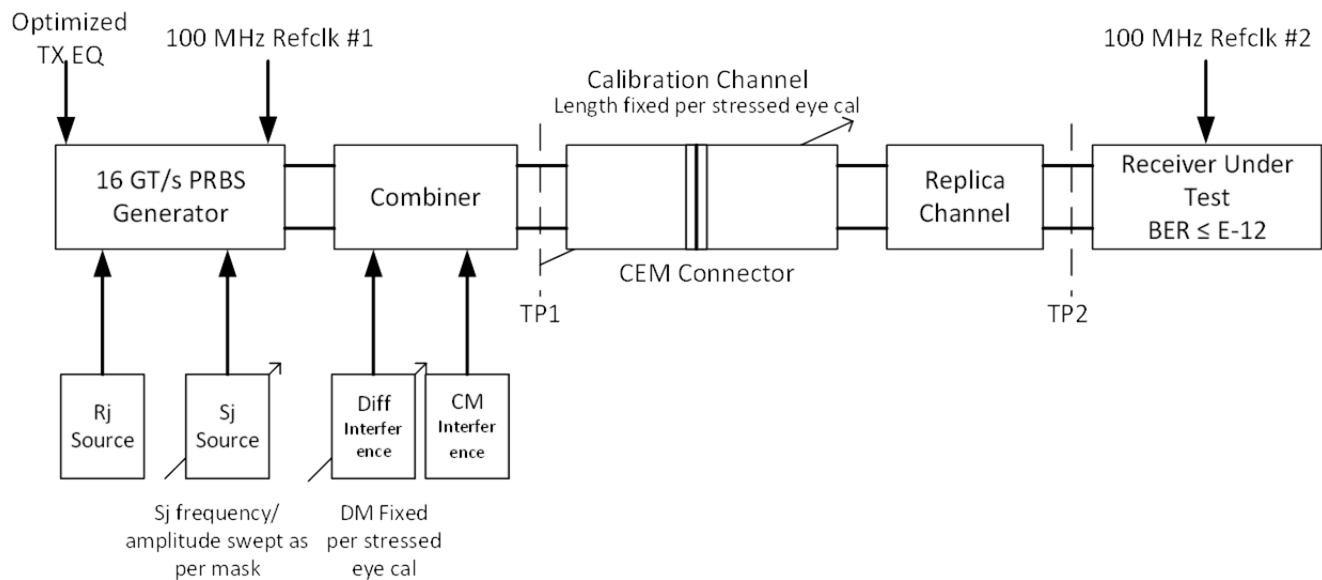


Figure 8-54 Layout for Jitter Testing for Independent Refclk Rx at 16.0 GT/s §

8.4.3 Common Receiver Parameters §

§ Table 8-12 lists the common Receiver parameters that are not directly associated with stressed eye tolerancing. Values are separately defined for the four data rates.

Table 8-12 Common Receiver Parameters §

Symbol	Parameter	2.5 GT/s value	5.0 GT/s value	8.0 GT/s value	16.0 GT/s value	32.0 GT/s value	64.0 GT/s value	Units	Notes
UI (Rx)	Unit Interval	(min) 399.88 (max) 400.12 (300 PPM)	(min) 199.94 (max) 200.06 (300 PPM)	(min) 124.9625 (max) 125.0375 (300 PPM)	(min) 62.48125 (max) 62.51875 (300 PPM)	(min) 31.246875 (max) 31.253125 (100 PPM)	(min) 31.246875 (max) 31.253125 (100 PPM)	ps	UI tolerance does not include SSC effects
BW_{RX}-PKG-PLL1	Rx PLL bandwidth corresponding to <u>PKG_{RX}-PLL1</u>	(max) 22 (min) 1.5	(max) 16.0 (min) 8	(max) 4.0 (min) 0.5	(max) 4.0 (min) 0.5	(max) 1.8 (min) 0.5	(max) 1.0 (min) 0.5	MHz	Second order PLL transfer bounding function. See Note 1.
BW_{RX}-PKG-PLL2	Rx PLL bandwidth corresponding to <u>PKG_{RX}-PLL2</u>	Not Specified	(max) 16.0 (min) 5.0	(max) 5.0 (min) 0.5	(max) 5.0 (min) 0.5	N/A	N/A	MHz	Second order PLL transfer bounding function. See Note 1.
PKG_{RX}-PLL1	Maximum Rx PLL peaking corresponding to <u>BW_{RX}-PKG-PLL1</u>	(max) 3.0	3.0	2.0	2.0	2.0	2.0	dB	Second order PLL transfer bounding function. See Note 1.
PKG_{RX}-PLL2	Maximum Rx PLL peaking corresponding to <u>BW_{RX}-PKG-PLL2</u>	Not specified	1.0	1.0	1.0	N/A	N/A	dB	Second order PLL transfer bounding function. See Note 1.
RL_{RX-DIFF}	Differential receiver return loss	See § Figure 8-24					See § Figure 8-26	dB	Note 2
RL_{RX-CM}	Common mode receiver return loss	See § Figure 8-25					See § Figure 8-27	dB	Note 2
T_{RX-GND-FLOAT}	Rx termination float time			(max) 500	(max) 500	(max) 500	(max) 500	μs	Note 5
V_{RX-CM-AC-P}	Rx AC common Mode Voltage	(max) 150	(max) 150	(max) 75	(max) 75	(max) 75	(max) 37.5	mVP	Measured at Rx pins into a pair of 50 Ω terminations to ground
Z_{RX-DC}	Receiver DC single ended impedance	(min) 40 (max) 60	(min) 40 (max) 60	Not specified	Not specified	Not specified	Not specified	Ω	DC impedance limits are needed to guarantee Receiver detect. For 8.0,

Symbol	Parameter	2.5 GT/s value	5.0 GT/s value	8.0 GT/s value	16.0 GT/s value	32.0 GT/s value	64.0 GT/s value	Units	Notes
									16.0, 32.0, and 64.0 GT/s it is bounded by RL_{RX-CM} . See Note 3.
$Z_{RX-HIGH-IMP-DC-POS}$	DC input CM input impedance for $V \geq 0$ during Reset or power-down	$\geq 10K$ (0-200 mV) $\geq 20K$ (> 200 mV)	$\geq 10K$ (0-200 mV) $\geq 20K$ (> 200 mV)	$\geq 10K$ (0-200 mV) $\geq 20K$ (> 200 mV)	$\geq 10K$ (0-200 mV) $\geq 20K$ (> 200 mV)	$\geq 10K$ (0-200 mV) $\geq 20K$ (> 200 mV)	$\geq 10K$ (0-200 mV) $\geq 20K$ (> 200 mV)	Ω	Voltage measured wrt. ground. Parameters may not scale with process technology. See Note 4.
$Z_{RX-HIGH-IMP-DC-NEG}$	DC input CM input impedance for $V < 0$ during Reset or power-down	1.0K (min)	1.0K (min)	1.0K (min)	1.0K (min)	1.0K (min)	1.0K (min)	Ω	Voltage measured over the range of -150 mV to 0 mV wrt. ground. Parameters may not scale with process technology. See Note 4.
$V_{RX-IDLE-DET-DIFF-PP}$	Electrical Idle Detect threshold	(min) 65 (max) 175	(min) 65 (max) 175	(min) 65 (max) 175	(min) 65 (max) 175	(min) 65 (max) 175	(min) 65 (max) 175	mV	$V_{RX-IDLE-DET-DIFF-p} = 2 \times V_{RX-D+} - V_{RX-D-} $
$T_{RX-IDLE-DET-DIFF-ENTERTIME}$	Unexpected Electrical Idle Enter Detect Threshold Integration Time	(max) 10	(max) 10	(max) 10	(max) 10	(max) 10	(max) 10	ms	An unexpected Electrical Idle ($V_{RX-DIFF-PP} < V_{RX-IDLE-DET-DIFF-p}$) must be recognized no longer than $T_{RX-IDLE-DET-DIFF-ENTERTIME}$ to signal an unexpected idle condition.
$L_{RX-SKEW}$	Lane to Lane skew	(max) 20	(max) 8	(max) 6	(max) 5	(max) 5	(max) 5	ns	Across all Lanes on a Port. $L_{RX-SKEW}$ comprehends Lane-Lane variations due to channel and repeater delay differences.

Notes:

- Two combinations of PLL BW and peaking are specified at 5.0 GT/s to permit designers to make tradeoffs between the two parameters. If the PLL's min BW is ≥ 8 MHz, then up to 3.0 dB of peaking is permitted. If the PLL's min BW is relaxed to ≥ 5.0 MHz, then a tighter peaking value of 1.0 dB must be met. Note: a PLL BW extends from zero up to the value(s) defined as the min or max in the above table. For 2.5 GT/s a single PLL bandwidth and peaking value of 1.5-22 MHz and 3.0 dB are defined.
- Measurements must be made for both common mode and differential return loss. In both cases the DUT must be powered up and DC isolated, and its D+/D- inputs must be in the low-Z state.

data sample location - but the actual margining method is implementation specific. For example - the timing margin could be implemented on the actual data sampler or an independent error sampler. Further the timing margin can be achieved by injecting an appropriate amount of stress/jitter to the data sample location, or by moving the data/error sample location. The parameters in § Table 8-13 are reported for 16.0, 32.0, or 64.0 GT/s and all are allowed to be different for each of those data rates, though some are required to be different for NRZ vs. PAM4 data rates.

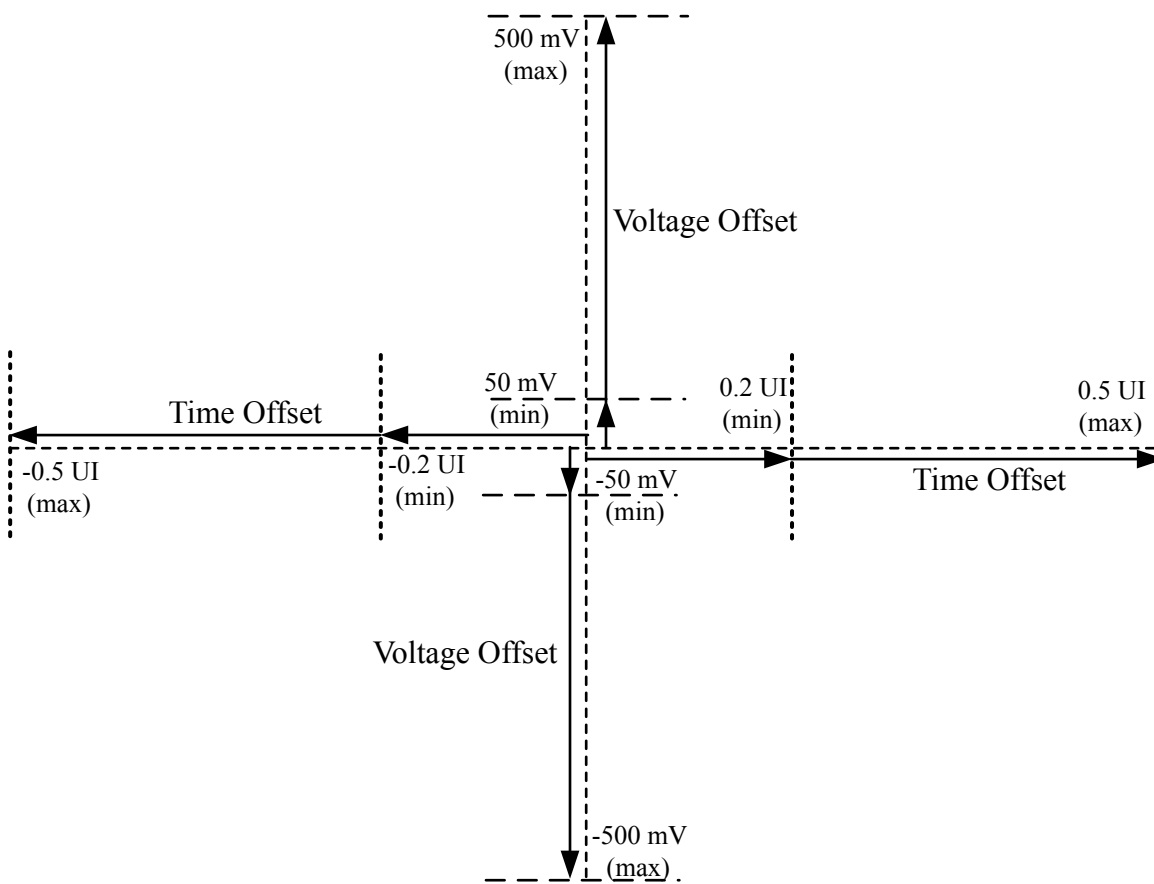


Figure 8-56 Allowed Ranges for Maximum NRZ Timing and Voltage Margin §

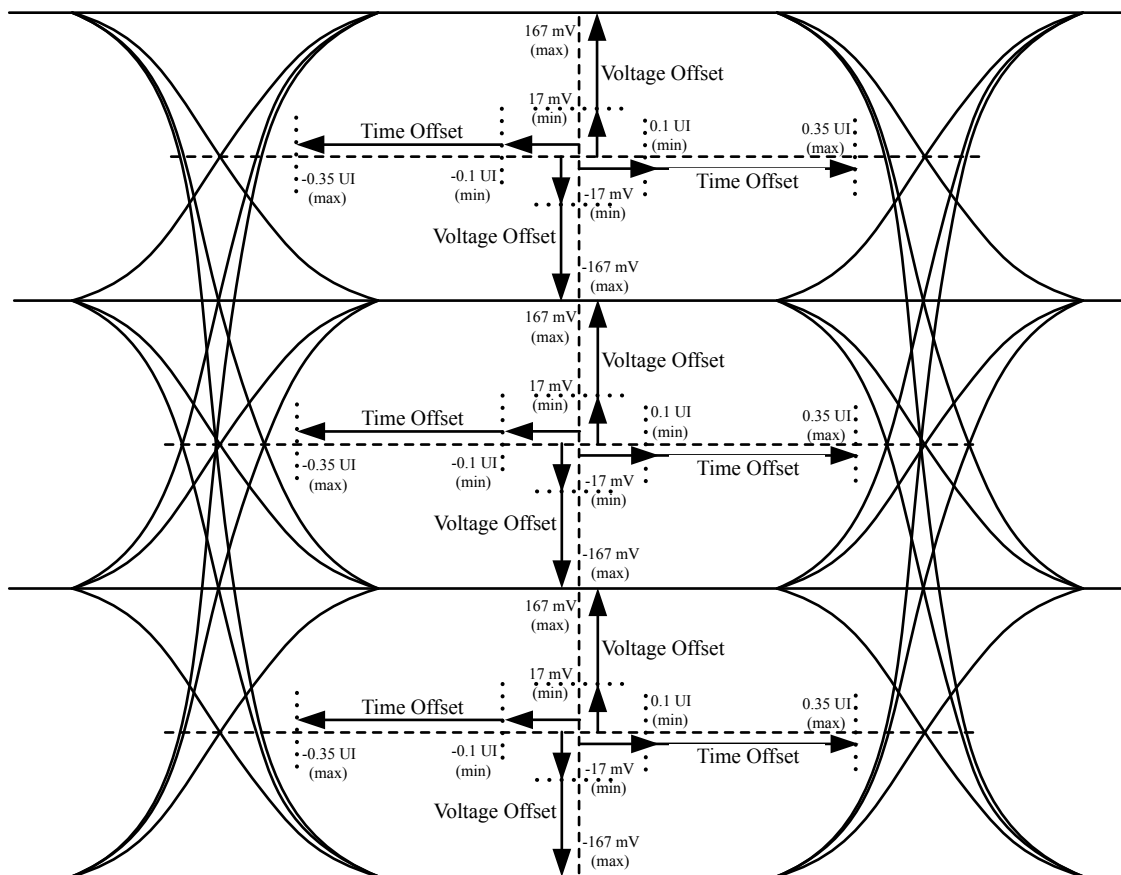


Figure 8-57 Allowed Ranges for Maximum PAM4 Timing and Voltage Margins §

Table 8-13 Lane Margining §

Parameter Name	Min	Max	Description
$M_{NumTimingSteps}$	6 (NRZ) 10 (PAM4)	63	Number of time steps from default (to either left or right), range must be at least ± 0.2 UI if operating in NRZ, or at least ± 0.1 UI if operating in PAM4 Timing offset must increase monotonically The number of steps in both positive (toward the end of the unit interval) and negative (toward the beginning of the unit interval) must be identical
$M_{MaxTimingOffset}$	20 % (NRZ) 10 % (PAM4)	50 % (NRZ) 35 % (PAM4)	Offset from default at maximum step value as percentage of a nominal UI A 0 value may be reported if the vendor chooses not to report the offset
$M_{NumVoltageSteps}$	32 (NRZ) 64 (PAM4)	127	Number of voltage steps from default (either up or down), minimum range ± 50 mV if operating in NRZ, or ± 17 mV if operating in PAM4, either as measured by the reference equalizer Voltage offset must increase monotonically

Parameter Name	Min	Max	Description
			The number of steps in both positive and negative direction from the default sample location must be identical This value is undefined if <u>M_{VoltageSupported}</u> is 0b
M_{MaxVoltageOffset}	5 %	50 %	Offset from default at maximum step value as percentage of one volt if operating in NRZ, or as a percentage of one third volt if operating in PAM4 A 0 value may be reported if the vendor chooses not to report the offset when <u>M_{VoltageSupported}</u> is 1b This value is undefined if <u>M_{VoltageSupported}</u> is 0b

Placeholder table row so Errata B677 is properly attributed in changebar Base 6.3 documents. Remove this table row for subsequent revisions.

Errata B677 combined:

- M_{NumTimingStepsPAM4} into M_{NumTimingSteps}
- M_{MaxTimingOffsetPAM4} into M_{MaxTimingOffset}
- M_{NumVoltageStepsPAM4} into M_{NumVoltageSteps}
- M_{MaxVoltageOffsetPAM4} into M_{MaxVoltageOffset}

M_{SamplingRateVoltage}	0	63	The ratio of bits tested to bits received during voltage margining. A value of 0 is a ratio of 1:64 (1 bit of every 64 bits received), and a value of 63 is a ratio of 64:64 (all bits received).
M_{SamplingRateTiming}	0	63	The ratio of bits tested to bits received during timing margining. A value of 0 is a ratio of 1:64 (1 bit of every 64 bits received), and a value of 63 is a ratio of 64:64 (all bits received).
M_{VoltageSupported}	0	1	1b indicates that voltage margining is supported
M_{IndLeftRightTiming}	0	1	1b indicates independent left/right timing margin supported
M_{IndUpDownVoltage}	0	1	1b independent up and down voltage margining supported
M_{IndErrorSampler}	0	1	1b Margining will not produce errors (change in the error rate) in data stream (i.e., error sampler is independent) 0b Margining may produce errors in the data stream
M_{MaxLanes}	0	31	Maximum number of Lanes minus 1 that can be margined at the same time. It is recommended that this value be greater than or equal to the number of Lanes in the Link minus 1. Encoding Behavior is undefined if software attempts to margin more than <u>M_{MaxLanes}</u>+1 at the same time. Note: This value is permitted to exceed the number of Lanes in the Link minus 1.

Parameter Name	Min	Max	Description
<i>M_{SampleReportingMethod}</i>	0	1	Indicates whether sampling rates ($M_{\text{SamplingRateVoltage}}$ and $M_{\text{SamplingRateTiming}}$) are supported (1) or a sample count is supported (0). One of the two methods is supported by each device.
<i>M_{ErrorCount}</i>	0	63	If $M_{\text{IndErrorSampler}}$ is 1b this is a count of the actual bit errors since margining started. If $M_{\text{IndErrorSampler}}$ is 0b this is the actual count of the logical errors since margining started. See the Physical Layer Logical Block chapter for the definition of what errors are counted. The count saturates at 63.
<i>M_{SampleCount}</i>	0	127	Value = $3 \cdot \log_2$ (number of bits margined). Where number of bits margined is a count of the actual number of bits tested during margining. The count stops when margining stops. The count saturates at 127 (after approximately 5.54×10^{12} bits). The count resets to zero when a new margin command is received.

8.4.5 Low Frequency and Miscellaneous Signaling Requirements §

8.4.5.1 ESD Standards §

All PCI Express signal and power supply pins must be tested for ESD protection levels to the Human Body Model (HBM) and the Charged Device Model (CDM) standards in accordance with [ESDA-JEDEC-JS-001-2010] (for HBM) and in accordance with [JEDEC-JESD22-C101] (for CDM). Pins must meet or exceed the minimum levels recommended in [JEDEC-JEP155-JEP157] (HBM/CDM) or JEDEC approved superseding documents.

8.4.5.2 Channel AC Coupling Capacitors §

Each Lane of a PCI Express Link must be AC coupled. The minimum and maximum values for the capacitance are given in § Table 8-7. Capacitors must be placed on the Transmitter side of an interface that permits adapters to be plugged and unplugged. In a topology where everything is located on a single substrate, the capacitors may be located anywhere along the channel. External capacitors are assumed because the values required are too large to feasibly construct on-chip.

8.4.5.3 Short Circuit Requirements §

All Transmitters and Receivers must support surprise hot insertion/removal without damage to the component. The Transmitter and Receiver must be capable of withstanding sustained short circuit to ground of D+ and D-.

8.4.5.4 Transmitter and Receiver Termination §

- The Transmitter is required to meet $RL_{TX-DIFF}$ and RL_{TX-CM} (see § Figure 8-24, § Figure 8-26, § Figure 8-25 and § Figure 8-27) any time functional differential signals are being transmitted.
- The Transmitter is required only to meet $I_{TX-SHORT}$ (see § Table 8-7) any time functional differential signals are not being transmitted.
- Note: The differential impedance during this same time is not defined.
- The Receiver is required to meet $RL_{RX-DIFF}$ and RL_{RX-CM} (see § Table 8-12) during all LTSSM states excluding only times during when the device is powered down, Fundamental Reset is asserted, or when explicitly specified.
- The Receiver is required to meet $Z_{RX-HIGH-IMP-DC-NEG}$ and $Z_{RX-HIGH-IMP-DC-POS}$ (see § Table 8-12) any time adequate power is not provided to the Receiver, Fundamental Reset is asserted, or when explicitly specified.

8.4.5.5 Electrical Idle §

Electrical Idle is a steady state condition where the Transmitter D+ and D- voltages are held constant at the same value. Electrical Idle is primarily used in power saving and inactive states (e.g., Disabled).

Before a Transmitter enters Electrical Idle, it must always send the required number of EIOs except for the LTSSM substates explicitly exempted from this requirement. After sending the last Symbol of the last of the required number of EIOs, the Transmitter must be in a valid Electrical Idle state within the time as specified by $T_{TX-IDLE-SET-TO-IDLE}$ in § Table 8-7.

The successful reception of an EIO occurs based on the rules defined in the Physical Layer Logical Block chapter. It should be noted that in substates (e.g., Loopback.Active for a Loopback Follower) where multiple consecutive EIOs are expected, the Receiver must receive the appropriate number of EIO sequences comprising of COM, IDL, IDL, IDL.

The low impedance common mode and differential Receiver termination values (see § Table 8-7 and § Table 8-12) must be met in Electrical Idle. The Transmitter can be in either a low or high impedance mode during Electrical Idle.

Any time a Transmitter enters Electrical Idle it must remain in Electrical Idle for a minimum of $T_{TX-IDLE-MIN}$. The Receiver should expect the last EIO followed by a minimum amount of time in Electrical Idle ($T_{TX-IDLE-MIN}$) to arm its Electrical Idle Exit detector.

When the Transmitter transitions from Electrical Idle to a valid differential signal level it must meet the output return loss specifications described in § Figure 8-24, § Figure 8-26, § Figure 8-25, and § Figure 8-27.

Electrical Idle Exit shall not occur if a signal smaller than $V_{RX-IDLE-DET-DIFF-p}$ minimum is detected at a Receiver. Electrical Idle Exit shall occur if a signal larger than $V_{RX-IDLE-DET-DIFF-p}$ maximum is detected at a Receiver. Electrical Idle may be detected on the received signal regardless of its frequency components, or it may be detected only when the received signal is switching at a frequency of 125 MHz or higher.

8.4.5.6 DC Common Mode Voltage §

The Receiver DC common mode voltage is nominally 0 V when operating at 2.5 GT/s or 5.0 GT/s.

Transmitter DC common mode voltage is held at the same value during all states unless otherwise specified. The range of allowed Transmitter DC common mode values is specified in § Table 8-7 ($V_{TX-DC-CM}$).

8.4.5.7 Receiver Detection §

The Receiver Detection circuit is implemented as part of a Transmitter and must correctly detect whether a load impedance equivalent to a DC impedance implied by the Z_{RX-DC} parameter (40 Ω -60 Ω) is present. Note: Support for Rx detect, which only occurs at 2.5 GT/s, is the reason why 2.5 GT/s Receivers impedance at DC is specified.

The recommended behavior of the Receiver Detection sequence is described below:

Step 1. A Transmitter must start at a stable voltage prior to the detect common mode shift.

Step 2. A Transmitter changes the common mode voltage on D+ and D- consistent with meeting the $V_{TX-RCV-DETECT}$ parameter and consistent with detection of Receiver high impedance which is bounded by parameters $Z_{RX-HIGH-IMP-DC-POS}$, $Z_{RX-HIGH-IMP-DC-NEG}$ in § Table 8-12. Receiver is detected based on the rate that the lines change to the new voltage.

- a. The Receiver is not present if the voltage at the Transmitter charges at a rate dictated only by the Transmitter impedance and the capacitance of the interconnect and series capacitor.
- b. The Receiver is present if the voltage at the Transmitter charges at a rate dictated by the Transmitter impedance, the series capacitor, the interconnect capacitance, and the Receiver termination.

If the Receiver Detection circuit performs the detect sequence on each conductor of the differential pair (both D+ and D-) and detects a load impedance greater than Z_{RX-DC} on either conductor, the Receiver Detection circuit shall interpret this as no termination load present and respond as if neither load were present.

It is required that the detect sequence be performed on both conductors of a differential pair.

8.5 Channel Tolerancing §

8.5.1 Channel Compliance Testing §

This section of the specification is relevant only for those cases where a platform design comprehends the relevant channel between Transmitter device pins and Receiver device pins. These types of platform designs are called “captive channels”. Designs that are not captive channels shall refer to the appropriate form factor (CEM is one example) specification, since in this case the form factor specification takes precedence over this specification and splits the channel between two different types of components.

The key components and processes of channel tolerancing are illustrated in § Figure 8-58 and § Figure 8-59. The major difference lies in the complexity of the Behavioral Tx and Rx equalization, which depends on the data rate. 2.5 and 5.0 GT/s utilize fixed Tx presets and assume no Rx equalization, whereas 8.0, 16.0, 32.0, and 64.0 GT/s assume multi-valued, adjustable Tx presets and a combination of Rx DFE and CTLE.

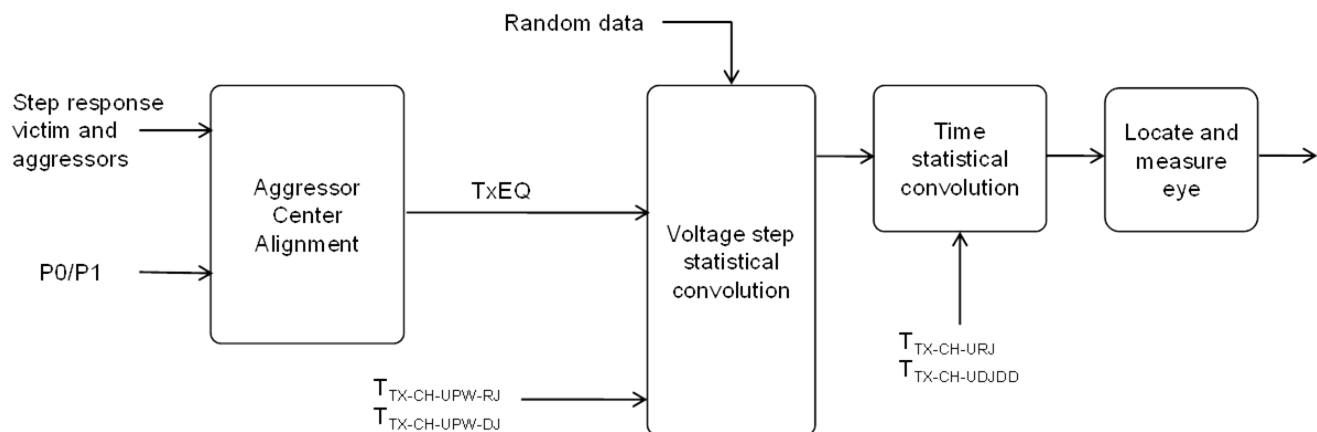


Figure 8-58 Flow Diagram for Channel Tolerancing at 2.5 and 5.0 GT/s §

The basic channel compliance approach is to first acquire the channel's characteristics, usually by means of s-parameters or equivalent model. Behavioral Tx and Rx package models are then appended to the channel model to yield a die pad to die pad topology. The model shall include both victim path and a sufficient number of aggressor paths to accurately capture channel crosstalk effects. Using the Tx voltage and jitter limits defined in the Transmitter specification section it is possible to transform these parameters to what would appear at the die pad of a Tx.

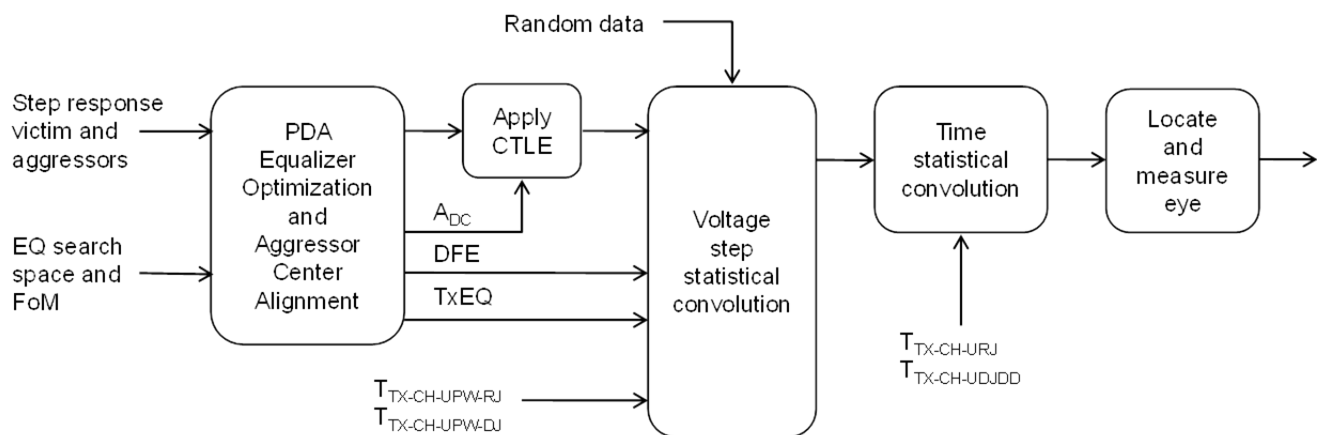


Figure 8-59 Flow Diagram for Channel Tolerancing at 8.0 and 16.0 GT/s §

The resulting model is analyzed via simulation, yielding voltage and jitter at a point equivalent to the input latch of the Receiver. The signal observed at the Receiver's latch is referenced to a recovered data clock from which an eye diagram may be constructed.

For 8.0, 16.0, 32.0, and 64.0 GT/s testing the simulation process must properly account for Tx and Rx equalization optimization as must be supported by a minimum capability Tx/Rx pair. This means the simulation process must be able to select the optimum values for the Tx presets or coefficients and Rx equalization settings based upon:

- a 1st order CTLE and a 1-tap DFE for 8.0 GT/s,
- a 1st order CTLE and a 2-tap DFE for 16.0 GT/s,

- a 2nd order CTLE and a 3-tap DFE for 32.0 GT/s, and
- a CTLE transfer function with six poles and three zeros and a 16-tap DFE for 64.0 GT/s.

8.5.1.1 Behavioral Transmitter and Receiver Package Models §

Behavioral package models are defined in this specification to represent the combined die and package loss that is expected to interoperate with the targeted range of channels. Note that at 16.0 GT/s, the behavioral packages represent a high, but not worst-case, loss for many devices. (see § Section 8.3.3.11 and § Section 8.4.1.5)

A separate pair of package models are defined for 8.0, 16.0, 32.0, and 64.0 GT/s. At 8.0 GT/s, separate package models are defined for TX and RX ports to reflect the smaller CPAD capacitance typical in most Receiver implementations. At 16.0, 32.0, and 64.0 GT/s, separate package models are defined for devices containing Root Ports and all other devices to reflect the large and socketed nature of most devices containing Root Complexes. Channel testing for all three data rates typically requires testing in both directions.

The package models are included with the specification as design collateral. Each model for 8.0 and 16.0 GT/s comprehends C_{PIN} and C_{PAD} parasitic capacitances plus a differential t-line element as illustrated in § Figure 8-60.

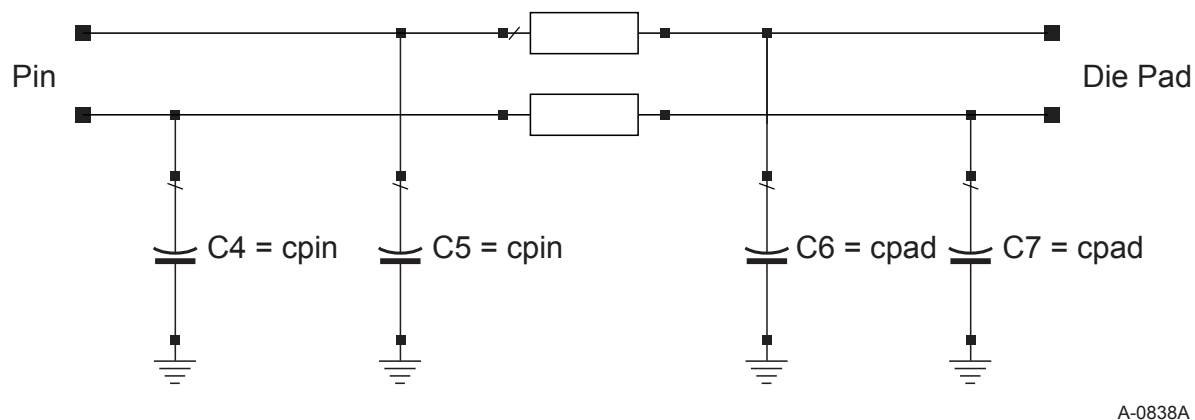


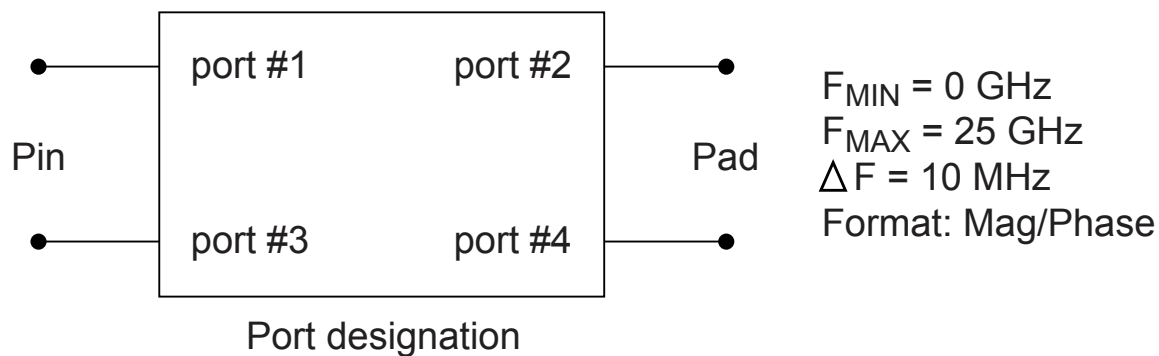
Figure 8-60 Tx/Rx Behavioral Package Models §

The C_{PIN} and C_{PAD} values used in the package model generation are provided for informative purposes only.

Table 8-14 Package Model Capacitance Values §

	8.0 GT/s Tx	8.0 GT/s Rx	16.0 GT/s
C_{PIN}	0.25 pF	0.25 pF	0.25 pF
C_{PAD}	1.0 pF	0.8 pF	0.5 pF

For ease of incorporation into the post processing flow the 8.0 and 16.0 behavioral package models are specified as 4-port s-parameter files. The files are specified with port designations, frequency range and granularity as listed below. The reference impedance for the s-parameters is 50 Ω . File format is Touchstone.



A-0839A

Figure 8-61 Behavioral Tx and Rx S-Port Designation for 8.0 and 16.0 GT/s Packages §

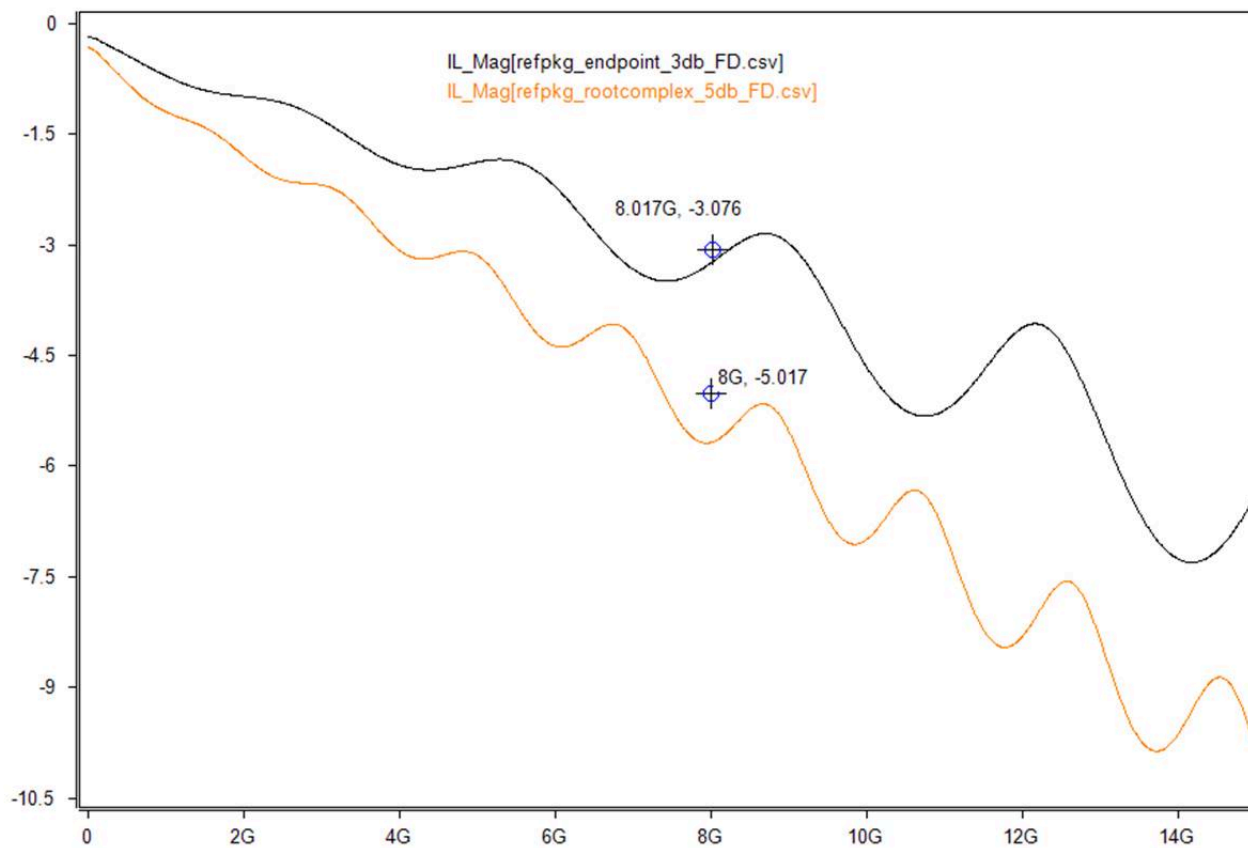


Figure 8-62 SDD21 Plots for Root and Non-Root Packages for 16.0 GT/s §

For 32.0 and 64.0 GT/s the package models are based on real package and socket models for the root package and package and BGA models for the non-root package.

For 32.0 GT/s, reference package models the die capacitive loads are included in the models. For 64.0 GT/s, the effective die cap has been replaced by the S-parameter representation of an on-die network that represents the insertion and

return loss characteristics of on-die pad for a typical design. The reference impedance for the 32.0 and 64.0 GT/s reference package model s-parameters is 50 Ω . Figures below show 32.0 and 64.0 GT/s reference package insertion loss, return loss, FEXT, and NEXT S-parameter plots based on 50 Ω impedance. To account for the on-die inductive coil DC loss of about 3 Ω , the Tx and Rx DC termination should be set to 47 Ω in 64.0 GT/s channel compliance simulations.

Insertion loss

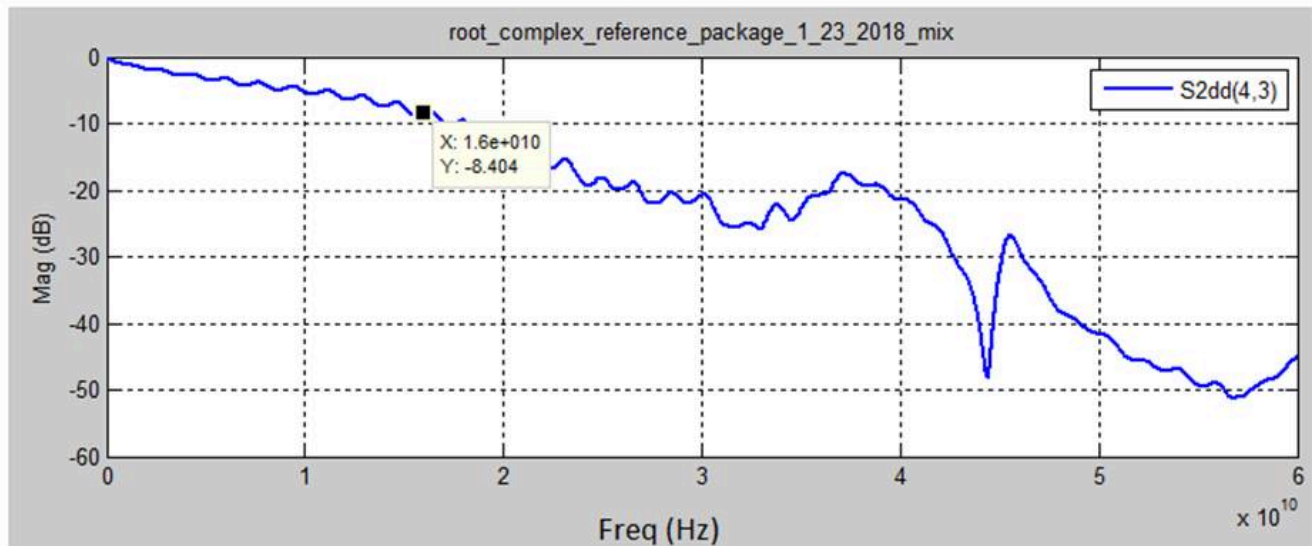


Figure 8-63 Insertion Loss for Root Reference Package for 32.0 GT/s §

Return Loss (Board-side)

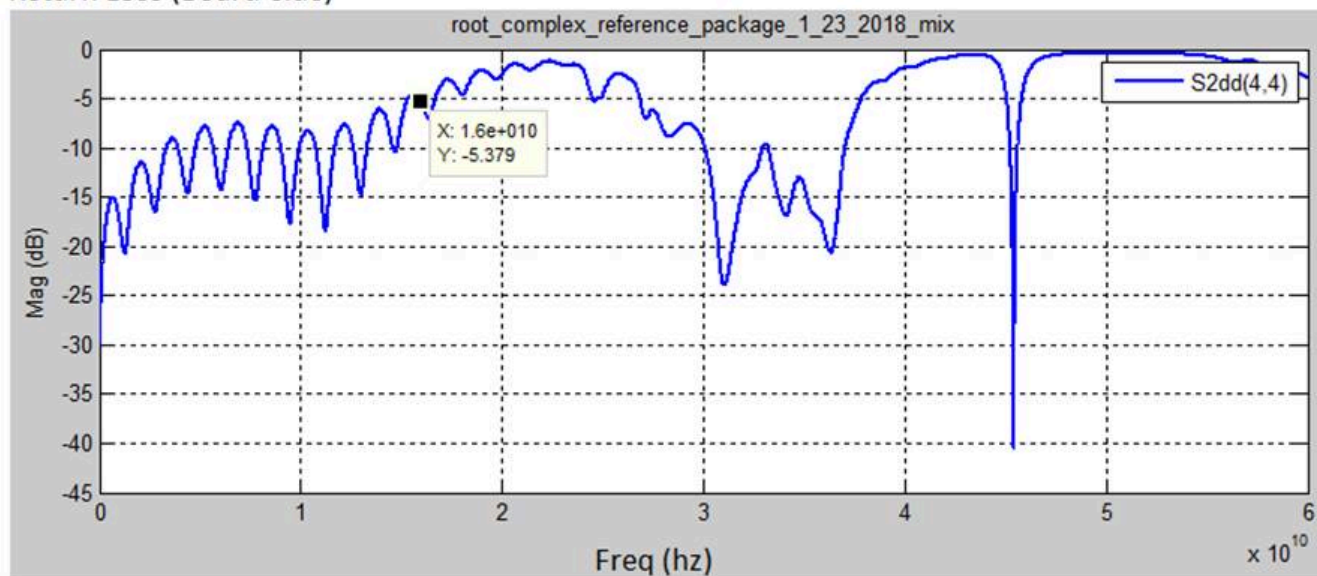


Figure 8-64 Return Loss for Root Reference Package for 32.0 GT/s §

NEXT 2

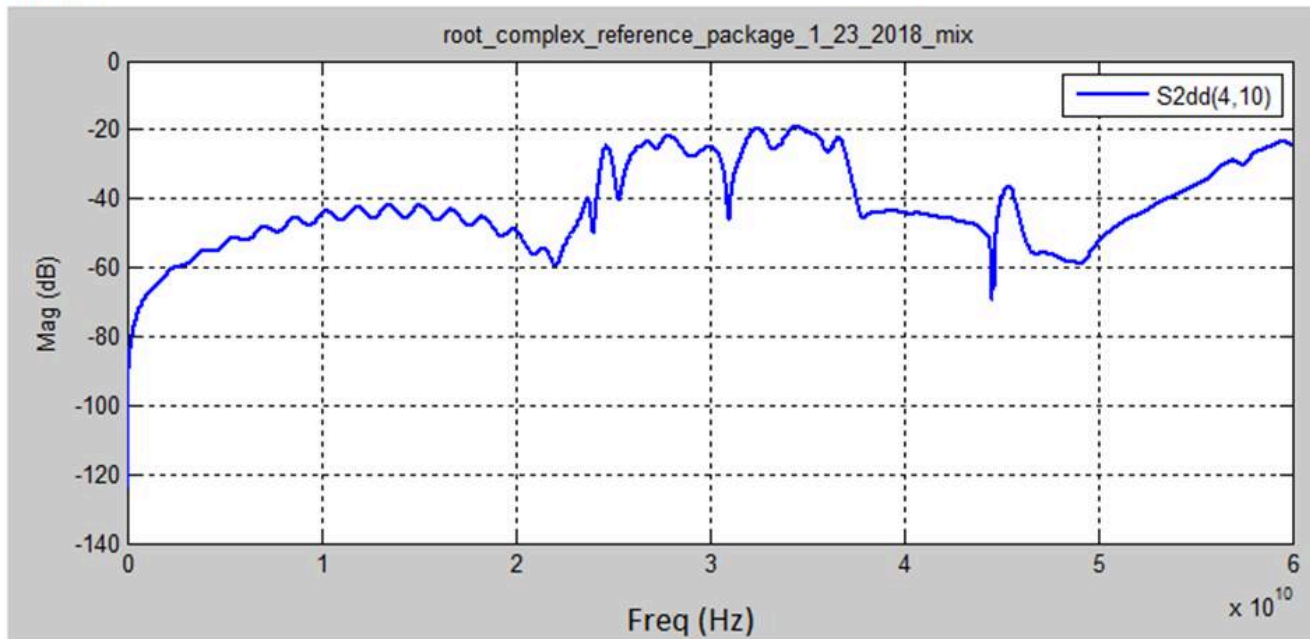


Figure 8-65 NEXT for Root Reference Package (Worst-Case) for 32.0 GT/s §

FEXT 1

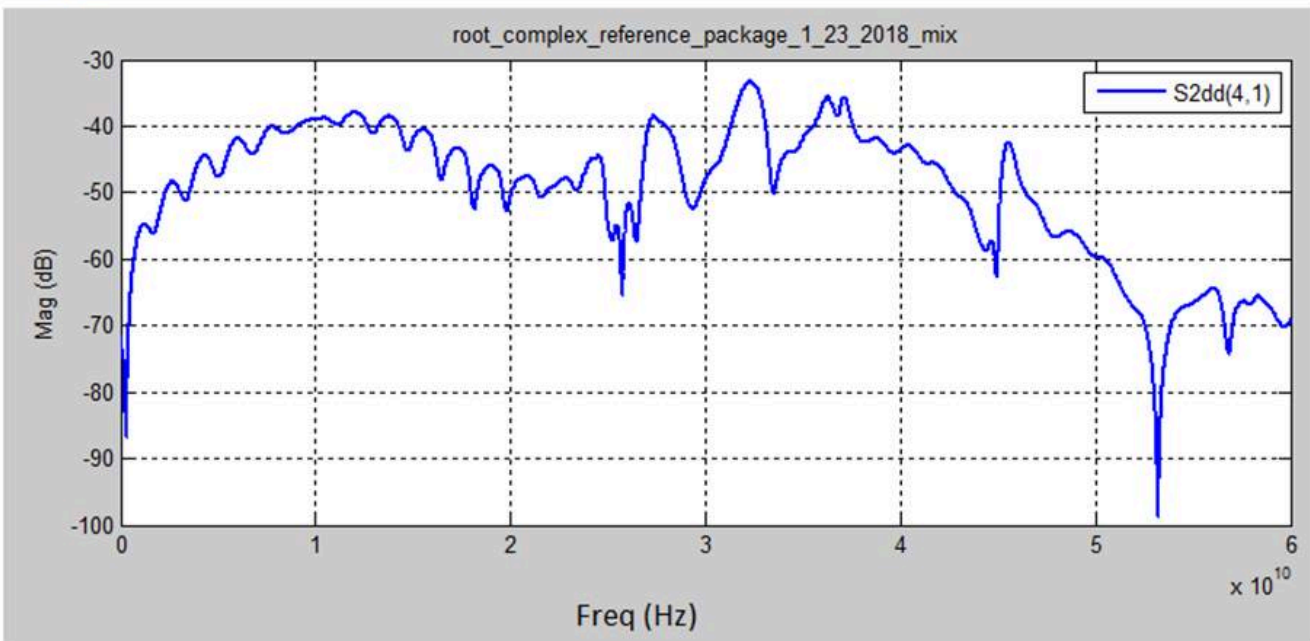


Figure 8-66 FEXT for Root Reference Package (Worst-Case) for 32.0 GT/s §

Insertion loss

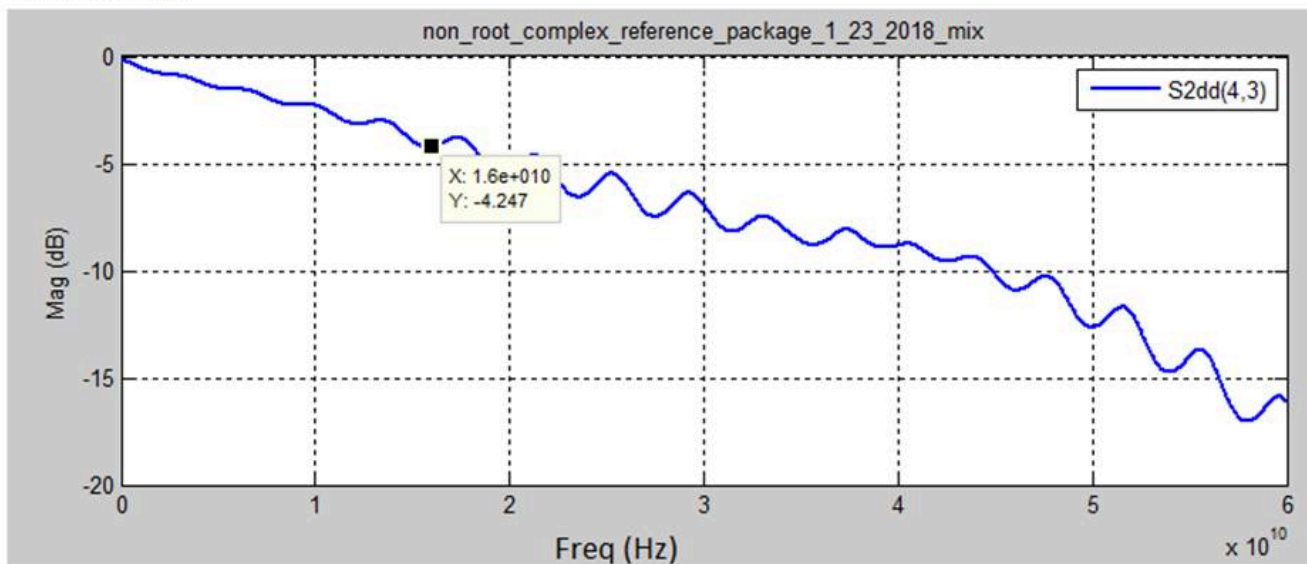


Figure 8-67 Insertion Loss for Non-Root Reference Package for 32.0 GT/s §

Return Loss (Board-side)

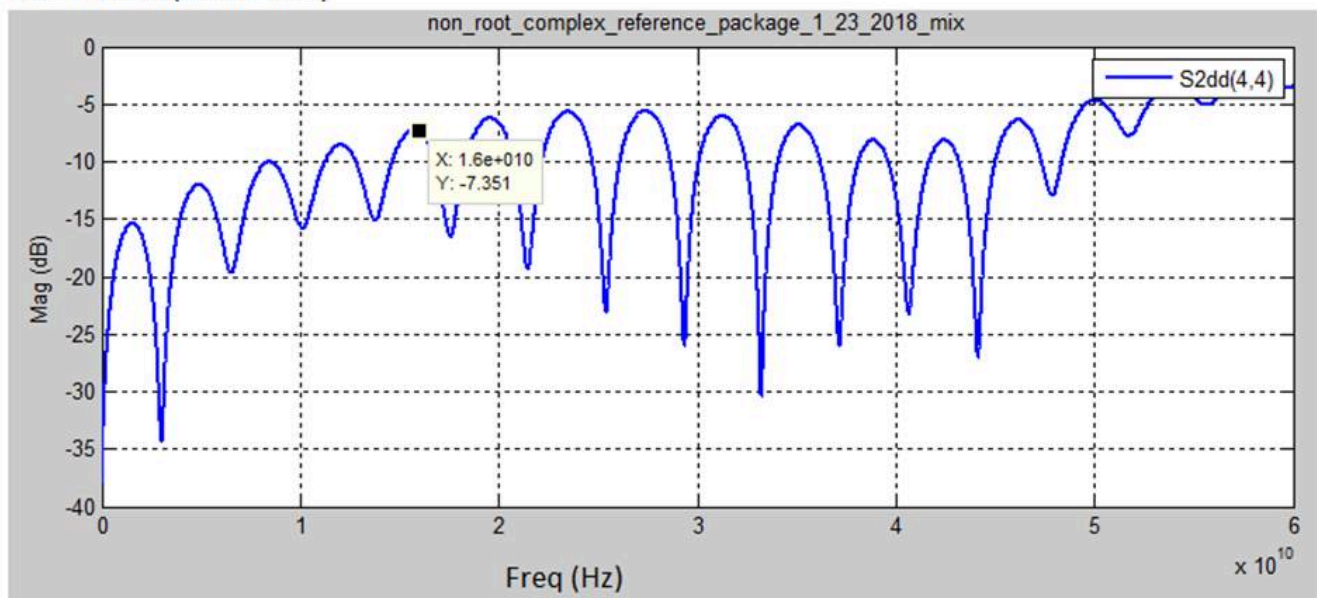


Figure 8-68 Return Loss for Non-Root Reference Package for 32.0 GT/s §

NEXT 1

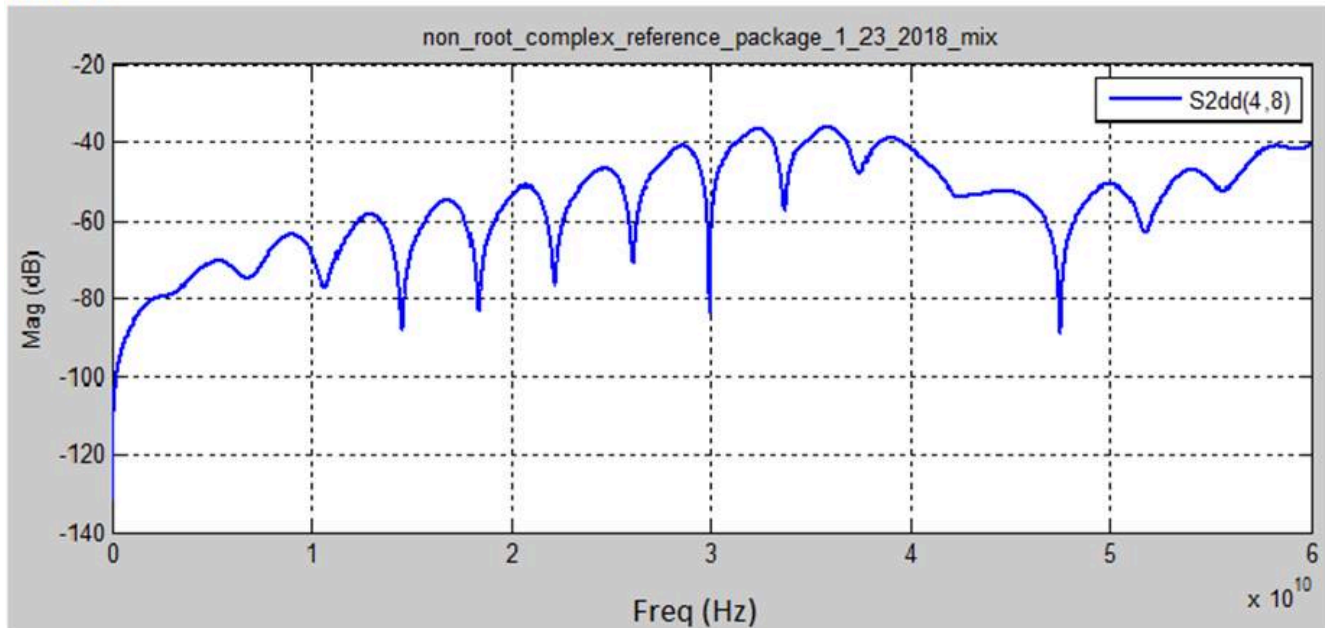


Figure 8-69 NEXT for Non-Root Reference Package (Worst-Case) for 32.0 GT/s §

FEXT 2 (Die-side)

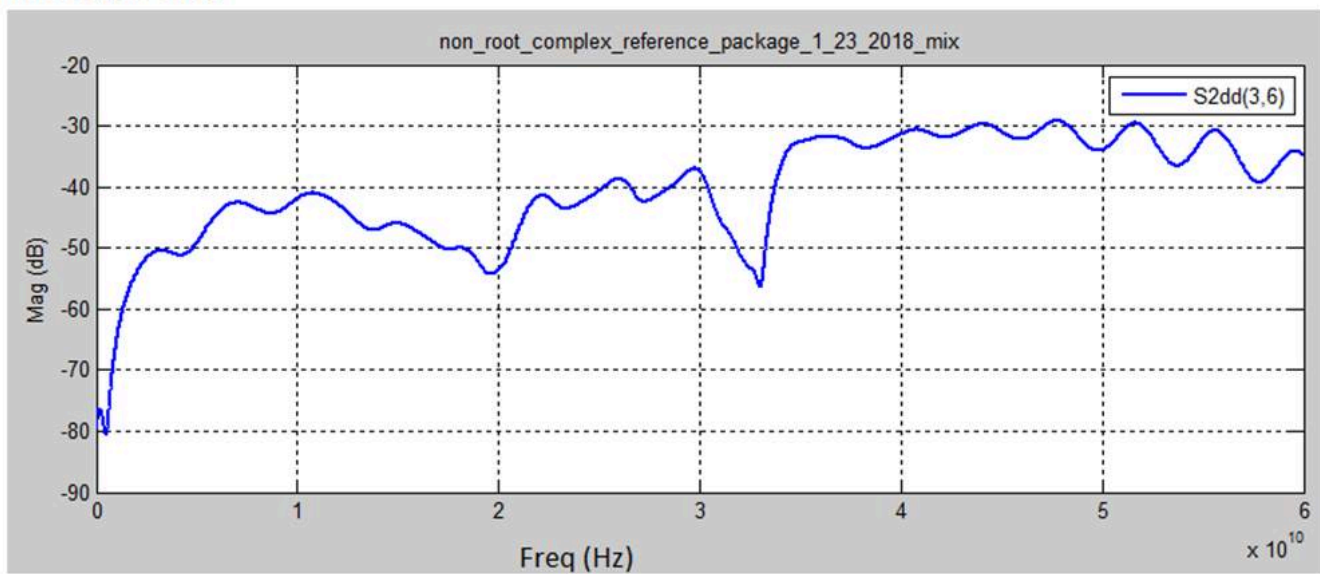


Figure 8-70 FEXT for Non-Root Reference Package (Worst-Case) for 32.0 GT/s §

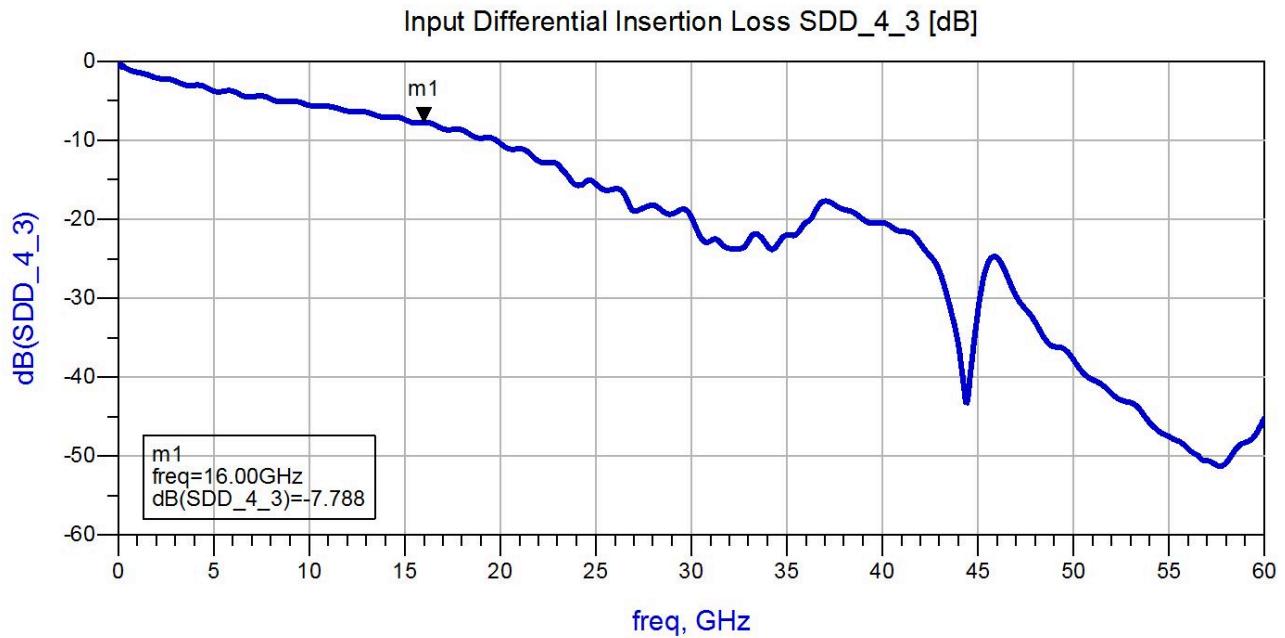


Figure 8-71 Insertion Loss for Root Reference Package for 64.0 GT/s §

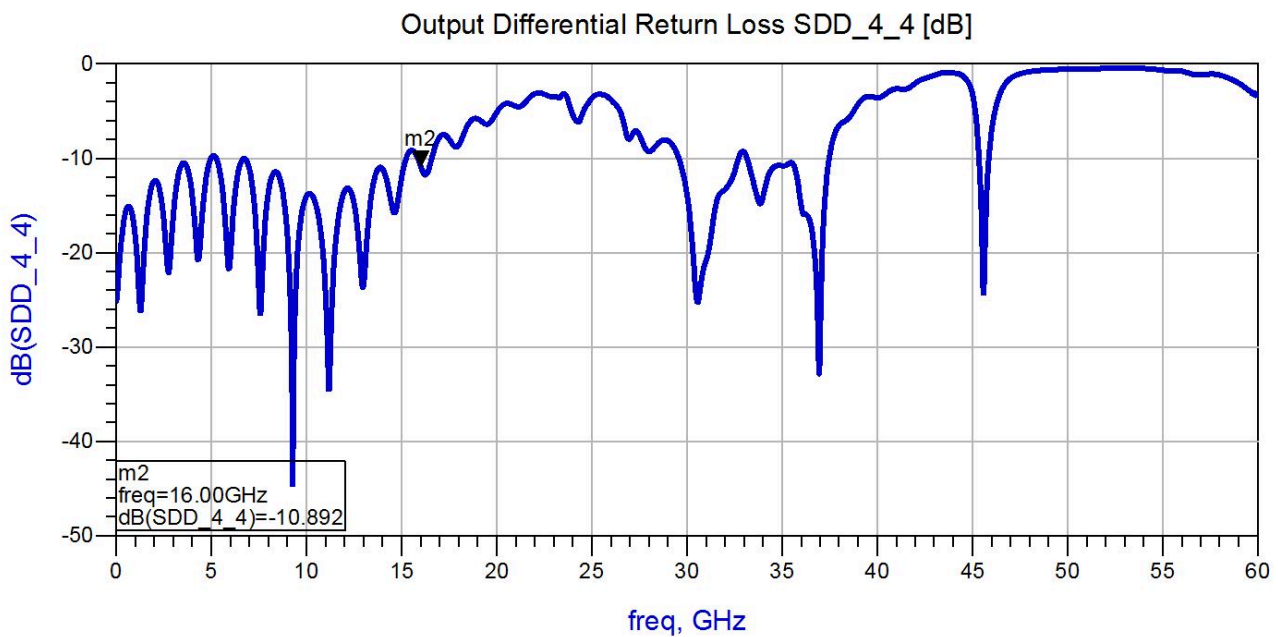


Figure 8-72 Return Loss for Root Reference Package for 64.0 GT/s §

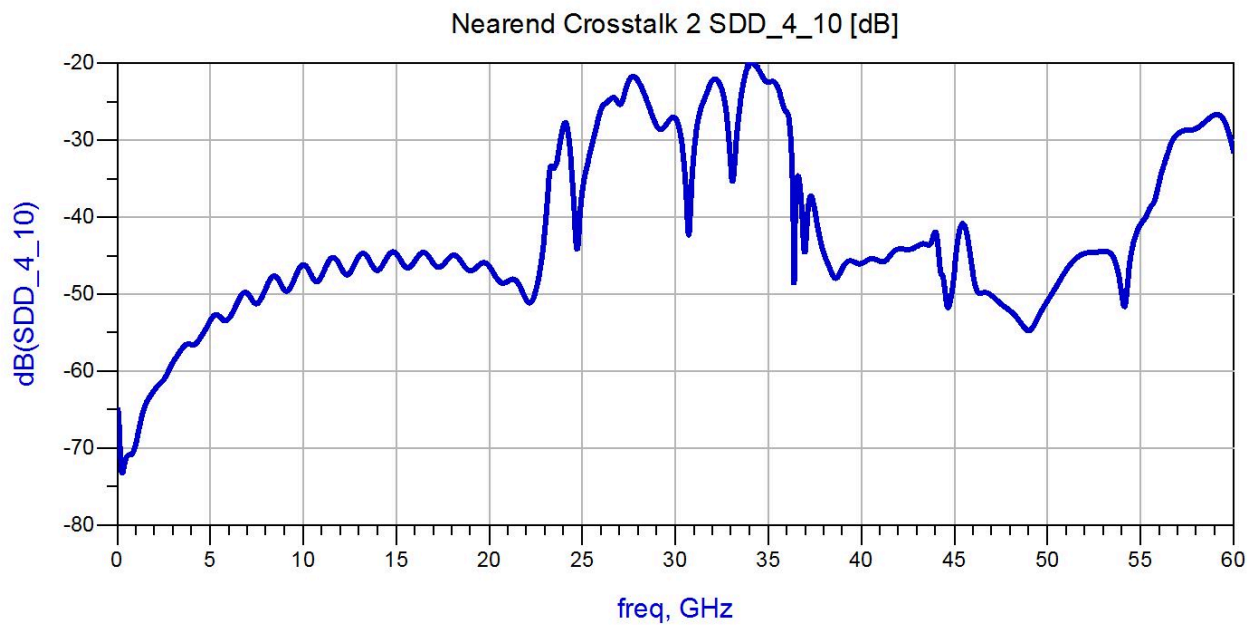


Figure 8-73 NEXT for Root Reference Package (Worst Case) for 64.0 GT/s §

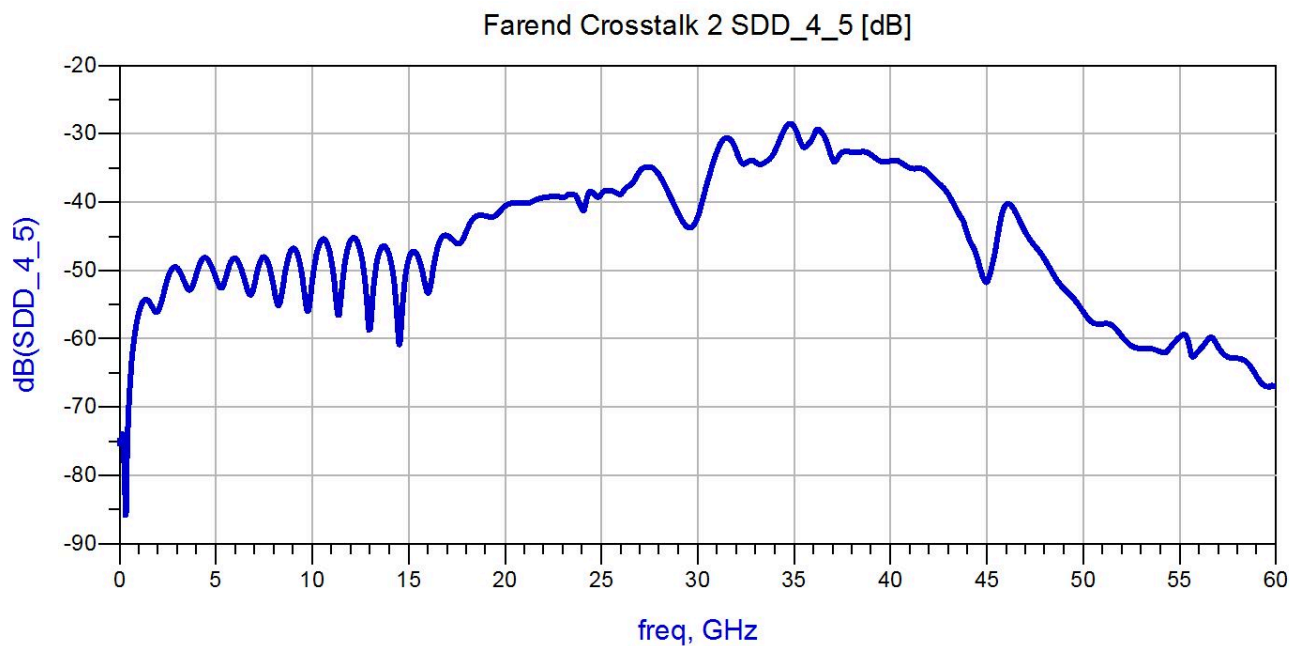


Figure 8-74 FEXT for Root Reference Package (Worst Case) for 64.0 GT/s §

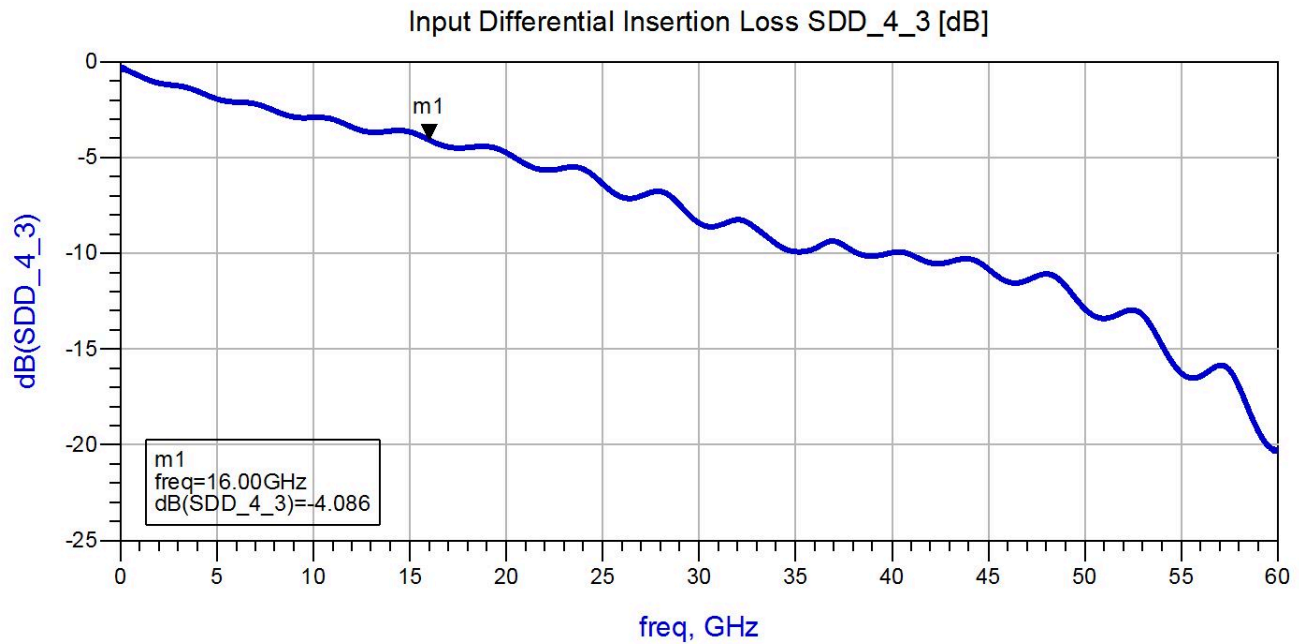


Figure 8-75 Insertion Loss for Non-Root Reference Package for 64.0 GT/s §

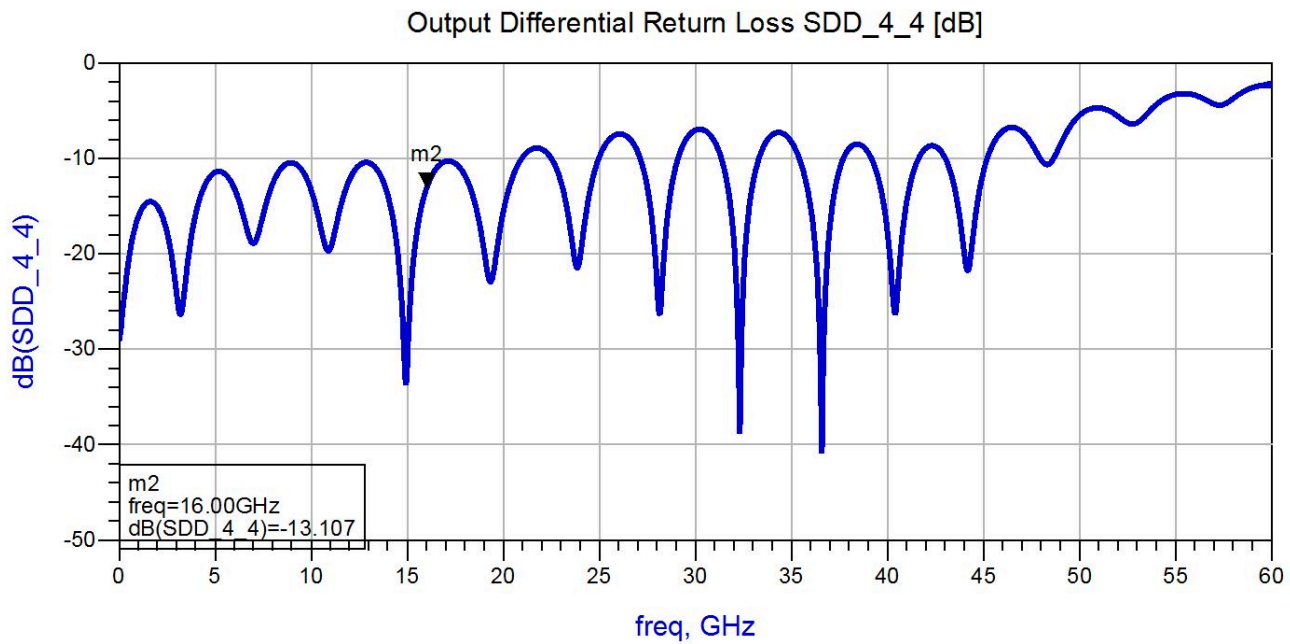


Figure 8-76 Return Loss for Non-Root Reference Package for 64.0 GT/s §

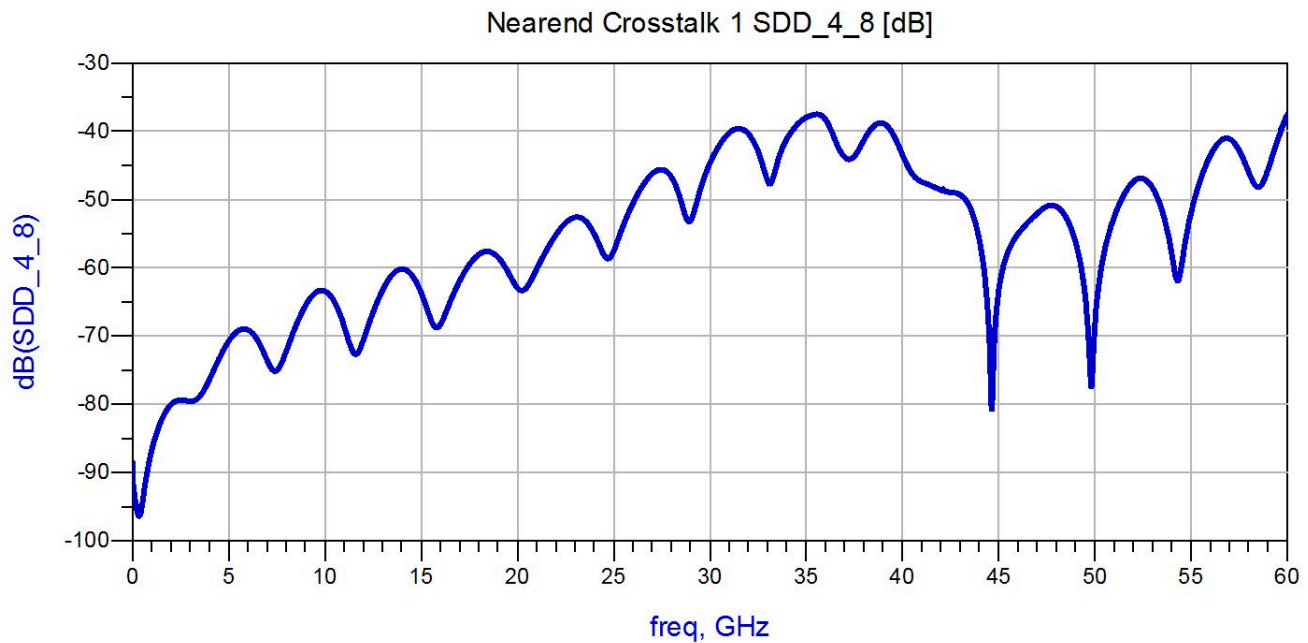


Figure 8-77 NEXT for Non-Root Reference Package (Worst Case) for 64.0 GT/s §

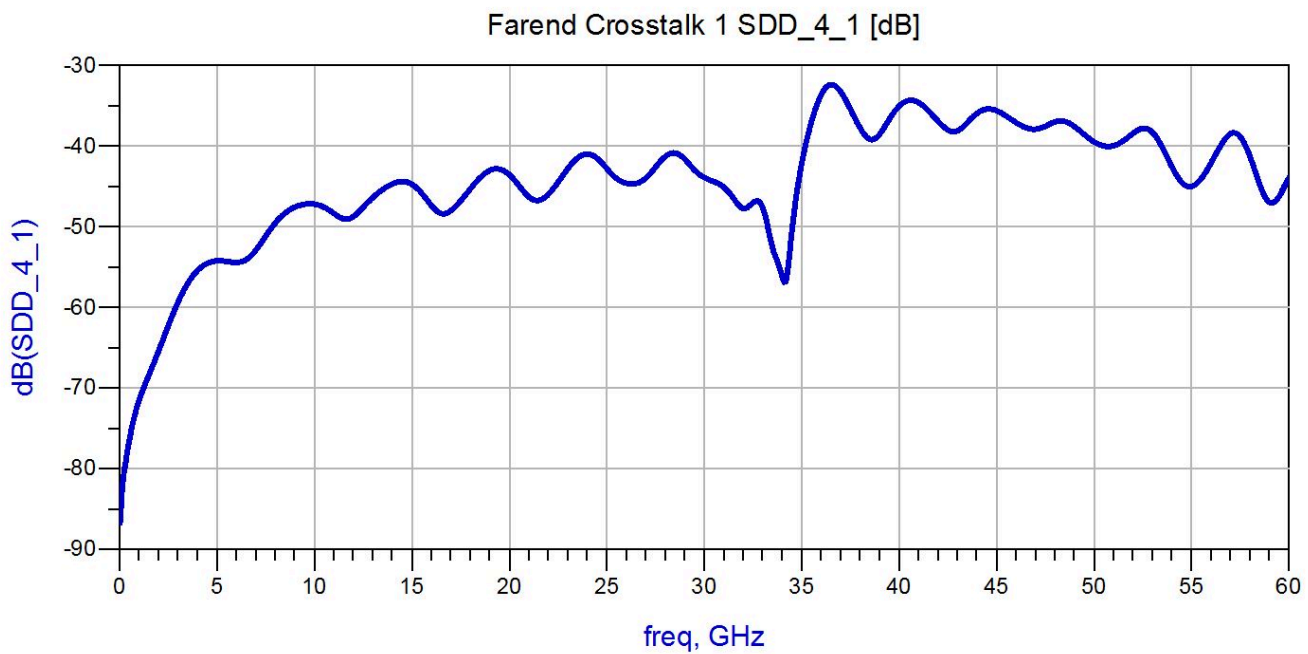


Figure 8-78 FEXT for Non-Root Reference Package (Worst Case) for 64.0 GT/s §

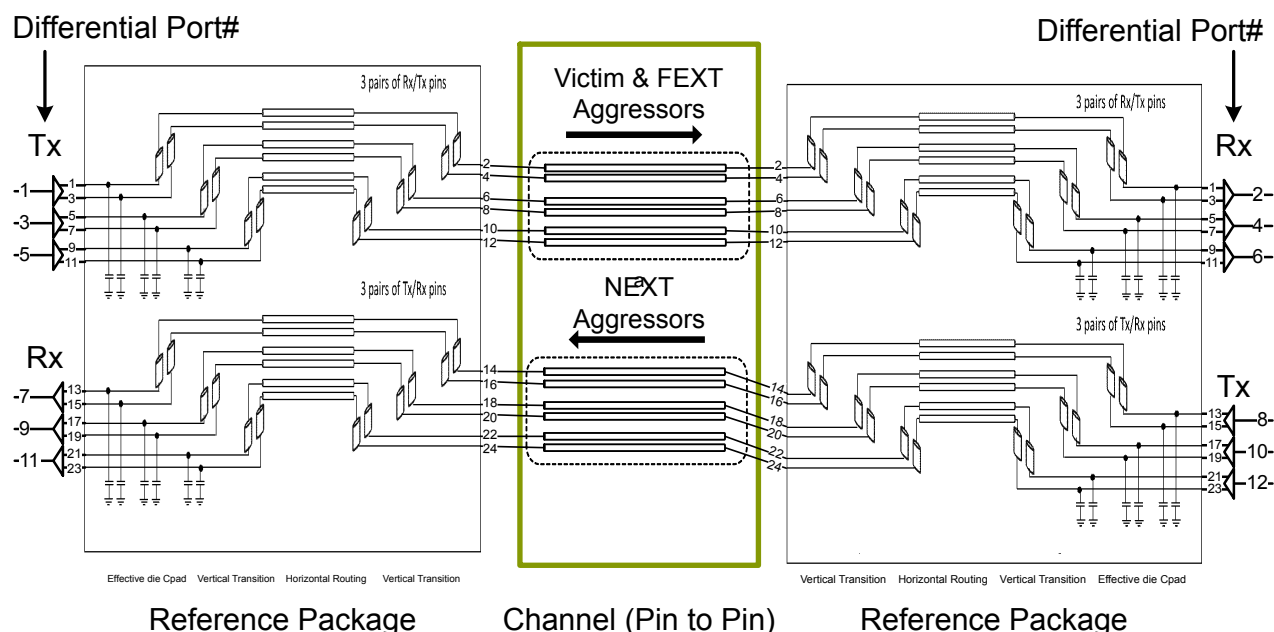


Figure 8-79 32.0 and 64.0 GT/s Reference Package Port Connections for Pin to Pin Channel Evaluation §

§ Figure 8-79 shows the port connections for using the 32.0 and 64.0 GT/s reference packages to evaluate a pin to pin channel using the channel compliance methodology. Both directions (root transmitting and non-root transmitting) must be evaluated. The lane labeled channel 3 to channel 4 is intended as the worst-case victim channel and must be evaluated in both directions. Other channels in the reference package model are intended only for use as crosstalk aggressors (not victim channels).

8.5.1.2 Measuring Package Performance (16.0 GT/s only) §

Package insertion loss at 16.0 GT/s is an informative spec parameter. Some implementations at 16.0 GT/s (see § Section 8.3.3.11) are allowed to have packages that exceed reference packages in insertion loss and/or crosstalk. Actual package performance must be assessed by performing channel compliance with reference channels provided with the specification on the PCI-SIG website. A set of channel compliance simulations are run on the reference channels with one of the reference packages being replaced by the package that is being evaluated. If the eye height or eye width is smaller for any of the channels with the package that is being evaluated, then the package is considered to have worse performance than the reference package. An implementation with a package that has worse performance than the reference package must use the implementation package model in the channel compliance methodology and may optionally use the implementation package in the Receiver stressed eye calibration. Note that form factor and channel compliance (for a captive channel) overall requirements still need to be met regardless of package characteristics.

8.5.1.3 Simulation Tool Requirements §

Channel tolerancing is implemented by means of simulation, where the pass/fail criteria are defined in terms of a time domain eye diagram. The simulation tool must accept a prescribed set of inputs, including the channel under

consideration and then simulate based upon a set of post processing requirements. This specification does not stipulate the use of any specific tool for simulating channels. However, any simulation tool must meet the following requirements.

8.5.1.3.1 Simulation Tool Chain Inputs §

- Channel characteristics defined as S-parameters or equivalent model: The model must include the victim differential Lane plus as many aggressors as required to accurately capture crosstalk. In most cases this will be between 2 and 4 additional differential Lanes. Note that 32.0 and 64.0 GT/s are most likely to require additional aggressors to accurately capture worst-case crosstalk and most likely to require several NEXT aggressors to capture crosstalk accurately.
- Behavioral Root and Non-Root package models: The models will be included as part of the specification in the form of s-parameter files (see § Section 8.5.1.1).
- Transmitter Jitter and voltage: The voltage and jitter parameters input to the simulator may be directly obtained from a combination of the Transmitter and Refclk jitter. Since these parameters are fixed the simulation tool may choose to hard code their values.
- Transmitter and Receiver Termination Impedance: The simulator shall use a $2 \times 50 \Omega$ termination for both the Transmitter and Receiver. This value matches the assumptions which are implicit in generating and measuring the stressed eye for Rx tolerancing.

8.5.1.3.2 Processing Steps §

- Time domain representation of the end-to-end connectivity: Included are the behavioral Tx and Rx packages and the channel under test.
- Tx voltage and jitter: Voltage and jitter parameters defined for the Transmitter but have been recalculated to properly comprehend high and low frequency jitter components, and also include Refclk jitter contributions.
- Behavioral Transmitter Equalization: The simulator shall replicate the Transmitter equalization capabilities defined in the Transmitter section.
- Behavioral Rx CTLE: The simulation tool shall implement a behavioral CTLE that replicates the CTLE function employed for Rx tolerancing.
- Behavioral DFE: The simulation tool shall implement a 1-tap (8.0 GT/s), a 2-tap (16.0 GT/s), a 3 tap DFE (32.0 GT/s), and a 16-tap DFE (64.0 GT/s), where the dynamic range for the feedback coefficient is defined in § Section 8.4.1.10.
- Optimizing Tx equalization and Rx DFE/CTLE settings: The simulation tool shall implement an optimization algorithm that selects the combination of Tx equalization and Rx CTLE and DFE settings that yields a maximum value for the eye height (at the data sample point) multiplied by the eye width at the far end of the channel. For details refer to § Section 8.4.1.8.
- Statistical Treatment of jitter: In order to avoid overestimating the effect of channel-data and channel-jitter interactions, the tool shall use a statistical analysis of these parameters to generate voltage/jitter eye margins.

8.5.1.3.3 Simulation Tool Outputs §

Output eye parameters: The simulator shall generate a statistically defined output that displays the eye width and eye height. EH will be measured as the peak eye height at the data sample location, while EW shall be measured at the zero-crossing line. Additionally the simulator shall have the capability to adjust the data sample point by $\pm 0.1 \text{ UI}$ from the mean center of the UI for 8.0 and 16.0 GT/s as shown in § Figure 8-81. For 32.0 GT/s the simulator shall adjust the sample point up to 0.30 UI to the left of the mean center of the UI sample position in 0.05 UI increments, computing the

DFE coefficients for each sample location and selecting the result producing the maximum value for the eye height (at the data sample point) multiplied by the eye width. For 64.0 GT/s the simulator shall adjust the sample point up to 0.30 UI to the left of the mean center of the UI sample position in 0.015 UI increments, computing the DFE coefficients for each sample location and selecting the result producing the maximum value for the eye height (at the data sample point) multiplied by the eye width.

8.5.1.3.4 Open Source Simulation Tool §

An open source simulation tool shall be provided with the specification as design collateral. The tool will provide a turnkey capability, where the user provides the channel characteristics at the Receiver's die pad as step responses, and the tool calculates a statistical eye showing pass/fail.

8.5.1.4 Behavioral Transmitter Parameters §

8.5.1.4.1 Deriving Voltage and Jitter Parameters §

This section is for informative purposes. The voltage and jitter parameters may be derived from the Transmitter voltage and jitter parameters but are referenced to the die pad. This is necessary to allow the channel simulation to include a behavioral Tx package and drive the package from the die pads. Additionally, the Tj terms must be decomposed into separate Rj and DjDD terms.

- $V_{TX-CH-FS-NO-EQ}$ and $V_{TX-CH-RS-NO-EQ}$: These two parameters define the minimum peak-peak voltage corresponding to Vd in § Figure 8-6.
- The jitter parameters are derived based on the following set of equations. Algebraic manipulation is used to extract the Rj implicitly defined by the combination of Tj and DjDD terms. The following numbers are based on 8.0 GT/s Tx jitter parameters based on values specified in § Table 8-6. The same approach is used to extract jitter parameters for 2.5, 5.0, 16.0, 32.0, and 64.0 GT/s where it is assumed that maximum data rate supported by the PCIe 6.0 device is 64.0 GT/s and the jitter values in § Table 8-6 are used in the extraction process.

$$\begin{aligned} \text{jit_hfrj_nui} &= (T_{TX-UTJ} - T_{TX-UDJ-DD})/14.06 = 1.11\text{ps} \\ T_{TX-CH-UPW-RJ} &= (T_{TX-UPWJ-TJ} - T_{TX-UPWJ-DJDD})/14.06 = 1.00\text{ps} \\ T_{TX-CH-UPW-DJ} &= T_{TX-UPWJ-DJDD} = 10.0\text{ps} \\ T_{TX-CH-URJ} &= \sqrt{(\text{jit_hfrj_nui})^2 - (T_{TX-CH-UPW-RJ} \cdot 0.707)^2} + T_{REFCLK-RMS} = 1.31\text{ps} \\ T_{TX-CH-UDJDD} &= T_{TX-UDJ-DD} - (T_{TX-UPWJ-DJDD})/2 = 7.00\text{ps} \end{aligned}$$

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Figure 8-80 Example Derivation of 8.0 GT/s Jitter Parameters for § Table 8-15 §

A channel must be tested at all data rates, with the corresponding Tx jitter parameters, that it is intended to support during normal operation. For example, a channel intended to support a maximum data rate of 8.0 GT/s must be tested at 2.5, 5.0, and 8.0 GT/s.

Table 8-15 Jitter/Voltage Parameters for Channel Tolerancing §

Symbol	Parameter	Value	Units	Notes
$V_{TX-CH-FS-NO-EQ}$	Full swing Tx voltage	804	mVPP	Full swing, No Tx Eq.

Symbol	Parameter	Value	Units	Notes
$V_{TX-CH-RS-NO-EQ}$	Reduced swing Tx voltage	402	mVPP	Reduced swing, No Tx Eq.

2.5 GT/s Jitter Parameters and Voltage Parameters

$T_{TX-CH-URJ-2.5G}$	Tx uncorrelated Rj	3.45	ps RMS	See Note 1
$T_{TX-CH-UDJDD-2.5G}$	Tx uncorrelated DjDD	20	ps PP	
$T_{TX-CH-UPW-RJ-2.5G}$	Uncorrelated PW Rj	1.42	ps RMS	See Note 2
$T_{TX-CH-UPW-DJ-2.5G}$	PW DDj	80	ps PP	
$T_{TX-DIEPAD-EDGERATE-2.5G}$	Signal edge rate at behavioral Tx die pad	140	ps	Measured 10% to 90% using a gaussian lowpass filter to shape the edge. See Note 3.

5.0 GT/s Jitter Parameters and Voltage Parameters

$T_{TX-CH-URJ-5G}$	Tx uncorrelated Rj	3.45	ps RMS	See Note 1
$T_{TX-CH-UDJDD-5G}$	Tx uncorrelated DjDD	20-	ps PP	
$T_{TX-CH-UPW-RJ-5G}$	Uncorrelated PW Rj	1.42	ps RMS	
$T_{TX-CH-UPW-DJ-5G}$	PW DDj	40	ps PP	See Note 2.
$T_{TX-DIEPAD-EDGERATE-5G}$	Signal edge rate at behavioral Tx die pad	70	ps	Measured 10% to 90% using a gaussian lowpass filter to shape the edge. See Note 3.

8.0 GT/s Jitter Parameters and Voltage Parameters

$T_{TX-CH-URJ-8G}$	Tx uncorrelated Rj	1.31	ps RMS	No DDj of HF jitter. See Note 1.
$T_{TX-CH-UDJDD-8G}$	Tx uncorrelated DjDD	7.0	ps PP	No DDj of HF jitter
$T_{TX-CH-UPW-RJ-8G}$	Uncorrelated PW Rj	1.0	ps RMS	
$T_{TX-CH-UPW-DJ-8G}$	PW DDj	10	ps PP	See Note 2.
$T_{TX-DIEPAD-EDGERATE-8G}$	Signal edge rate at behavioral Tx die pad	43.75	ps	Measured 10% to 90% using a gaussian lowpass filter to shape the edge. See Note 3.

16.0 GT/s Jitter Parameters and Voltage Parameters

$T_{TX-CH-URJ-16G}$	Tx uncorrelated Rj	0.71	ps RMS	See Note 1.
$T_{TX-CH-UDJDD-16G}$	Tx uncorrelated DjDD	3.75	ps PP	
$T_{TX-CH-UPW-RJ-16G}$	Uncorrelated PW Rj	0.54	ps RMS	
$T_{TX-CH-UPW-DJ-16G}$	PW DDj	5.0	ps PP	See Note 2.
$T_{TX-DIEPAD-EDGERATE-16G}$	Signal edge rate at behavioral Tx die pad	21.875	ps	Measured 10% to 90% using a gaussian lowpass filter to shape the edge. See Note 3.

Symbol	Parameter	Value	Units	Notes
32.0 GT/s Jitter Parameters and Voltage Parameters				
$T_{TX-CH-URJ-32G}$	Tx uncorrelated Rj	0.276	ps RMS	See Note 1.
$T_{TX-CH-UDJDD-32G}$	Tx uncorrelated DjDD	1.875	ps PP	
$T_{TX-CH-UPW-RJ-32G}$	Uncorrelated PW Rj	0.27	ps RMS	
$T_{TX-CH-UPW-DJ-32G}$	PW DDj	2.5	ps PP	See Note 2.
$T_{TX-DIEPAD-EDGERATE-32G}$	Signal edge rate at behavioral Tx die pad	10.94	ps	Measured 10% to 90% using a gaussian lowpass filter to shape the edge. See Note 3.

64.0 GT/s Jitter Parameters and Voltage Parameters

$T_{TX-CH-URJ-64G}$	Tx uncorrelated Rj	0.215	ps RMS	See Note 1.
$T_{TX-CH-UDJDD-64G}$	Tx uncorrelated DjDD	0.938	ps PP	
$T_{TX-CH-UPW-RJ-64G}$	Uncorrelated PW Rj	0.289	ps RMS	
$T_{TX-CH-UPW-DJ-64G}$	PW DDj	1.25	ps PP	See Note 2.
$T_{TX-DIEPAD-EDGERATE-64G}$	Signal edge rate at behavioral Tx die pad	10.94	ps	Measured 10% to 90% using a gaussian lowpass filter to shape the edge. See Note 3.

Notes:

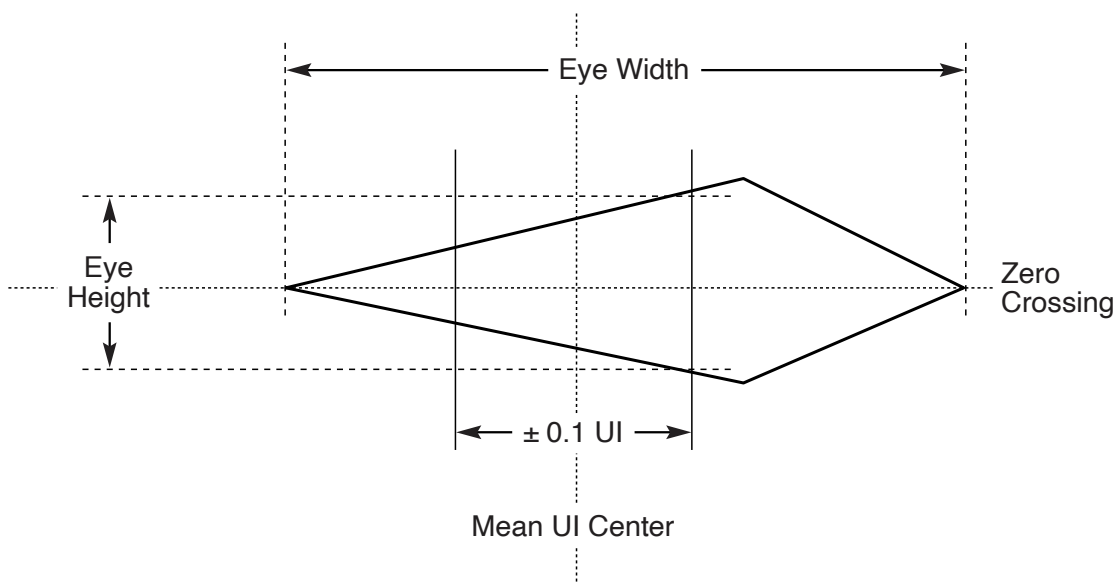
1. Includes low frequency (non F/2) Rj components from the Transmitter and Rj from the Refclk.
2. Applied on a per edge basis as a dual Dirac model.
3. Does not include parasitic die pad capacitance. See § Figure 8-60 for details of behavioral package.

8.5.1.4.2 Optimizing Tx/Rx Equalization (8.0, 16.0, 32.0, and 64.0 GT/s only) §

The behavioral Receiver selects the combination of Transmitter Equalization, CTLE, DFE and sample location (32.0 and 64.0 GT/s only) that produces the optimal eye area (eye width multiplied by eye height).

8.5.1.4.3 Pass/Fail Eye Characteristics §

The output of the simulation tool shall be in the form of pass/fail characteristics as defined by an eye mask as shown in § Figure 8-81. EH and EW must meet respectively the voltage and jitter parameters defined in § Table 8-16. Eye margins are defined at the die pad of the Receiver after the appropriate Tx and Rx equalization algorithms have been applied. In the case where the channel is being designed for a specific pair of silicon devices and the package models of these silicon devices including crosstalk aggressors are known the actual device packages may be used instead of the reference packages in running the pad to pad channel pass/fail compliance simulations.



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Figure 8-81 EH, EW Mask §

Note that the pass/fail EH and EW limits shown in § Figure 8-81 are identical to the limits defined for Rx testing in § Table 8-11 for 8.0 and 16.0 GT/s. For 32.0 and 64.0 GT/s, the pass/fail EH and EW limits are still identical but they are computed at the optimal sample location. For 2.5 and 5.0 GT/s the limits for Rx testing are referenced to the package pins and the limits for channel tolerancing in this table are referenced to the Receiver pad after applying the same reference package used for 8.0 GT/s channel tolerancing.

Table 8-16 Channel Tolerancing Eye Mask Values §

Symbol	Parameter	Value	Units	Comments
2.5 GT/s Eye Margins				
$V_{RX-CH-EH-2.5G}$	Eye height	<130 (min)	mVPP	Eye height at BER= 10^{-12} . Note 1
$T_{RX-CH-EW-2.5G}$	Eye width at zero-crossing	<0.35 (min)	UI	Eye width at BER= 10^{-12}
$T_{RX-DS-OFFSET-2.5G}$	Peak EH offset from UI center	± 0.1	UI	See § Figure 8-81 for details.
5.0 GT/s Eye Margins				
$V_{RX-CH-EH-5G}$	Eye height	< 85 (min)	mVPP	Eye height at BER= 10^{-12} . Note 1
$T_{RX-CH-EW-5G}$	Eye width at zero-crossing	<0.30 (min)	UI	Eye width at BER= 10^{-12}
$T_{RX-DS-OFFSET-5G}$	Peak EH offset from UI center	± 0.1	UI	See § Figure 8-81 for details.
8.0 GT/s Eye Margins				

Symbol	Parameter	Value	Units	Comments
$V_{RX-CH-EH-8G}$	Eye height	25 (min)	mVPP	Eye height at BER=10 ⁻¹² . Note 1.
$T_{RX-CH-EW-8G}$	Eye width at zero-crossing	0.3 (min)	UI	Eye width at BER=10 ⁻¹²
$T_{RX-DS-OFFSET-8G}$	Peak EH offset from <u>UI</u> center	±0.1	UI	See § Figure 8-81 for details.
$V_{RX-DFE-D1-8G}$	Range for DFE d ₁ coefficient	±30	mV	

16.0 GT/s Eye Margins

$V_{RX-CH-EH-16G}$	Eye height	15 (min)	mVPP	Eye height at BER=10 ⁻¹² . Note 1.
$T_{RX-CH-EW-16G}$	Eye width at zero-crossing	0.3 (min)	UI	Eye width at BER=10 ⁻¹²
$T_{RX-DS-OFFSET-16G}$	Peak EH offset from <u>UI</u> center	±0.1	UI	See § Figure 8-81 for details.
$V_{RX-DFE-D1-16G}$	Range for DFE d ₁ coefficient	±30	mV	
$V_{RX-DFE-D2-16G}$	Range for DFE d ₂ coefficient	±20	mV	

32.0 GT/s Eye Margins

$V_{RX-CH-EH-32G}$	Eye height	15 (min)	mVPP	Eye height at BER=10 ⁻¹² . Note 1.
$T_{RX-CH-EW-32G}$	Eye width at zero-crossing	0.3 (min)	UI	Eye width at BER=10 ⁻¹²
$T_{RX-DS-OFFSET-32G}$	Peak EH offset from <u>UI</u> center	N/A	UI	See § Figure 8-81 for details.
$T_{RX-SAMPLE-OFFSET-32G}$	Max sample location offset to the left from <u>UI</u> center	0.30	UI	Note 2
$T_{RX-SAMPLE-GRANULARITY-32G}$	Granularity for sample location offset	0.05	UI	Note 2
$V_{RX-DFE-D1-32G}$	Range for DFE d ₁ coefficient	d ₁ / d ₀ (cursor amplitude) ≤ 0.8		
$V_{RX-DFE-D2-32G}$	Range for DFE d ₂ coefficient	±20	mV	
$V_{RX-DFE-D3-32G}$	Range for DFE d ₃ coefficient	±20	mV	

64.0 GT/s Eye Margins

$V_{RX-CH-TOP-EH-64G}$	Top Eye height	6 (min)	mVPP	Eye height at BER=10 ⁻⁶ . Note 1.
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Symbol	Parameter	Value	Units	Comments
<i>T_{RX-CH-TOP-EW-64G}</i>	Top Eye width at zero-crossing	0.1 (min)	UI	Eye width at BER=10 ⁻⁶
<i>T_{RX-DS-OFFSET-64G}</i>	Peak EH offset from UI center	N/A	UI	
<i>T_{RX-SAMPLE-OFFSET-64G}</i>	Max sample location offset to the left from UI center	0.30	UI	Note 2
<i>T_{RX-SAMPLE-GRANULARITY-64G}</i>	Granularity for sample location offset	0.015	UI	Note 2
<i>V_{RX-DFE-D1-64G}</i>	Range for DFE d1 coefficient	$ d_1 / d_0$ (cursor amplitude) < 0.55		
<i>V_{RX-DFE-TAPS-WEIGHTED-SUM-64G}</i>	Range for weighted sum of absolute values of DFE d1-d16 coefficients	$(d_1 + d_2 + 0.85 \times d_3 + 0.6 \times d_4 + 0.25 \times d_5 + 0.1 \times d_6 + 0.05 \times \{ d_7 + d_8 + d_9 + d_{10} + d_{11} + d_{12} + d_{13} + d_{14} + d_{15} + d_{16} \}) / d_0 < 0.85$	mV	

Notes:

1. ***V_{RX-CH-EH}*** is defined as max EH within an aperture of ± 0.1 UI from mean UI center.
2. The optimal eye area is computed at each offset from mean UI center -0.30 UI to mean UI center with the specified granularity.

8.5.1.4.4 Characterizing Channel Common Mode Noise §

A channel must meet the common mode requirements as they are defined in the Receiver specification. In general, it is not possible to accurately simulate all the channel's common mode noise contributions due to the large number of mechanisms that can generate CM noise, including the Transmitter. Typically channel common mode noise is a budgeted parameter, and the limits defined below assume a budgeting process. The channel's CM limit is defined as the amount of CM noise that a channel can add and still meet the Rx CM limits assuming the worst-case Tx CM.

Note that the Tx and channel CM noise parameters cannot simply be added to obtain the Rx CM limit. This is due to the fact that a channel will attenuate some of high frequency Tx CM noise while propagating Tx LF CM noise through with little loss. The channel may also contribute both high and low frequency CM components of its own.

8.5.1.4.5 Verifying $V_{CH-IDLE-DET-DIFF-pp}$ §

$V_{CH-IDLE-DET-DIFF-pp}$ is defined to guarantee that, when a Transmitter issues an EIEOS sequence, the Receiver is guaranteed to detect it. Potentially larger Transmitter equalization boost ratios at 8.0, 16.0, 32.0, and 64.0 GT/s necessitate that this parameter be verified; this procedure was not necessary for 2.5 or 5.0 GT/s, where the max Transmitter equalization boost is smaller. Defining the launch and detect voltages at the Tx/Rx die pad permits $V_{CH-IDLE-DET-DIFF-pp}$ to be verified with the same channel and Tx/Rx package models used to determine eye margins. It is also acceptable to simulate from Tx pin to Rx pin (excluding the Tx and Rx behavioral package models), in which case the EIEOS and idle detect parameters defined in the Tx and Rx sections are applicable.

Long channels, where $V_{TX-EIEOS-FS}$ is applicable, are characterized by driving the channel under test with the EIEOS pattern and -11.0 dB de-emphasis and zero dB preshoot. For short channels, where $V_{TX-EIEOS-RS}$ is applicable, -4.5 dB of de-emphasis and zero dB of preshoot are applied.

Table 8-17 EIEOS Signaling Parameters §

Parameter	Description	Value	Units	Comments
$V_{CH-IDLE-EXIT-pp}$	Idle detect voltage seen at the Rx die pad	172	mVPP	Assuming Rx RTERM of $2 \times 50 \Omega$
$V_{CH-EIEOS-FS-Vb}$	PP voltage during Vb interval at behavioral Tx die pad for full swing signaling	255	mVPP	Assuming Tx RS of $2 \times 50 \Omega$
$V_{CH-EIEOS-RS-Vb}$	PP voltage during Vb interval at behavioral Tx die pad for reduced swing signaling	237	mVPP	Assuming Tx RS of $2 \times 50 \Omega$

8.6 Refclk Specifications §

This version of the specification consolidates and streamlines the Refclk requirements. The 2.5 GT/s Refclk parameters are moved from the CEM spec to this spec so that all Refclk parameters for all data rates (2.5, 5.0, 8.0, 16.0, 32.0, and 64.0 GT/s) are now contained in this section.

8.6.1 Refclk Test Setup §

The test setup for the Refclk assumes that only the Refclk generator itself is present. Provision is made in the test setup to account for signal degradation that occurs between the pins of the Refclk generator and the Transmitter or Receiver in an actual system. The above described setup emulates the worst case signal degradation that is likely to occur at the pins of a PCI Express device. Note that the Refclk signal is tested into a load that represents the series (open) termination appearing at the Refclk input pins of a PCIe device for all requirements except 32.0 and 64.0 GT/s reference clock jitter. For 32.0 and 64.0 GT/s, the reference clock jitter is measured with an oscilloscope, and is tested with the reference clock terminated by 50 Ohm terminations without a channel.

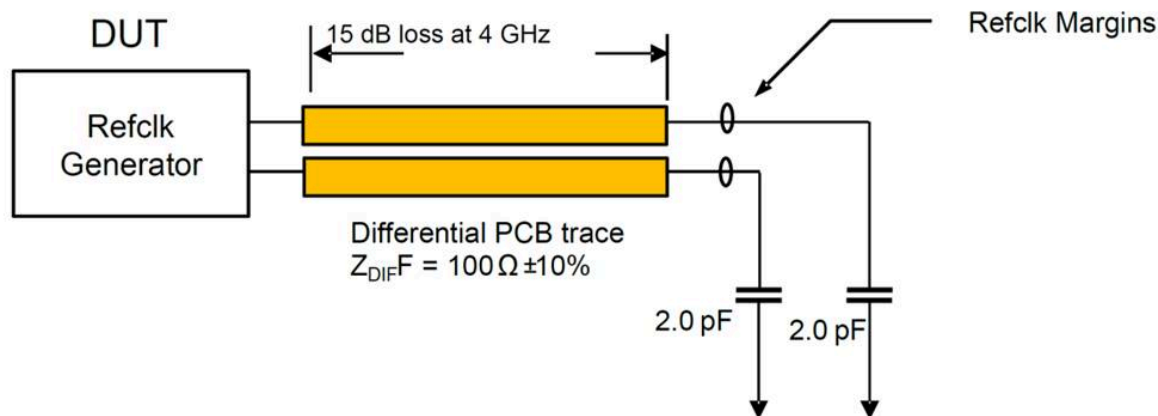


Figure 8-82 Oscilloscope Refclk Test Setup for All Cases Except Jitter at 32.0 and 64.0 GT/s §

8.6.2 REFCLK AC Specifications §

All specifications in § Table 8-18 are to be measured using a test configuration as described in Note 11 with a circuit as shown in § Figure 8-82.

Table 8-18 REFCLK DC Specifications and AC Timing Requirements §

Symbol	Parameter	100 MHz Input		Unit	Note
		Min	Max		
Rising Edge Rate	Rising Edge Rate	0.6	4.0	V/ns	2, 3
Falling Edge Rate	Falling Edge Rate	0.6	4.0	V/ns	2, 3
V_{IH}	Differential Input High Voltage	+150		mV	2
V_{IL}	Differential Input Low Voltage		-150	mV	2
V_{CROSS}	Absolute crossing point voltage	+250	+550	mV	1, 4, 5
V_{CROSS DELTA}	Variation of <u>V_{CROSS}</u> over all rising clock edges		+140	mV	1, 4, 9
V_{RB}	Ring-back Voltage Margin	-100	+100	mV	2, 12
T_{STABLE}	Time before <u>V_{RB}</u> is allowed	500		ps	2, 12
T_{PERIOD AVG}	Average Clock Period Accuracy	-300	+2800	ppm	2, 10, 13
T_{PERIOD AVG_32G_64G_CC}	Average Clock Period Accuracy for devices that support 32.0 and 64.0 GT/s in CC Mode at any speed	-100	+2600	ppm	2, 10, 13

Symbol	Parameter	100 MHz Input		Unit	Note
		Min	Max		
<i>T_{PERIOD}</i> <i>AVG_32G_64G_SRIS</i>	Average Clock Period Accuracy for devices that support 32.0 and 64.0 GT/s in SRIS Mode at any speed	-100	+1600	ppm	2, 10, 13
<i>T_{PERIOD}</i> <i>ABS</i>	Absolute Period (including Jitter and Spread Spectrum modulation)	9.847	10.203	ns	2, 6
<i>T_{PERIOD}</i> <i>ABS_32G_64G_CC</i>	Absolute Period (including Jitter and Spread Spectrum modulation) for devices that support 32.0 and 64.0 GT/s in CC Mode at any speed	9.849	10.201	ns	2, 6
<i>T_{PERIOD}</i> <i>ABS_32G_64G_SRIS</i>	Absolute Period (including Jitter and Spread Spectrum modulation) for devices that support 32.0 and 64.0 GT/s in SRIS Mode at any speed	9.849	10.181	ns	2, 6
<i>T_{CCJITTER}</i>	Cycle to Cycle jitter		150	ps	2
<i>V_{MAX}</i>	Absolute Max input voltage		+1.15	V	1, 7
<i>V_{MIN}</i>	Absolute Min input voltage		-0.3	V	1, 8
<i>Duty Cycle</i>	Duty Cycle	40	60	%	2
<i>Rise-Fall Matching</i>	Rising edge rate (REFCLK+) to falling edge rate (REFCLK-) matching		20	%	1, 14
<i>Z_{C-DC}</i>	Clock source DC impedance	40	60	Ω	1, 11

Notes:

1. Measurement taken from single ended waveform.
2. Measurement taken from differential waveform.
3. Measured from -150 mV to +150 mV on the differential waveform (derived from REFCLK+ minus REFCLK-). The signal must be monotonic through the measurement region for rise and fall time. The 300 mV measurement window is centered on the differential zero-crossing. See § Figure 8-87.
4. Measured at crossing point where the instantaneous voltage value of the rising edge of REFCLK+ equals the falling edge of REFCLK-. See § Figure 8-83.
5. Refers to the total variation from the lowest crossing point to the highest, regardless of which edge is crossing. Refers to all crossing points for this measurement. See § Figure 8-83.
6. Defines as the absolute minimum or maximum instantaneous period. This includes cycle to cycle jitter, relative PPM tolerance, and spread spectrum modulation. See § Figure 8-86.
7. Defined as the maximum instantaneous voltage including overshoot. See § Figure 8-83.
8. Defined as the minimum instantaneous voltage including undershoot. See § Figure 8-83.
9. Defined as the total variation of all crossing voltages of Rising REFCLK+ and Falling REFCLK-. This is the maximum allowed variance in V_{CROSS} for any system. See § Figure 8-84.
10. Note deleted.
11. REFCLK+ and REFCLK- are to be measured at the load capacitors C_L . Single ended probes must be used for measurements requiring single ended measurements. Either single ended probes with math or differential probe can be used for differential measurements. Test load $C_L = 2$ pF.
12. T_{STABLE} is the time the differential clock must maintain a minimum ± 150 mV differential voltage after rising/falling edges before it is allowed to droop back into the $V_{RB} \pm 100$ mV differential range. See § Figure 8-88.

Symbol	Parameter	100 MHz Input		Unit	Note
		Min	Max		

13. PPM refers to parts per million and is a DC absolute period accuracy specification. 1 PPM is $1/1,000,000^{\text{th}}$ of 100.000000 MHz exactly or 100 Hz. For example, for 300 PPM, then we have an error budget of 100 Hz/
 $\text{PPM} \times 300 \text{ PPM} = 30 \text{ kHz}$. The period is to be measured with a frequency counter with measurement window set to 100 ms or greater.
14. Matching applies to rising edge rate for REFCLK+ and falling edge rate for REFCLK-. It is measured using a $\pm 75 \text{ mV}$ window centered on the median cross point where REFCLK+ rising meets REFCLK- falling. The median cross point is used to calculate the voltage thresholds the oscilloscope is to use for the edge rate calculations. The Rise Edge Rate of REFCLK+ should be compared to the Fall Edge Rate of REFCLK-; the maximum allowed difference should not exceed 20% of the slowest edge rate. See § Figure 8-85.

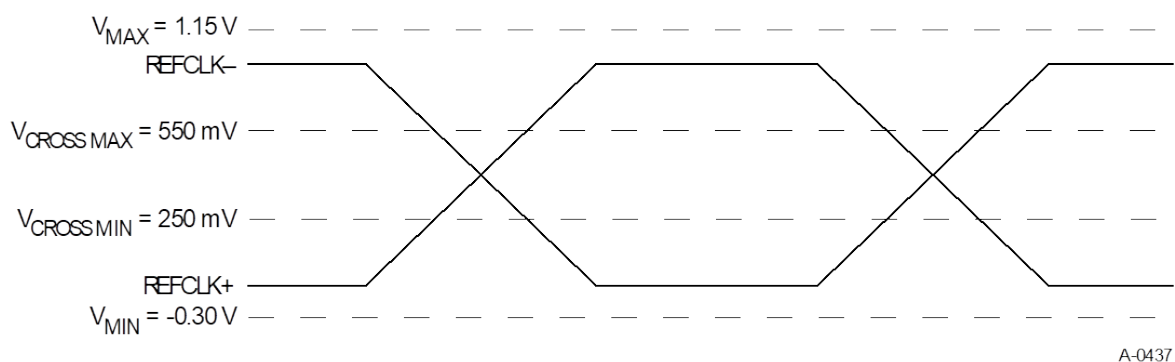


Figure 8-83 Single-Ended Measurement Points for Absolute Cross Point and Swing §

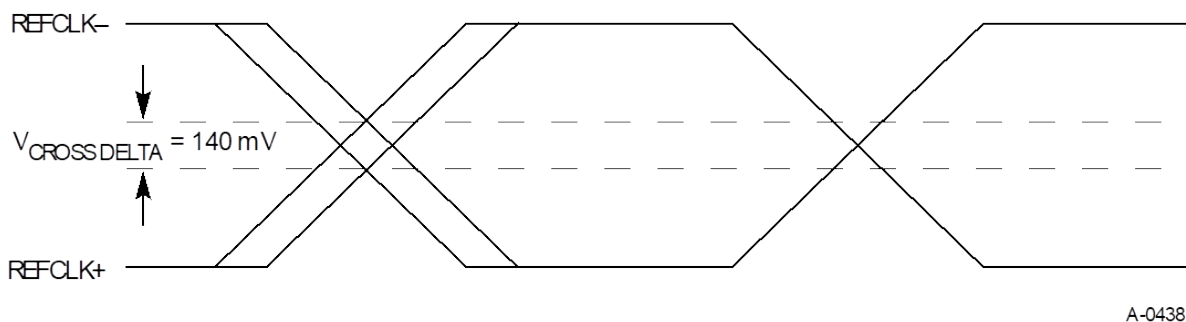


Figure 8-84 Single-Ended Measurement Points for Delta Cross Point §

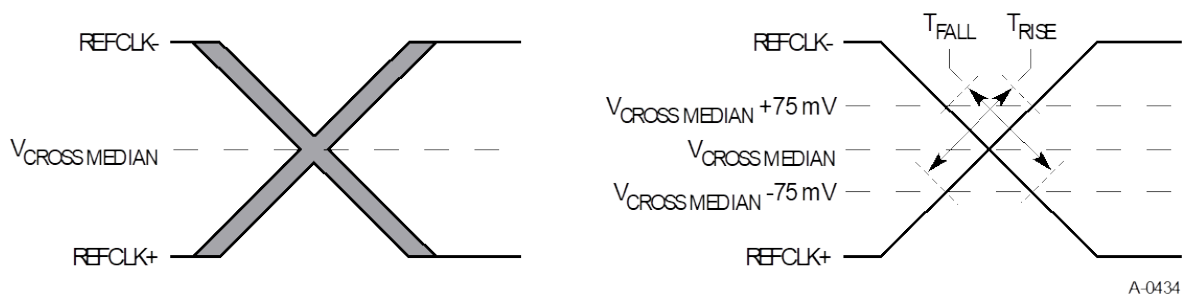
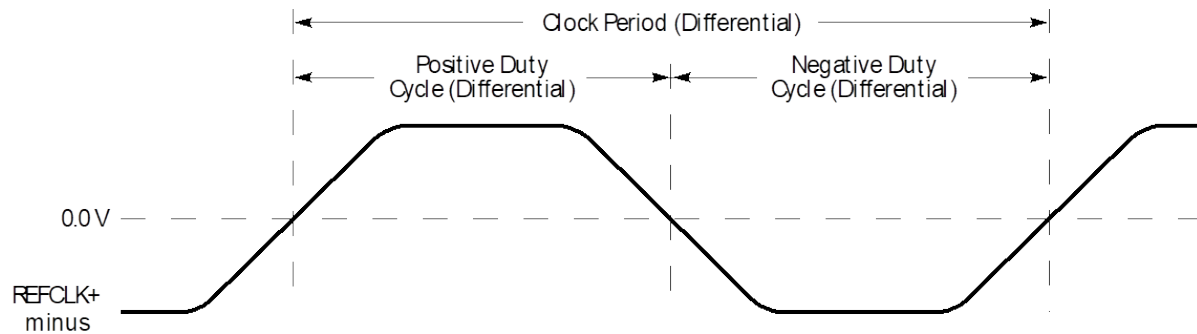
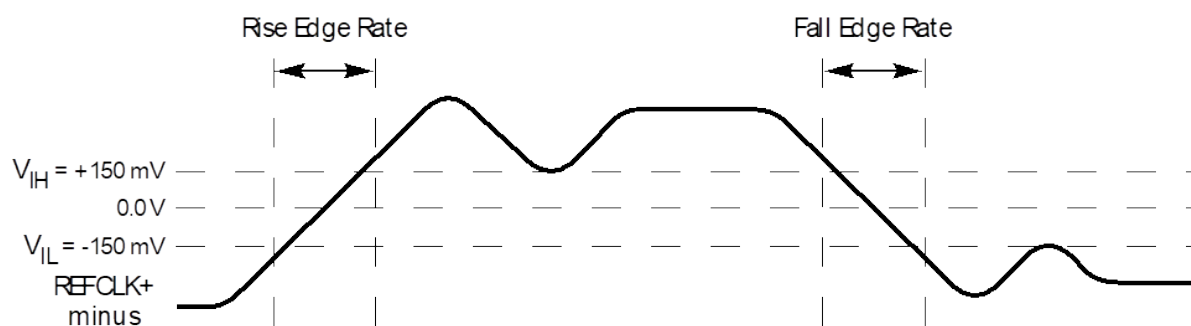


Figure 8-85 Single-Ended Measurement Points for Rise and Fall Time Matching §



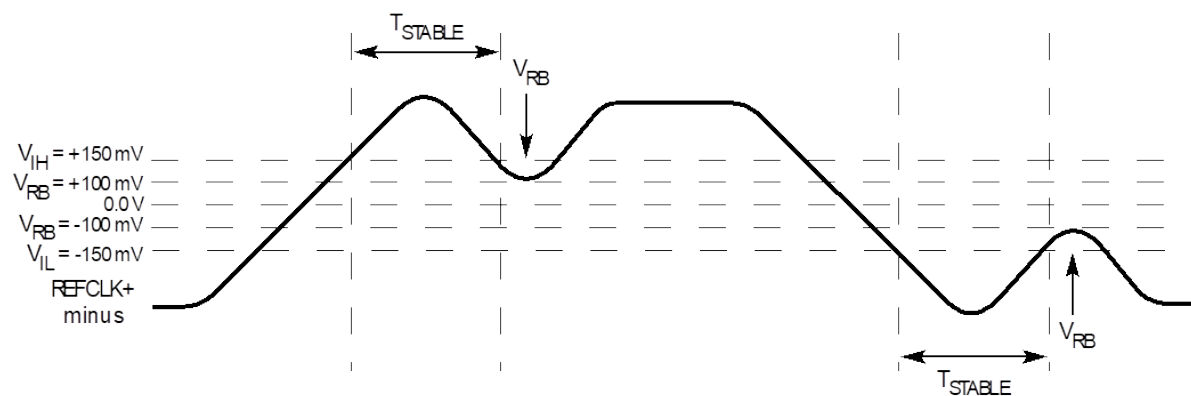
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Figure 8-86 Differential Measurement Points for Duty Cycle and Period §



A-0436

Figure 8-87 Differential Measurement Points for Rise and Fall Time §



A-0432

Figure 8-88 Differential Measurement Points for Ringback §

8.6.3 Data Rate Independent Refclk Parameters §

A number of Refclk parameters are data rate independent and are listed in the table below. $T_{\text{TRANSPORT-DELAY}}$ is defined in § Section 8.6.6 and illustrated in § Figure 8-91. It is relevant only for the Common Refclk architecture. For the SRIS mode the source of the SSC modulation is implementation dependent.

Table 8-19 Data Rate Independent Refclk Parameters §

Symbol	Description	Limits	Units	Notes
F_{REFCLK}	Refclk Frequency	99.97 (min) 100.03 (max)	MHz	
$F_{REFCLK_32G_64G}$	Refclk Frequency for devices that support 32.0 and 64.0 GT/s	99.99 (min) 100.01 (max)	MHz	
F_{SSC}	SSC frequency range	30 (min) 33 (max)	kHz	3
$T_{SSC-FREQ-DEVIATION}$	SSC deviation	-0.5 (min) 0.0 (max)	%	3
$T_{SSC-FREQ-DEVIATION_32G_64G_SRIS}$	SSC deviation for devices that support 32.0 and 64.0 GT/s and SRIS when operating in SRIS mode at all speeds	-0.3 (min) 0.0 (max)	%	3
$T_{TRANSPORT-DELAY}$	Tx-Rx transport delay	12 (max)	ns	1, 4
$T_{SSC-MAX-FREQ-SLEW}$	Max SSC df/dt	1250	ppm/ μ s	2, 3

Notes:

1. Parameter is relevant only for Common Refclk architecture.
2. Measurement is made over 0.5 μ s time interval with a 1st order LPF with an f_c of 60x the modulation frequency.
3. When testing the a device configured for the IR reference clock architecture the SSC related parameters must be tested with the Tx output data instead of the reference clock.
4. There are form factors (for example topologies including long cables) that may exceed the transport delay limit. Extra jitter from the large transport delay must be accounted by these form factor specifications.

8.6.3.1 Low Frequency Refclk Jitter Limits §

Refclks supporting SSC must meet an additional jitter limit over a range of low frequencies. Low frequency Refclk jitter limits are defined as a continuous, piece-wise linear graph from 30 kHz to 500 kHz as shown below. Unfiltered Refclk phase jitter must fall below this graph over the frequency range of interest.

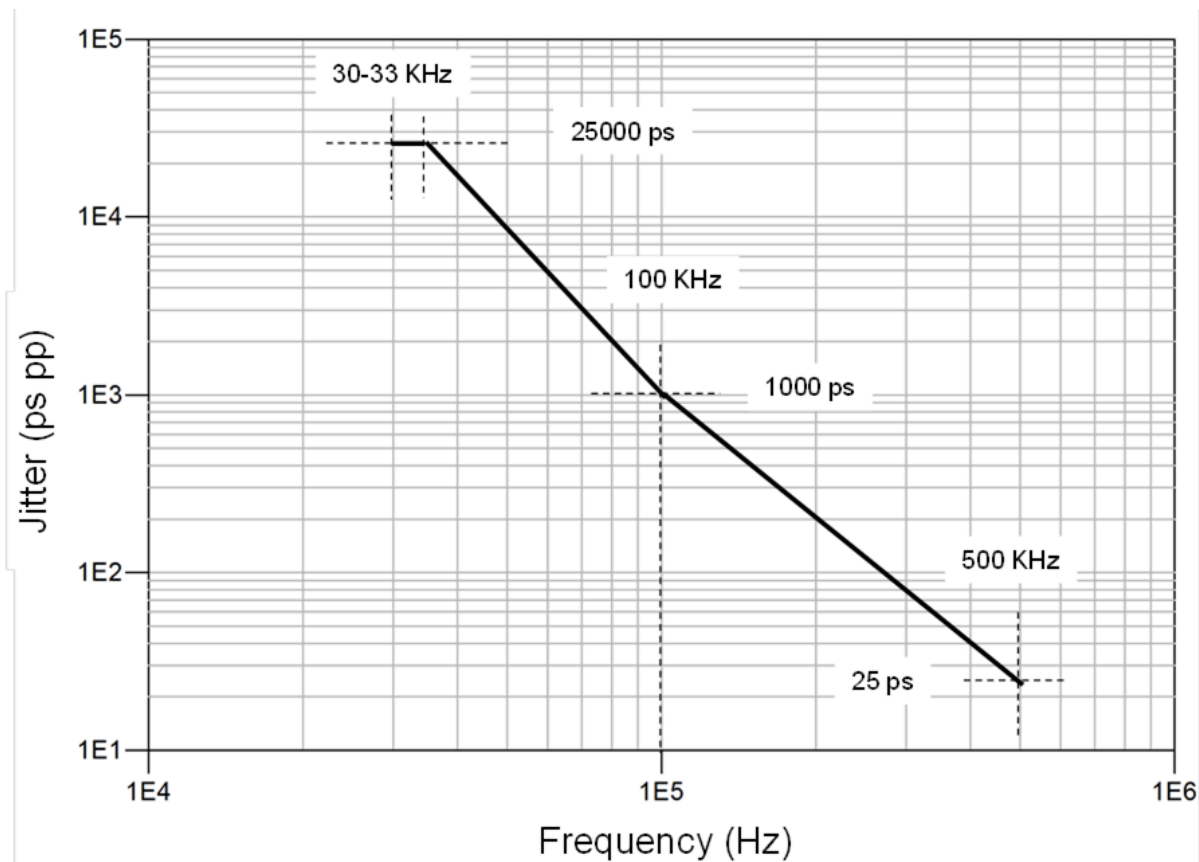


Figure 8-89 Limits for phase jitter from the Reference with 5000 ppm SSC §

8.6.4 Refclk Architectures Supported §

Two Refclk architectures are supported: **Common Refclk (CC)** and **Independent Refclk (IR)**. The CC clock architecture is described in terms of its topology and its corresponding jitter transfer function based on PLL and CDR characteristics. Finally, the corresponding Refclk jitter limits are given for each data rate. The jitter transfer function and corresponding jitter limits are not defined for the IR clock architecture. It is up to the implementer to trade off reference clock jitter and PLL characteristics to ensure that Transmitter requirements are met in the IR clocking mode.

8.6.5 Filtering Functions Applied to Raw Data §

Two types of filtering are applied to the raw Refclk data. The first, edge filtering, minimizes the measurement-induced jitter due to the finite sampling rate of the test equipment. The second, replicates the jitter filtering that is inherent in the combination of Tx/Rx PLLs, the Rx CDR and (where applicable) the transport delay. The combination of the preceding filter functions yields the effective Refclk jitter as it appears at the sample latch of the Receiver.

Note that the PLL and CDR filter functions represent minimally capable approximations to actual Receiver implementations and are not intended to define actual PLL or CDR implementations.

8.6.5.1 PLL Filter Transfer Function Example §

All PLLs are behaviorally modeled with a second order transfer function, $H(s)$, as defined in § Figure 8-91. The parameters defining the transfer function include the damping factor ζ and the natural frequency ω_n .

The relation between the 2nd order PLL (lowpass) natural frequency, ω_n and the 3 dB point ω_{3dB} , is given by the following expression:

$$\frac{\omega_{3dB}}{\omega_n} = \sqrt{\sqrt{(2\zeta^2 + 1)^2 + 1} + 2\zeta^2 + 1}$$

Equation 8-28 Relationship between 2nd order PLL natural frequency and 3 dB point §

The following plot of a 2nd order PLL illustrates the transfer function with an f_{3dB} of 5.0 MHz and 1.0 dB of peaking. This corresponds to $\zeta = 1.15$, and $\omega_n = 11.55$ Mrad/sec.

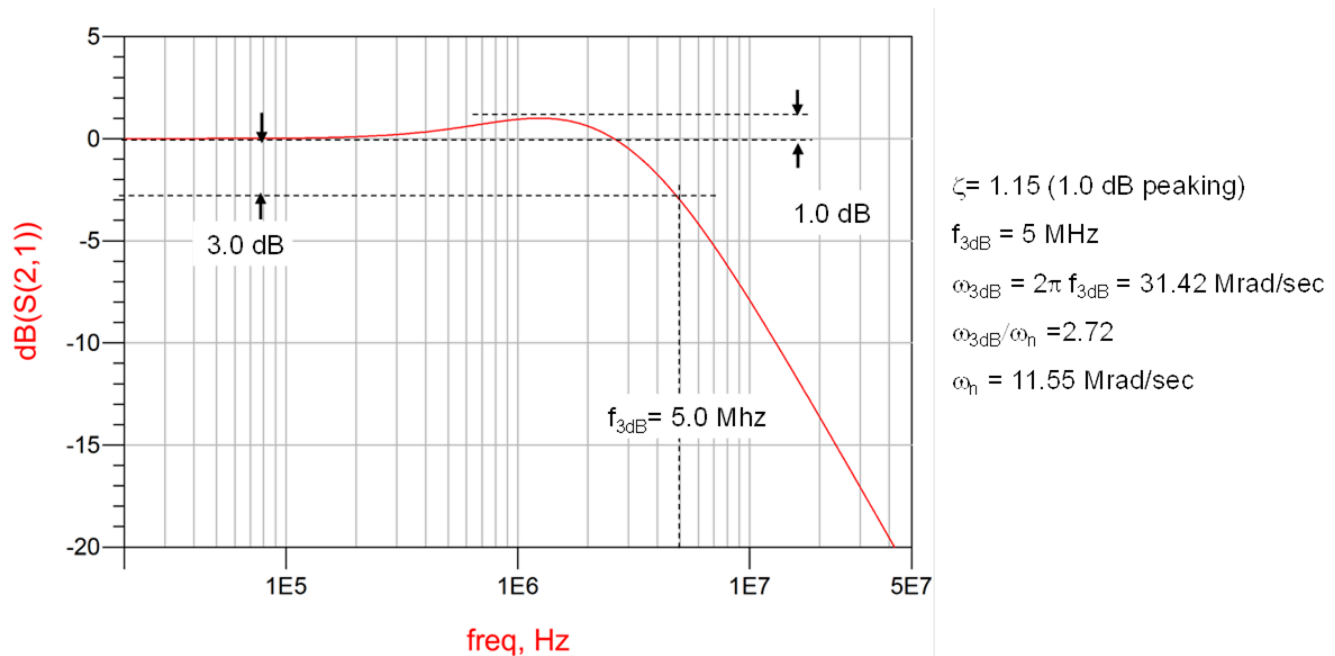


Figure 8-90 5 MHz PLL Transfer Function Example §

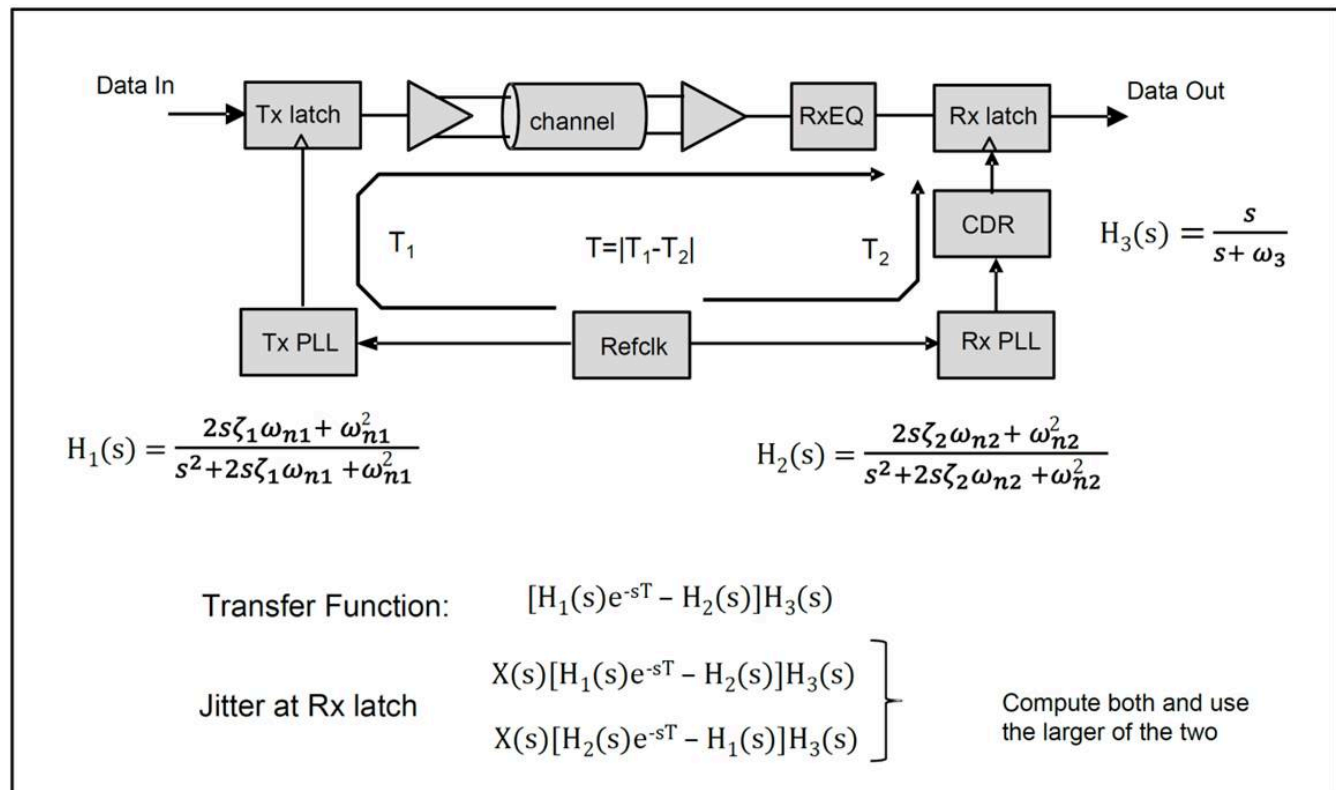
8.6.5.2 CDR Transfer Function Examples §

Depending on the Refclk architecture and data rate, either a first or higher order transfer function shall be used as a behavioral CDR bounding limit.

For behavioral CDR functions refer to § Section 8.3.5.5

8.6.6 Common Refclk Rx Architecture (CC) §

This architecture utilizes a single Refclk source that is distributed to both the Tx and Rx. Most of the SSC jitter sourced by the Refclk is propagated equally through Tx and Rx PLLs, and so intrinsically tracks LF jitter. This is particularly true for SSC which tends to be low frequency. § Figure 8-91 illustrates the Common Refclk Rx architecture, showing key jitter, delay, and PLL and CDR transfer function sources for all data rates except 32.0 and 64.0 GT/s. At 32.0 and 64.0 GT/s the only difference in the figure is Behavioral CDR transfer function as defined in § Section 8.3.5.5 . The amount of jitter appearing at the CDR is then defined by the difference function between the Tx and Rx PLLs multiplied by the CDR highpass characteristic.



Based on the above clock architecture, it is possible to define a difference function that corresponds to the worst case mismatch between Tx and Rx PLLs. Second order PLL transfer functions are assumed, (even though most PLL transfer functions are 3rd order or higher), since a 2nd order function tends to yield a slightly conservative difference function vis-a-vis most actual PLL implementations.

In the Common Refclk Rx architecture it is also necessary to comprehend a maximum Transmitter to Receiver transport delay difference. This delay delta is illustrated in § Figure 8-91 and represents the delay difference between the Transmitter data and recovered Receiver clock as seen at the inputs to the receiver's data latch.

8.6.6.1 Determining the Number of PLL BW and peaking Combinations §

A Tx or Rx PLL is defined by the combination of min/max bandwidth and peaking, making for a total of four possible limits. In the CC architecture both the Tx and Rx PLLs contribute to the jitter transfer function. At 2.5 GT/s only one set of BW/peaking limits is defined. If the combinations for the Tx and Rx PLLs limits are defined by the sets $(A_{TX}, B_{TX}, C_{TX}, D_{TX})$, $(A_{RX}, B_{RX}, C_{RX}, D_{RX})$ then it's easily demonstrated that there are a total of ten ways of selecting one element from each set. The delay term e^{-sT} can be applied to either $H_1(s)$ or $H_2(s)$, adding another six possibilities. Only six, as opposed to ten, terms are added when the delay term is considered because terms like A_{TX}, A_{RX} , are identical to A_{RX}, A_{TX} .

At 5.0 GT/s and higher data rates, two possible sets of limits of PLL bandwidth and peaking are defined, which may be defined by the sets $(A_{TX}, B_{TX}, C_{TX}, D_{TX})$ and $(E_{RX}, F_{RX}, G_{RX}, H_{RX})$. In this case the number of unique 2-element combinations from the above 4-element sets is 36, which increases to 64 when the delay term is considered.

8.6.6.2 CDR and PLL BW and Peaking Limits for Common Refclk §

The Common Refclk architecture filter function is dependent on the difference function defined by the BW and peaking of Tx and the Rx PLLs, plus the CDR high-pass characteristic. It is necessary to consider all corner case combinations of Tx and Rx PLL peaking and bandwidth plus the CDR characteristics at their minimum and maximum peaking values.

This procedure must be applied to all six data rates.

§ Figure 8-92 lists the PLL BW and CDR BW/peaking values that need to be applied as filter functions for 2.5 GT/s data rates. It is necessary to assign a min and a max value for peaking for the 2.5 GT/s PLL. The values of 0.01 dB and 3.0 dB represent best estimates of realistic PLL implementations.

The minimum peaking for 2.5 and 5.0 GT/s has been reduced to 0.01 dB to bring it into line with the 8.0 and 16.0 GT/s cases. Note that the Rx CDR is 1st order, so its natural frequency, ω_n can be directly obtained from its BW, unlike the 2nd order CDRs, where ω_n is a function of both BW and peaking. For 32.0 GT/s, a 2nd-order CDR filter with 20 MHz BW described by § Equation 8-25 must be used.

PLL #1, PLL #2	0.01 dB peaking	3.0 dB peaking	BW_{CDR}(min) = 1.5 MHz, 1st order CDR
BW _{PLL} (min) = 1.5 MHz	$\omega_{n1} = .336 \text{ Mrad/s}$ $\zeta_1 = 14$	$\omega_{n1} = 5.09 \text{ Mrad/s}$ $\zeta_1 = 0.54$	
BW _{PLL} (max) = 22 MHz	$\omega_{n1} = 4.93 \text{ Mrad/s}$ $\zeta_1 = 14$	$\omega_{n1} = 74.68 \text{ Mrad/s}$ $\zeta_1 = 0.54$	
16 combinations			2.5 GT/s

Figure 8-92 Common Refclk PLL and CDR Characteristics for 2.5 GT/s §

PLL and CDR jitter and peaking characteristics for 5.0, 8.0, and 16.0 GT/s yield a larger number of possible combinations because two sets of PLL BW and peaking limits are given. This choice to support two sets of BW and peaking was made to give designers as much latitude as possible when designing PLL circuits.

PLL #1			PLL #2		
0.01 dB peaking			0.01 dB peaking		
1.0 dB peaking			3.0 dB peaking		
$BW_{PLL}(\min) = 5.0$ MHz	$\omega_{n1} = 1.12$ Mrad/s $\zeta_1 = 14$	$\omega_{n1} = 11.01$ Mrad/s $\zeta_1 = 1.16$	$BW_{PLL}(\min) = 8.0$ MHz	$\omega_{n2} = 1.79$ Mrad/s $\zeta_2 = 14$	$\omega_{n2} = 26.86$ Mrad/s $\zeta_2 = 0.54$
$BW_{PLL}(\max) = 16$ MHz	$\omega_{n1} = 3.58$ Mrad/s $\zeta_1 = 14$	$\omega_{n1} = 35.26$ Mrad/s $\zeta_1 = 1.16$	$BW_{PLL}(\max) = 16$ MHz	$\omega_{n2} = 3.58$ Mrad/s $\zeta_2 = 14$	$\omega_{n2} = 53.73$ Mrad/s $\zeta_2 = 0.54$
$BW_{CDR}(\min) = 5$ MHz, 1 st order			64 combinations		
			5 GT/s		

Figure 8-93 Common Refclk PLL and CDR Characteristics for 5.0 GT/s §

PLL #1			PLL #2		
0.01 dB peaking			0.01 dB peaking		
2.0 dB peaking			1.0 dB peaking		
$BW_{PLL}(\min) = 2.0$ MHz	$\omega_{n1} = 0.448$ Mrad/s $\zeta_1 = 14$	$\omega_{n1} = 6.02$ Mrad/s $\zeta_1 = 0.73$	$BW_{PLL}(\min) = 2.0$ MHz	$\omega_{n2} = 0.448$ Mrad/s $\zeta_2 = 14$	$\omega_{n2} = 4.62$ Mrad/s $\zeta_2 = 1.15$
$BW_{PLL}(\max) = 4.0$ MHz	$\omega_{n1} = 0.896$ Mrad/s $\zeta_1 = 14$	$\omega_{n1} = 12.04$ Mrad/s $\zeta_1 = 0.73$	$BW_{PLL}(\max) = 5.0$ MHz	$\omega_{n2} = 1.12$ Mrad/s $\zeta_2 = 14$	$\omega_{n2} = 11.53$ Mrad/s $\zeta_2 = 1.15$
$BW_{CDR}(\min) = 10$ MHz, 1 st order			64 combinations		
			8.0, 16.0 GT/s		

Figure 8-94 Common Refclk PLL and CDR Characteristics for 8.0 and 16.0 GT/s §

PLL #1, PLL #2		0.01 dB peaking	2.0 dB peaking	32.0 GT/s CC	CDR
BW _{PLL} (min) = 0.5 MHz		ω _{n1} = 0.112 Mrad/s ζ ₁ = 14	ω _{n1} = 1.50 Mrad/s ζ ₁ = 0.73		
BW _{PLL} (max) = 1.8 MHz		ω _{n1} = 0.403 Mrad/s ζ ₁ = 14	ω _{n1} = 5.42 Mrad/s ζ ₁ = 0.73		
16 combinations				32.0 GT/s	

Figure 8-95 Common Refclk PLL and CDR Characteristics for 32.0 GT/s §

PLL #1, PLL #2		0.01 dB peaking	2.0 dB peaking	64.0 GT/s CC	CDR
BW _{PLL} (min) = 0.5 MHz		ω _{n1} = 0.112 Mrad/s ζ ₁ = 14	ω _{n1} = 1.50 Mrad/s ζ ₁ = 0.73		
BW _{PLL} (max) = 1.0 MHz		ω _{n1} = 0.224 Mrad/s ζ ₁ = 14	ω _{n1} = 3.00 Mrad/s ζ ₁ = 0.73		
16 combinations				64.0 GT/s	

Figure 8-96 Common Refclk PLL and CDR Characteristics for 64.0 GT/s §

8.6.7 Jitter Limits for Refclk Architectures §

§ Table 8-20 lists the jitter limits for the CC Refclk architecture at each of the four data rates.

Jitter at 2.5 GT/s is measured as a peak to peak jitter value, because a substantial proportion of the jitter is SSC harmonics which appears at the receiver as Dj. The combination of the 2.5 GT/s PLL and CDR bandwidths passes a significant amount of SSC residual, where it appears Dj.

For 5.0, 8.0, and 16.0 GT/s jitter is specified as an RMS (Rj) value. These signaling speeds utilize a lower PLL BW and a higher CDR BW, and the effect is to suppress SSC harmonics such that almost all the jitter appears as Rj.

Table 8-20 Jitter Limits for CC Architecture §

Data Rate	CC jitter Limit	Notes
2.5 GT/s	86 ps pp	1, 2
5.0 GT/s	3.1 ps RMS	1, 2
8.0 GT/s	1.0 ps RMS	1, 2
16.0 GT/s	0.5 ps RMS	1, 2, 3, 4
32.0 GT/s	0.15 ps RMS	1, 2, 3, 5
64.0 GT/s	0.1 ps RMS	1, 2, 3, 6

Notes:

1. The Refclk jitter is measured after applying the filter function in § Figure 8-91
2. Jitter measurements shall be made with a capture of at least 100,000 clock cycles captured by a real time oscilloscope (RTO) with a sample rate of 20 GS/s or greater. Broadband oscilloscope noise must be minimized in the measurement. The measured PP jitter is used (no extrapolation) for RTO measurements. Alternately - Jitter measurements may be used with a Phase Noise Analyzer (PNA) extending (flat) and integrating and folding the frequency content up to an offset from the carrier frequency of at least 200 MHz (at 300 MHz absolute frequency) below the Nyquist frequency. For PNA measurements for the 2.5 GT/s data rate the RMS jitter is converted to peak to peak jitter using a multiplication factor of 8.83. In the case where real time oscilloscope and PNA measurements have both been done and produce different results the RTO result must be used.
3. For the 16.0, 32.0, and 64.0 GT/s CC measurements SSC spurs from the fundamental and harmonics are removed up to a cutoff frequency of 2 MHz taking care to minimize removal of any non-SSC content.
4. Note that 0.7 ps RMS is to be used in channel simulations to account for additional noise in a real system.
5. Note that 0.25 ps RMS is to be used in channel simulations to account for additional noise in a real system.
6. Note that 0.15 ps RMS is to be used in channel simulations to account for additional noise in a real system.

8.6.8 Form Factor Requirements for RefClock Architectures §

Each form factor specification must include the following table (see § Table 8-21) to provide a clear summary of the clocking architecture requirements for devices that support the form factor specification. For each clocking architecture the table indicates whether that architecture is required, optional, or not allowed for this form factor. Note that this refers to the operation of the device not the underlying silicon capabilities.

A form factor must provide the CLKREQ# signal if it supports L1 PM Substates. Form factor specifications must indicate if the CLKREQ# signal is required, optional, or not allowed.

Table 8-21 Form Factor Clocking Architecture Requirements §

Clock Architecture	System Board (Motherboard)	Add-in Card (Module)	Retimer
Common	**	**	**

Clock Architecture	System Board (Motherboard)	Add-in Card (Module)	Retimer
SRNS	**	**	**
SRIS	**	**	**

** Each entry in the table must be filled in with one of: Required, Optional, or Not Allowed

If the Common Reference Clock architecture is required or optional for the form factor, then there must be an additional table (see § Table 8-22) providing details for the common clock. Each entry in the table is marked required, optional, not allowed, or NA. “Clock Source” indicates the source of the common reference clock, if applicable. “SSC” indicates whether the clock source is spread.

Table 8-22 Form Factor Common Clock Architecture Details §

Common Clock Details	System Board (Motherboard)	Add-in Card (Module)	Retimer
Clock Source			
SSC			

If a form factor has clocking requirements that cannot be provided in this simple one or two table form then careful consideration must be given to ensure that the form factor requirements are supported by this specification.

As an example the populated tables are shown for a hypothetical form factor that requires all components use the common clock architecture and does not allow the use of any other clocking architecture (see § Table 8-23 and § Table 8-24). The common clock source is required to be provided by the motherboard component and may optionally have SSC. L1 PM Substates are not supported and therefore CLKREQ# is not defined as a connector signal for this example form factor.

Table 8-23 Form Factor Clocking Architecture Requirements Example §

Clock Architecture	System Board (Motherboard)	Add-in Card (Module)	Retimer
Common	Required	Required	Required
SRNS	Not Allowed	Not Allowed	Not Allowed
SRIS	Not Allowed	Not Allowed	Not Allowed

Table 8-24 Form Factor Common Clock Architecture Details Example §

Common Clock Details	System Board (Motherboard)	Add-in Card (Module)	Retimer
Clock Source	Required	Not Allowed	Not Allowed
SSC	Optional	N/A	N/A

It is important for form factor specifications to recognize that the CLKREQ# signal is required if L1 PM Substates are to be supported, and that for L1 PM Substates the CLKREQ# signal is used even if there is no common reference clock.

If a form factor has clocking requirements that cannot be provided in this simple one or two table form then careful consideration must be given to ensure that the form factor requirements are supported by this specification.